

```
In [49]: import matplotlib.pyplot as plt
import numpy as np
import pandas as pd
%matplotlib inline
import seaborn as sns
%matplotlib inline

from plotly.subplots import make_subplots
import plotly.graph_objects as go
import plotly.graph_objects as px
```

```
In [50]: df=pd.read_csv(r"C:\Users\HP\Downloads\netflix_titles.csv\netflix_titles.csv")
```

```
In [51]: df
```

Out[51]:

	show_id	type	title	director	cast	country	date_added	release_year
0	s1	Movie	Dick Johnson Is Dead	Kirsten Johnson	NaN	United States	September 25, 2021	2020
1	s2	TV Show	Blood & Water	NaN	Ama Qamata, Khosi Ngema, Gail Mabalane, Thaban...	South Africa	September 24, 2021	2021
2	s3	TV Show	Ganglands	Julien Leclercq	Sami Bouajila, Tracy Gotoas, Samuel Jouy, Nabi...	NaN	September 24, 2021	2021
3	s4	TV Show	Jailbirds New Orleans	NaN	NaN	NaN	September 24, 2021	2021
4	s5	TV Show	Kota Factory	NaN	Mayur More, Jitendra Kumar, Ranjan Raj, Alam K...	India	September 24, 2021	2021
...	...	...	...	...	...	...	...	...
8802	s8803	Movie	Zodiac	David Fincher	Mark Ruffalo, Jake Gyllenhaal, Robert Downey J...	United States	November 20, 2019	2007
8803	s8804	TV Show	Zombie Dumb	NaN	NaN	NaN	July 1, 2019	2018
8804	s8805	Movie	Zombieland	Ruben Fleischer	Jesse Eisenberg, Woody	United States	November 1, 2019	2009

show_id	type	title	director	cast	country	date_added	release_year	
8805	s8806	Movie	Zoom	Peter Hewitt	Harrelson, Emma Stone, ...	United States	January 11, 2020	2006
					Tim Allen, Courteney Cox, Chevy Chase, Kate Ma...			
8806	s8807	Movie	Zubaan	Mozez Singh	Vicky Kaushal, Sarah-Jane Dias, Raaghav Chanan...	India	March 2, 2019	2015

8807 rows × 12 columns

```
In [52]: df.head(15)
```

Out[52]:

	show_id	type	title	director	cast	country	date_added	release_yr
0	s1	Movie	Dick Johnson Is Dead	Kirsten Johnson	NaN	United States	September 25, 2021	2020
1	s2	TV Show	Blood & Water	NaN	Ama Qamata, Khosi Ngema, Gail Mabalane, Thaban...	South Africa	September 24, 2021	2020
2	s3	TV Show	Ganglands	Julien Leclercq	Sami Bouajila, Tracy Gotoas, Samuel Jouy, Nabi...	NaN	September 24, 2021	2020
3	s4	TV Show	Jailbirds New Orleans	NaN	NaN	NaN	September 24, 2021	2020
4	s5	TV Show	Kota Factory	NaN	Mayur More, Jitendra Kumar, Ranjan Raj, Alam K...	India	September 24, 2021	2020
5	s6	TV Show	Midnight Mass	Mike Flanagan	Kate Siegel, Zach Gilford, Hamish Linklater, H...	NaN	September 24, 2021	2020
6	s7	Movie	My Little Pony: A New Generation	Robert Cullen, José Luis Ucha	Vanessa Hudgens, Kimiko Glenn, James Marsden, ...	NaN	September 24, 2021	2020
7	s8	Movie	Sankofa	Haile Gerima	Kofi Ghanaba, Oyafunmike Ogunlano, Alexandra D...	United States, Ghana, Burkina Faso,	September 24, 2021	1990

show_id	type		title	director	cast	country	date_added	release_yr
						United Kin...		
8	s9	TV Show	The Great British Baking Show	Andy Devonshire	Mel Giedroyc, Sue Perkins, Mary Berry, Paul Ho...	United Kingdom	September 24, 2021	2021
9	s10	Movie	The Starling	Theodore Melfi	Melissa McCarthy, Chris O'Dowd, Kevin Kline, T...	United States	September 24, 2021	2021
10	s11	TV Show	Vendetta: Truth, Lies and The Mafia	NaN	NaN	NaN	September 24, 2021	2021
11	s12	TV Show	Bangkok Breaking	Kongkiat Komesiri	Sukollawat Kanarot, Sushar Manaying, Pavarit M...	NaN	September 23, 2021	2021
12	s13	Movie	Je Suis Karl	Christian Schwochow	Luna Wedler, Jannis Niewöhner, Milan Peschel, ...	Germany, Czech Republic	September 23, 2021	2021
13	s14	Movie	Confessions of an Invisible Girl	Bruno Garotti	Klara Castanho, Lucca Picon, Júlia Gomes, Marc...	NaN	September 22, 2021	2021
14	s15	TV Show	Crime Stories: India Detectives	NaN	NaN	NaN	September 22, 2021	2021

In [53]: `df.tail(15)`

Out[53]:

	show_id	type	title	director	cast	country	date_added	release_y
8792	s8793	Movie	Young Tiger	Mu Chu	Qiu Yuen, Charlie Chin, Jackie Chan, Hu Chin, ...	Hong Kong	November 1, 2016	1
8793	s8794	Movie	Yours, Mine and Ours	Raja Gosnell	Dennis Quaid, Rene Russo, Sean Faris, Katija P...	United States	November 20, 2019	2
8794	s8795	Movie	اشتباك	Mohamed Diab	Nelly Karim, Hany Adel, Tarek Abdel Aziz, Ahme...	Egypt, France	October 11, 2018	2
8795	s8796	TV Show	Yu-Gi-Oh! Arc-V	NaN	Mike Liscio, Emily Bauer, Billy Bob Thompson, ...	Japan, Canada	May 1, 2018	2
8796	s8797	TV Show	Yunus Emre	NaN	Gökhan Atalay, Payidar Tüfekçioğlu, Baran Akbu...	Turkey	January 17, 2017	2
8797	s8798	TV Show	Zak Storm	NaN	Michael Johnston, Jessica Gee-George, Christin...	United States, France, South Korea, Indonesia	September 13, 2018	2
8798	s8799	Movie	Zed Plus	Chandra Prakash Dwivedi	Adil Hussain, Mona Singh, K.K. Raina, Sanjay M...	India	December 31, 2019	2
8799	s8800	Movie	Zenda	Avadhoot Gupte	Santosh Juvekar, Siddharth Chandekar, Sachit P...	India	February 15, 2018	2
8800	s8801	TV Show	Zindagi Gulzar Hai	NaN	Sanam Saeed, Fawad	Pakistan	December 15, 2016	2

	show_id	type	title	director	cast	country	date_added	release_y
					Khan, Ayesha Omer, Mehreen ...			
8801	s8802	Movie	Zinzana	Majid Al Ansari	Ali Suliman, Saleh Bakri, Yasa, Ali Al- Jabri, ...	United Arab Emirates, Jordan	March 9, 2016	2016
8802	s8803	Movie	Zodiac	David Fincher	Mark Ruffalo, Jake Gyllenhaal, Robert Downey J...	United States	November 20, 2019	2007
8803	s8804	TV Show	Zombie Dumb	NaN	NaN	NaN	July 1, 2019	2019
8804	s8805	Movie	Zombieland	Ruben Fleischer	Jesse Eisenberg, Woody Harrelson, Emma Stone, ...	United States	November 1, 2019	2009
8805	s8806	Movie	Zoom	Peter Hewitt	Tim Allen, Courteney Cox, Chevy Chase, Kate Ma...	United States	January 11, 2020	2006
8806	s8807	Movie	Zubaan	Mozez Singh	Vicky Kaushal, Sarah-Jane Dias, Raaghav Chanan...	India	March 2, 2019	2012

In [54]: df.duplicated().sum()

Out[54]: np.int64(0)

In [55]: df.info()

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 8807 entries, 0 to 8806
Data columns (total 12 columns):
 #   Column          Non-Null Count  Dtype
---  -
 0   show_id         8807 non-null   object
 1   type            8807 non-null   object
 2   title           8807 non-null   object
 3   director        6173 non-null   object
 4   cast            7982 non-null   object
 5   country         7976 non-null   object
 6   date_added      8797 non-null   object
 7   release_year    8807 non-null   int64
 8   rating          8803 non-null   object
 9   duration        8804 non-null   object
10   listed_in       8807 non-null   object
11   description     8807 non-null   object
dtypes: int64(1), object(11)
memory usage: 825.8+ KB

```

```
In [56]: df.type.value_counts()
```

```

Out[56]: type
Movie      6131
TV Show    2676
Name: count, dtype: int64

```

```
In [57]: df.shape
```

```
Out[57]: (8807, 12)
```

```
In [58]: df.describe()
```

```

Out[58]:
      release_year
count  8807.000000
mean   2014.180198
std     8.819312
min    1925.000000
25%    2013.000000
50%    2017.000000
75%    2019.000000
max    2021.000000

```

```
In [59]: df.type.value_counts()
```

```
Out[59]: type
         Movie      6131
         TV Show    2676
         Name: count, dtype: int64
```

```
In [60]: df['type'] = df['type'].apply(lambda x: 1 if x == 'Movie' else 0)
```

```
In [61]: df
```

Out[61]:

	show_id	type	title	director	cast	country	date_added	release_year
0	s1	1	Dick Johnson Is Dead	Kirsten Johnson	NaN	United States	September 25, 2021	2020
1	s2	0	Blood & Water	NaN	Ama Qamata, Khosi Ngema, Gail Mabalane, Thaban...	South Africa	September 24, 2021	2021
2	s3	0	Ganglands	Julien Leclercq	Sami Bouajila, Tracy Gotoas, Samuel Jouy, Nabi...	NaN	September 24, 2021	2021
3	s4	0	Jailbirds New Orleans	NaN	NaN	NaN	September 24, 2021	2021
4	s5	0	Kota Factory	NaN	Mayur More, Jitendra Kumar, Ranjan Raj, Alam K...	India	September 24, 2021	2021
...	...	...	...	...	...	...	...	...
8802	s8803	1	Zodiac	David Fincher	Mark Ruffalo, Jake Gyllenhaal, Robert Downey J...	United States	November 20, 2019	2007
8803	s8804	0	Zombie Dumb	NaN	NaN	NaN	July 1, 2019	2018
8804	s8805	1	Zombieland	Ruben Fleischer	Jesse Eisenberg, Woody	United States	November 1, 2019	2009

show_id	type		title	director	cast	country	date_added	release_year
					Harrelson, Emma Stone, ...			
8805	s8806	1	Zoom	Peter Hewitt	Tim Allen, Courteney Cox, Chevy Chase, Kate Ma...	United States	January 11, 2020	2006
8806	s8807	1	Zubaan	Mozez Singh	Vicky Kaushal, Sarah- Jane Dias, Raaghav Chanan...	India	March 2, 2019	2015

8807 rows × 12 columns

```
In [62]: df['type'].unique()
```

```
Out[62]: array([1, 0])
```

```
In [63]: df.isnull().sum()
```

```
Out[63]: show_id      0
         type        0
         title       0
         director    2634
         cast       825
         country     831
         date_added   10
         release_year  0
         rating      4
         duration    3
         listed_in   0
         description  0
         dtype: int64
```

```
In [64]: df.columns
```

```
Out[64]: Index(['show_id', 'type', 'title', 'director', 'cast', 'country', 'date_added',
               'release_year', 'rating', 'duration', 'listed_in', 'description'],
              dtype='object')
```

```
In [65]: df['director'].replace(np.nan, "Unknown", inplace=True)
         df['cast'].replace(np.nan, "Unknown", inplace=True)
         df.isnull().sum()
```

C:\Users\HP\AppData\Local\Temp\ipykernel_21952\619762823.py:1: FutureWarning:

A value is trying to be set on a copy of a DataFrame or Series through chained assignment using an inplace method.

The behavior will change in pandas 3.0. This inplace method will never work because the intermediate object on which we are setting values always behaves as a copy.

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inplace=True)' or df[col] = df[col].method(value) instead, to perform the operation inplace on the original object.

C:\Users\HP\AppData\Local\Temp\ipykernel_21952\619762823.py:2: FutureWarning:

A value is trying to be set on a copy of a DataFrame or Series through chained assignment using an inplace method.

The behavior will change in pandas 3.0. This inplace method will never work because the intermediate object on which we are setting values always behaves as a copy.

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inplace=True)' or df[col] = df[col].method(value) instead, to perform the operation inplace on the original object.

```
Out[65]: show_id      0
         type        0
         title       0
         director    0
         cast        0
         country     831
         date_added  10
         release_year 0
         rating      4
         duration    3
         listed_in   0
         description 0
         dtype: int64
```

```
In [66]: df['country'].value_counts().to_frame().head(1)
```

```
Out[66]:
```

	count
country	
United States	2818

```
In [67]: df['country'].replace(np.nan, "United States", inplace=True)
```

C:\Users\HP\AppData\Local\Temp\ipykernel_21952\1862763933.py:1: FutureWarning:

A value is trying to be set on a copy of a DataFrame or Series through chained assignment using an inplace method.

The behavior will change in pandas 3.0. This inplace method will never work because the intermediate object on which we are setting values always behaves as a copy.

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inplace=True)' or df[col] = df[col].method(value) instead, to perform the operation inplace on the original object.

In [68]: `df.isnull().sum()`

```
Out[68]: show_id      0
         type        0
         title       0
         director    0
         cast        0
         country     0
         date_added  10
         release_year 0
         rating      4
         duration    3
         listed_in   0
         description 0
         dtype: int64
```

In [69]: `df['date_added'].value_counts().to_frame().head(1)`

```
Out[69]:
```

	count
date_added	
January 1, 2020	109

In [70]: `df['date_added'].replace(np.nan,"January 1, 2020",inplace=True)`

C:\Users\HP\AppData\Local\Temp\ipykernel_21952\2545193644.py:1: FutureWarning:

A value is trying to be set on a copy of a DataFrame or Series through chained assignment using an inplace method.

The behavior will change in pandas 3.0. This inplace method will never work because the intermediate object on which we are setting values always behaves as a copy.

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inplace=True)' or df[col] = df[col].method(value) instead, to perform the operation inplace on the original object.

In [71]: `df['season_count'] = df['duration'].str.extract(r'(\d+)\sSeason')`

```
In [72]: df['season_count'] = pd.to_numeric(df['season_count'], errors='coerce')
```

```
In [73]: df = df.replace(np.nan, 0, regex=True)
```

```
In [74]: df['season_count'] = pd.to_numeric(df['season_count'], downcast = 'signed')
```

```
In [75]: df
```

Out[75]:

	show_id	type	title	director	cast	country	date_added	release_year
0	s1	1	Dick Johnson Is Dead	Kirsten Johnson	Unknown	United States	September 25, 2021	2020
1	s2	0	Blood & Water	Unknown	Ama Qamata, Khosi Ngema, Gail Mabalane, Thaban...	South Africa	September 24, 2021	2021
2	s3	0	Ganglands	Julien Leclercq	Sami Bouajila, Tracy Gotoas, Samuel Jouy, Nabi...	United States	September 24, 2021	2021
3	s4	0	Jailbirds New Orleans	Unknown	Unknown	United States	September 24, 2021	2021
4	s5	0	Kota Factory	Unknown	Mayur More, Jitendra Kumar, Ranjan Raj, Alam K...	India	September 24, 2021	2021
...	...	...	...	...	...	...	...	...
8802	s8803	1	Zodiac	David Fincher	Mark Ruffalo, Jake Gyllenhaal, Robert Downey J...	United States	November 20, 2019	2007
8803	s8804	0	Zombie Dumb	Unknown	Unknown	United States	July 1, 2019	2018
8804	s8805	1	Zombieland	Ruben Fleischer	Jesse Eisenberg, Woody	United States	November 1, 2019	2009

show_id	type		title	director	cast	country	date_added	release_year
					Harrelson, Emma Stone, ...			
8805	s8806	1	Zoom	Peter Hewitt	Tim Allen, Courteney Cox, Chevy Chase, Kate Ma...	United States	January 11, 2020	2006
8806	s8807	1	Zubaan	Mozez Singh	Vicky Kaushal, Sarah- Jane Dias, Raaghav Chanan...	India	March 2, 2019	2015

8807 rows × 13 columns

```
In [76]: df['season_count'] = (df['duration'].str.extract(r'(\d+)\sSeason')[0].fillna(0).astype(int)
In [77]: df
```

Out[77]:

	show_id	type	title	director	cast	country	date_added	release_year
0	s1	1	Dick Johnson Is Dead	Kirsten Johnson	Unknown	United States	September 25, 2021	2020
1	s2	0	Blood & Water	Unknown	Ama Qamata, Khosi Ngema, Gail Mabalane, Thaban...	South Africa	September 24, 2021	2021
2	s3	0	Ganglands	Julien Leclercq	Sami Bouajila, Tracy Gotoas, Samuel Jouy, Nabi...	United States	September 24, 2021	2021
3	s4	0	Jailbirds New Orleans	Unknown	Unknown	United States	September 24, 2021	2021
4	s5	0	Kota Factory	Unknown	Mayur More, Jitendra Kumar, Ranjan Raj, Alam K...	India	September 24, 2021	2021
...	...	...	...	...	...	...	...	...
8802	s8803	1	Zodiac	David Fincher	Mark Ruffalo, Jake Gyllenhaal, Robert Downey J...	United States	November 20, 2019	2007
8803	s8804	0	Zombie Dumb	Unknown	Unknown	United States	July 1, 2019	2018
8804	s8805	1	Zombieland	Ruben Fleischer	Jesse Eisenberg, Woody	United States	November 1, 2019	2009

show_id	type		title	director	cast	country	date_added	release_year
					Harrelson, Emma Stone, ...			
8805	s8806	1	Zoom	Peter Hewitt	Tim Allen, Courteney Cox, Chevy Chase, Kate Ma...	United States	January 11, 2020	2006
8806	s8807	1	Zubaan	Mozez Singh	Vicky Kaushal, Sarah- Jane Dias, Raaghav Chanan...	India	March 2, 2019	2015

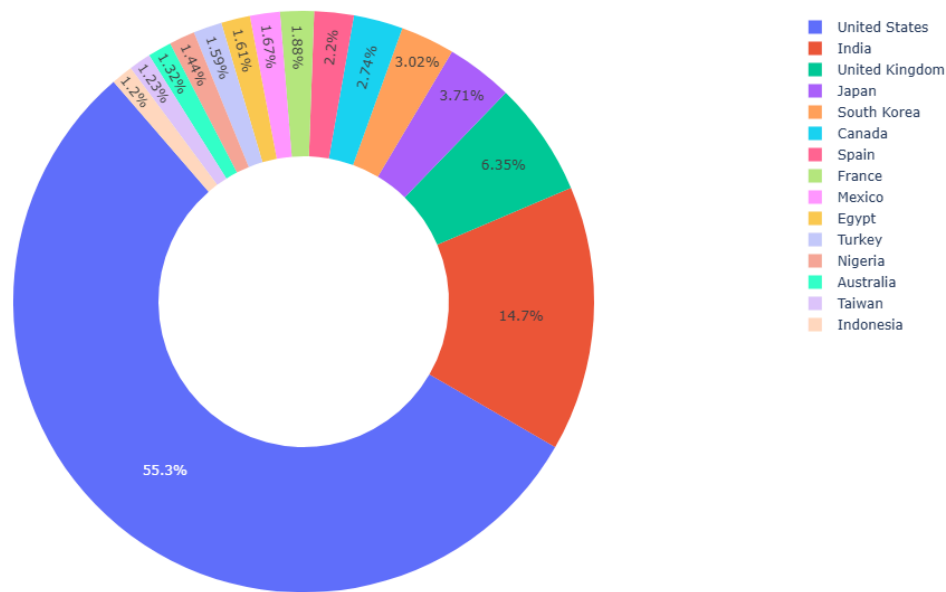
8807 rows × 13 columns

```
In [78]: labels = list(df['country'].value_counts().head(15).index)
values = list(df['country'].value_counts().head(15).values)

fig = go.Figure(
    data=[go.Pie(
        labels=labels,
        values=values,
        rotation=120,
        hole=0.5
    )],
    layout=go.Layout(
        title="Netflix's Worldwide Content Distribution",
        width=900,
        height=700
    )
)

fig.show()
```

Netflix's Worldwide Content Distribution



```
In [79]: df.loc[df['country']=='United States']['type'].value_counts()
```

```
Out[79]: type
1      2498
0      1151
Name: count, dtype: int64
```

```
In [80]: us_df = df.loc[df['country'] == 'United States']
```

```
In [81]: us_df
```

Out[81]:

	show_id	type	title	director	cast	country	date_added	release_year
0	s1	1	Dick Johnson Is Dead	Kirsten Johnson	Unknown	United States	September 25, 2021	2020
2	s3	0	Ganglands	Julien Leclercq	Sami Bouajila, Tracy Gotoas, Samuel Jouy, Nabi...	United States	September 24, 2021	2021
3	s4	0	Jailbirds New Orleans	Unknown	Unknown	United States	September 24, 2021	2021
5	s6	0	Midnight Mass	Mike Flanagan	Kate Siegel, Zach Gilford, Hamish Linklater, H...	United States	September 24, 2021	2021
6	s7	1	My Little Pony: A New Generation	Robert Cullen, José Luis Ucha	Vanessa Hudgens, Kimiko Glenn, James Marsden, ...	United States	September 24, 2021	2021
...	...	...	...	...	...	...	...	...
8793	s8794	1	Yours, Mine and Ours	Raja Gosnell	Dennis Quaid, Rene Russo, Sean Faris, Katija P...	United States	November 20, 2019	2005
8802	s8803	1	Zodiac	David Fincher	Mark Ruffalo, Jake Gyllenhaal, Robert Downey J...	United States	November 20, 2019	2007

	show_id	type	title	director	cast	country	date_added	release_year
8803	s8804	0	Zombie Dumb	Unknown	Unknown	United States	July 1, 2019	2018
8804	s8805	1	Zombieland	Ruben Fleischer	Jesse Eisenberg, Woody Harrelson, Emma Stone, ...	United States	November 1, 2019	2009
8805	s8806	1	Zoom	Peter Hewitt	Tim Allen, Courteney Cox, Chevy Chase, Kate Ma...	United States	January 11, 2020	2006

3649 rows × 13 columns

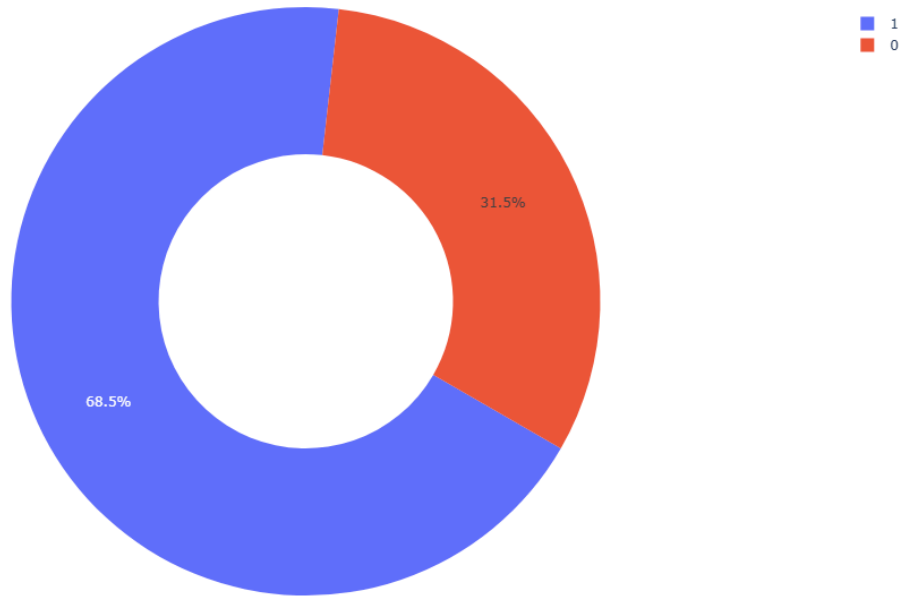
```
In [82]: # United States filter
us_df = df.loc[df['country'] == 'United States']

labels = list(us_df['type'].value_counts().index)
values = list(us_df['type'].value_counts().values)

fig = go.Figure(
    data=[go.Pie(labels=labels, values=values, rotation=120, hole=0.5)],
    layout=go.Layout(
        title="Netflix's United States Content",
        width=900,
        height=700
    )
)

fig.show()
```

Netflix's United States Content

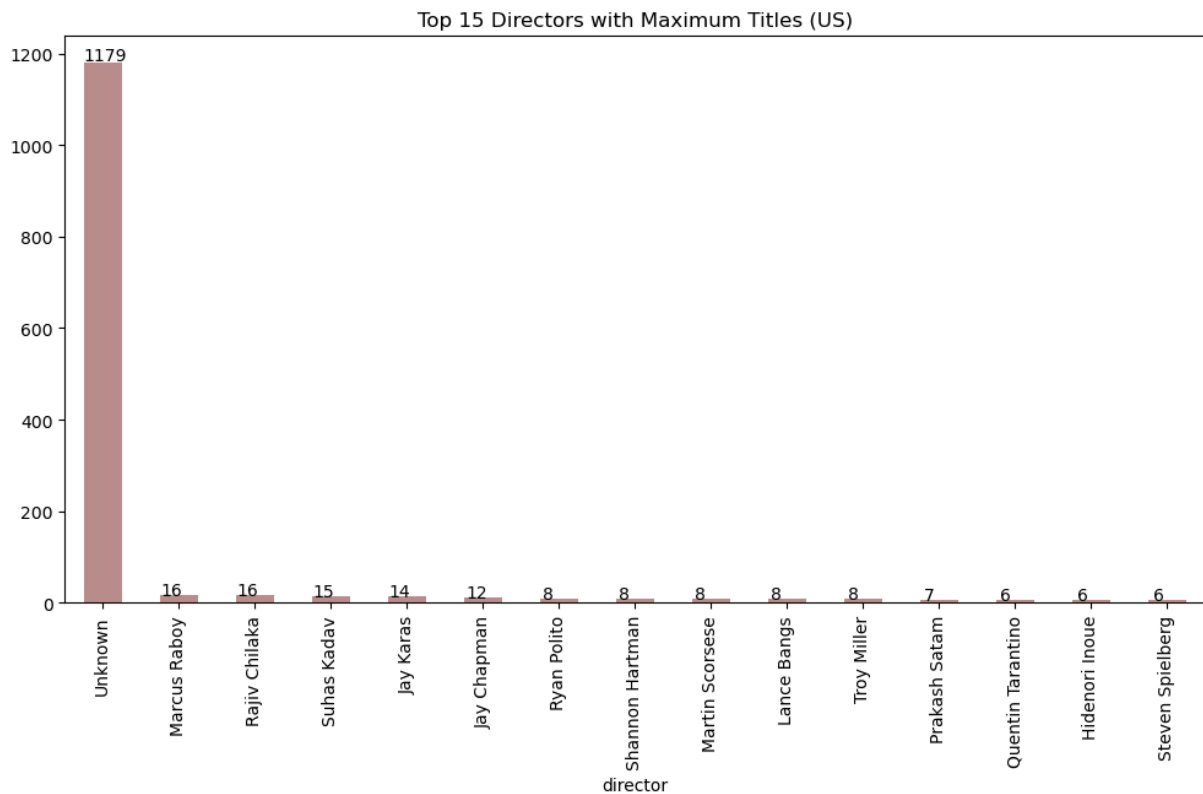


Analysis of Netflix United States

```
In [83]: us_df['type'].value_counts()
```

```
Out[83]: type
1      2498
0      1151
Name: count, dtype: int64
```

```
In [84]: # Top 15 Directors with maximum number of titles on Netflix (US)
ax = us_df['director'].value_counts().head(15).plot.bar(figsize=(12,6), color="rosy")
ax.set_title("Top 15 Directors with Maximum Titles (US)")
for i in ax.patches:
    ax.annotate(i.get_height(), (i.get_x() * 1.005, i.get_height() * 1.005))
plt.show()
```

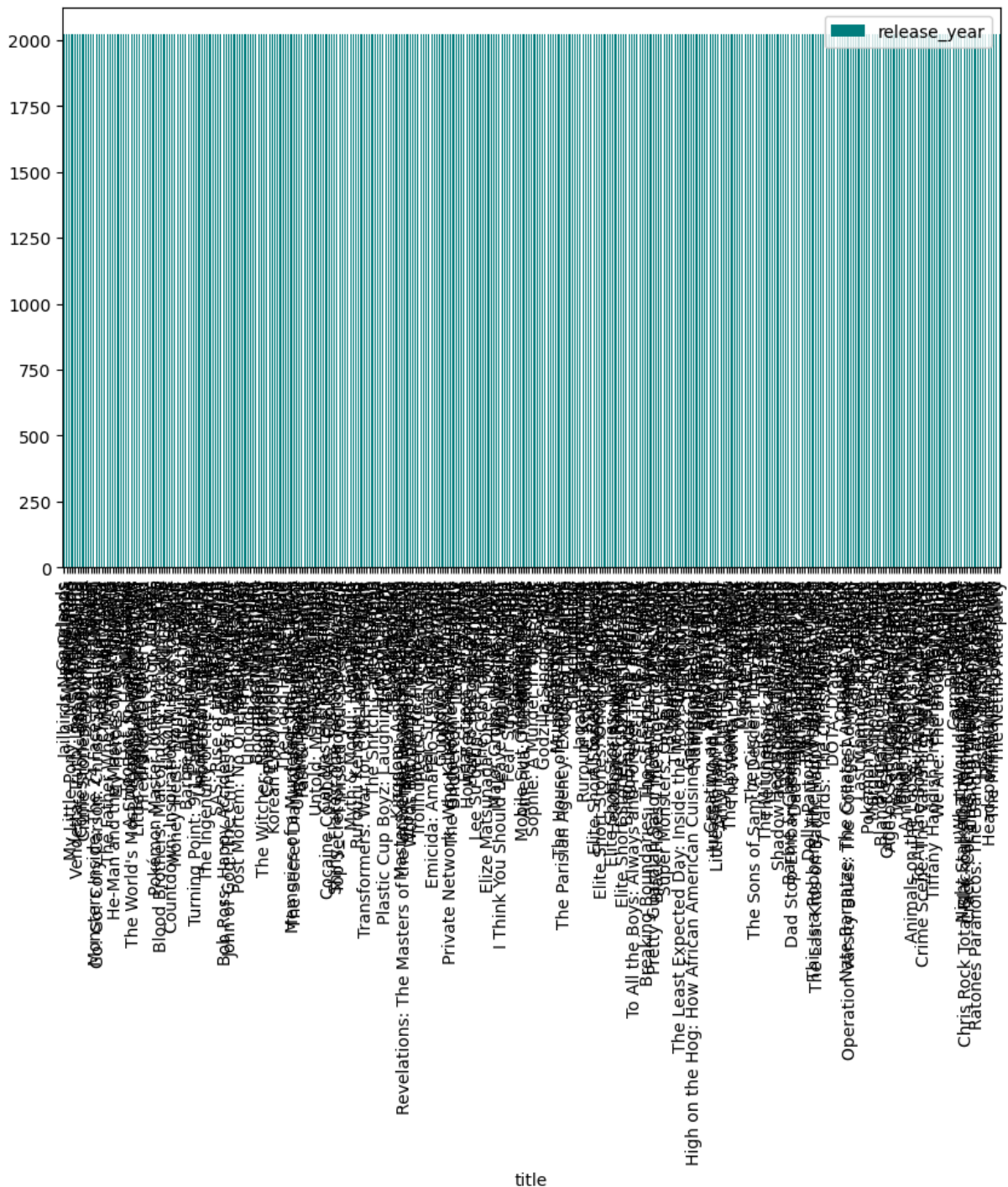


```
In [85]: # Top 10 latest released titles on Netflix (US)
max_votes = us_df['release_year'].sort_values(ascending=False).head(10)
max_votes
```

```
Out[85]: 1313    2021
1025    2021
483     2021
482     2021
481     2021
975     2021
479     2021
477     2021
993     2021
472     2021
Name: release_year, dtype: int64
```

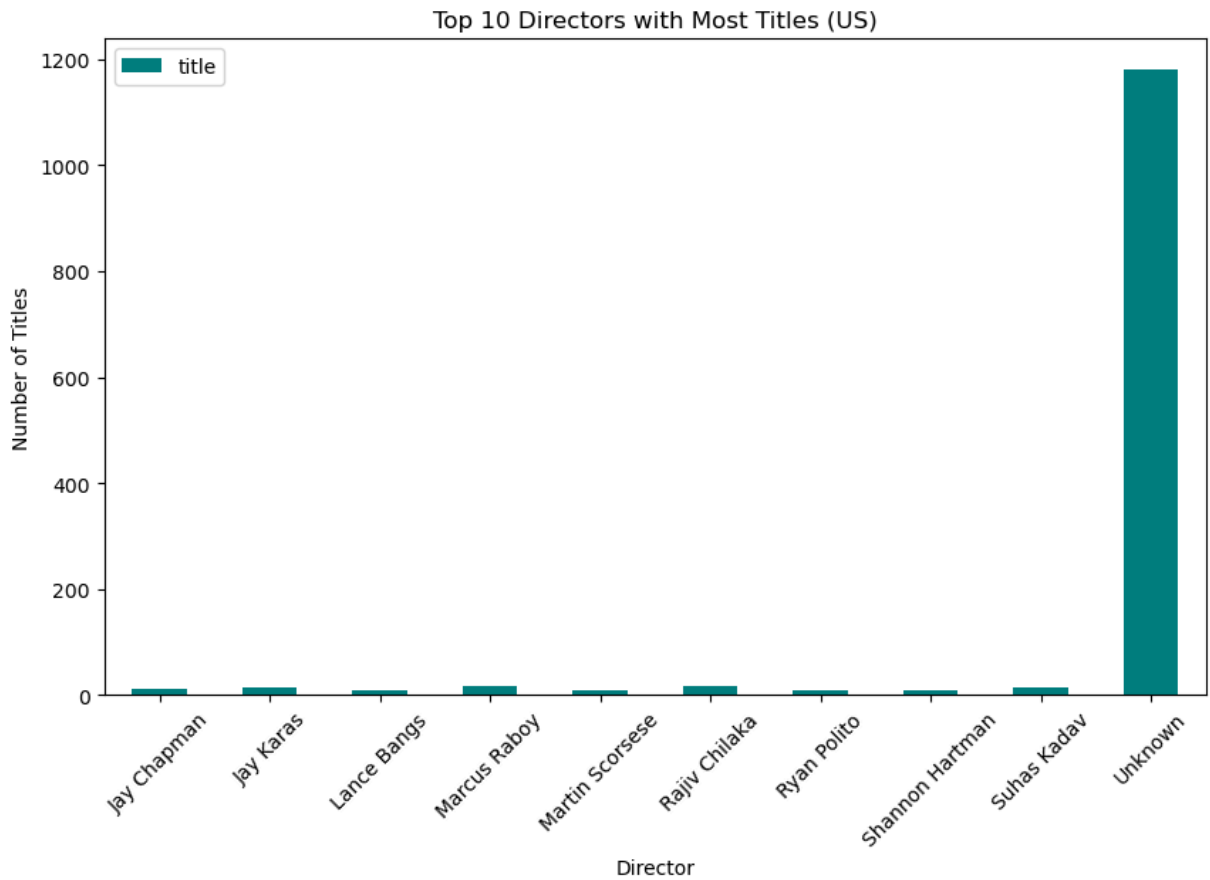
```
In [86]: # Top 10 Titles with highest release year (US)
max_year = us_df['release_year'].sort_values(ascending=False).head(10)

us_df.loc[us_df['release_year'].isin(max_year)][['title', 'release_year']].plot.bar
    x='title',
    y='release_year',
    figsize=(10,6),
    color="teal"
)
plt.show()
```



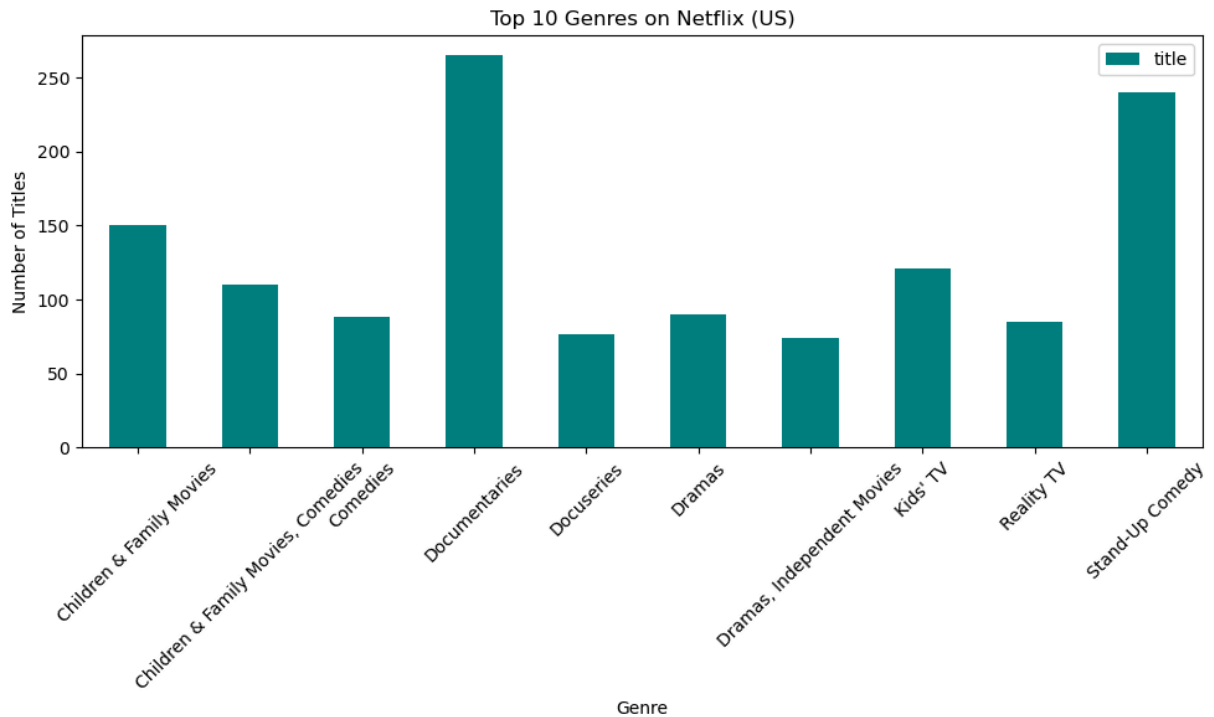
```
In [87]: # Top 10 Directors with most titles on Netflix (US)
top_directors = us_df['director'].value_counts().head(10)

us_df.loc[us_df['director'].isin(top_directors.index)][['title', 'director']].group
    figsize=(10,6),
    color="teal"
)
plt.title("Top 10 Directors with Most Titles (US)")
plt.xlabel("Director")
plt.ylabel("Number of Titles")
plt.xticks(rotation=45)
plt.show()
```



```
In [88]: # Top 10 Genres on Netflix (US)
max_genres = us_df['listed_in'].value_counts().sort_values(ascending=False).head(10)

us_df.loc[us_df['listed_in'].isin(max_genres.index)][['title', 'listed_in']].groupby(
    y='title',
    figsize=(10,6),
    color="teal"
)
plt.title("Top 10 Genres on Netflix (US)")
plt.xlabel("Genre")
plt.ylabel("Number of Titles")
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```



```
In [90]: us_movies = us_df[us_df['type'] == 1] # 1 = Movie
```

```
In [92]: us_df['duration_int'] = us_df['duration'].str.extract(r'(\d+)').astype(float)
us_movies = us_df[us_df['type'] == 1].copy()

print(us_movies[['title', 'duration', 'duration_int']].head())
```

	title	duration	duration_int
0	Dick Johnson Is Dead	90 min	90.0
6	My Little Pony: A New Generation	91 min	91.0
9	The Starling	104 min	104.0
13	Confessions of an Invisible Girl	91 min	91.0
16	Europe's Most Dangerous Man: Otto Skorzeny in ...	67 min	67.0

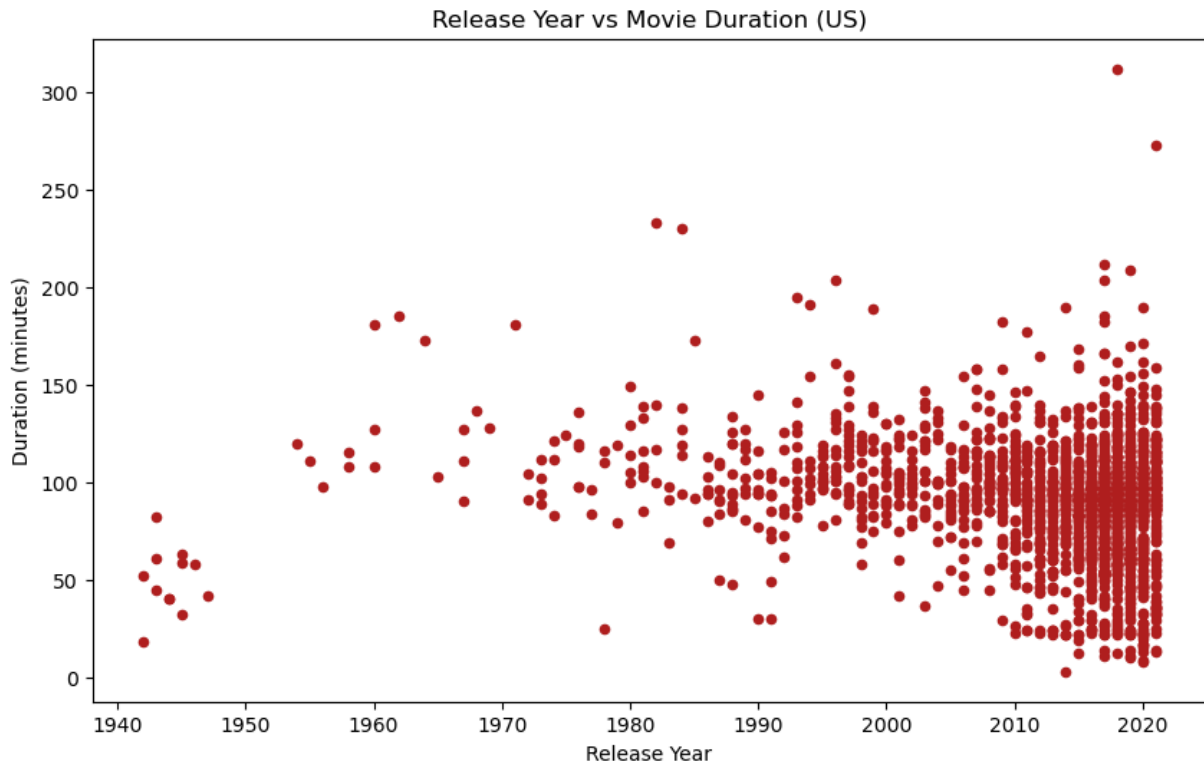
C:\Users\HP\AppData\Local\Temp\ipykernel_21952\1818660135.py:1: SettingWithCopyWarning:

A value is trying to be set on a copy of a slice from a DataFrame.
Try using `.loc[row_indexer,col_indexer] = value` instead

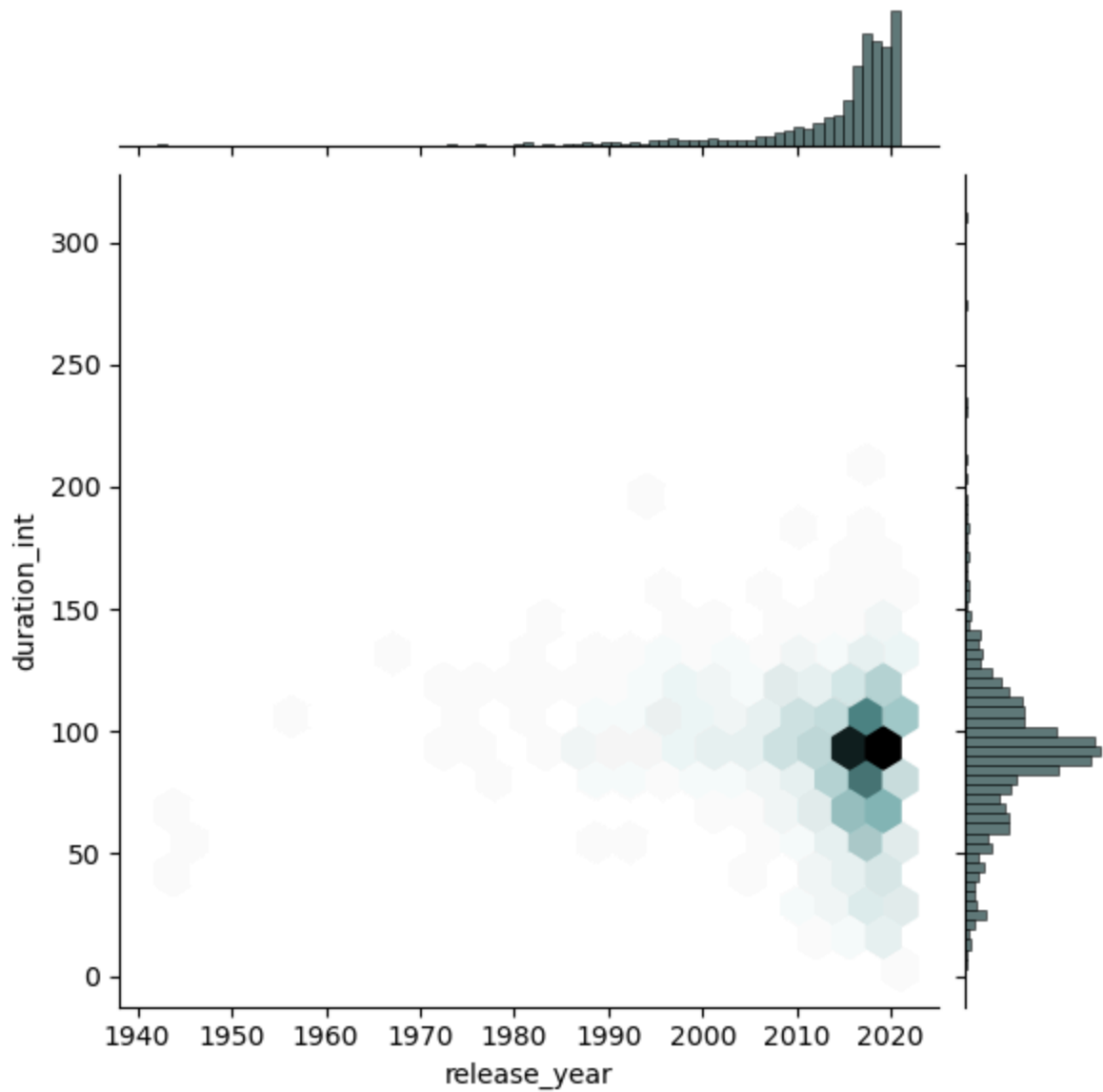
See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-view-versus-a-copy

```
In [93]: us_movies.plot.scatter(
    x='release_year',
    y='duration_int',
    figsize=(10,6),
    color='firebrick',
    title='Release Year vs Movie Duration (US)'
)
plt.xlabel("Release Year")
```

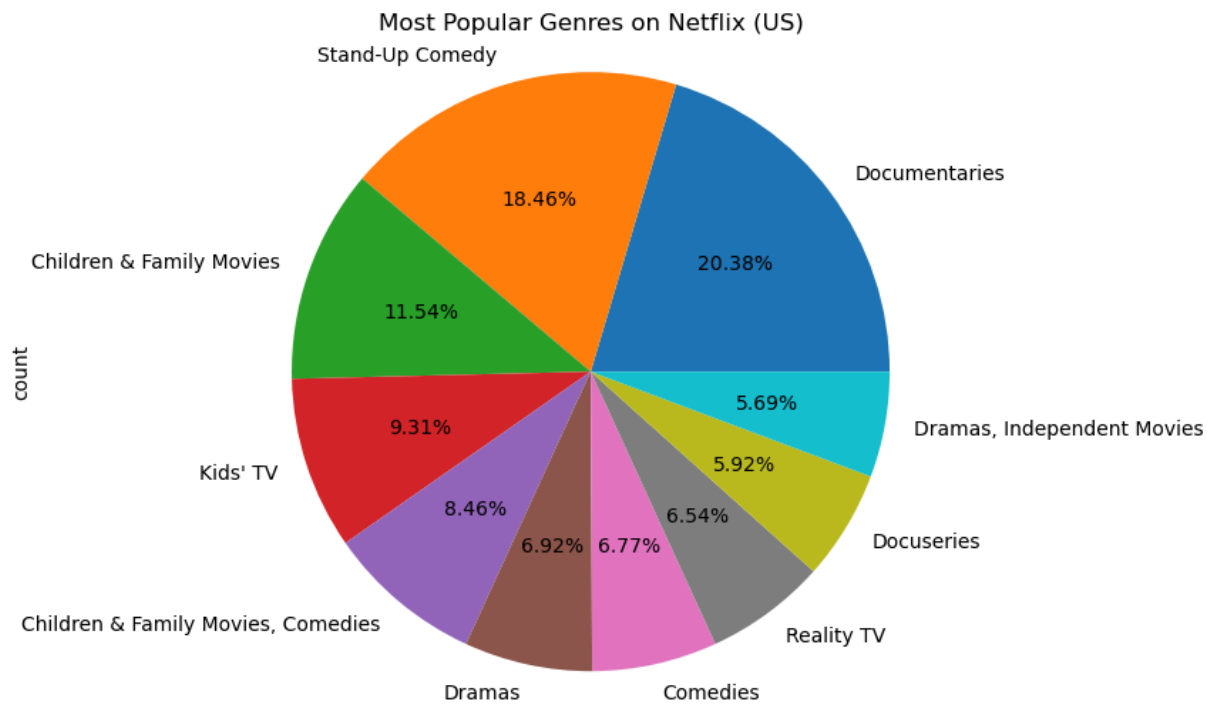
```
plt.ylabel("Duration (minutes)")  
plt.show()
```



```
In [94]: # Better view of relation between Release Year and Movie Duration (US)  
sns.jointplot(  
    x='release_year',  
    y='duration_int',  
    kind='hex',  
    gridsize=22,  
    data=us_movies,  
    color='darkslategrey'  
)  
plt.show()
```



```
In [95]: # Top 10 Genres on Netflix (US)
us_df['listed_in'].value_counts().sort_values(ascending=False).head(10).plot(
    kind='pie',
    figsize=(10,6),
    title="Most Popular Genres on Netflix (US)",
    autopct='%1.2f%%'
)
plt.axis('equal')
plt.show()
```



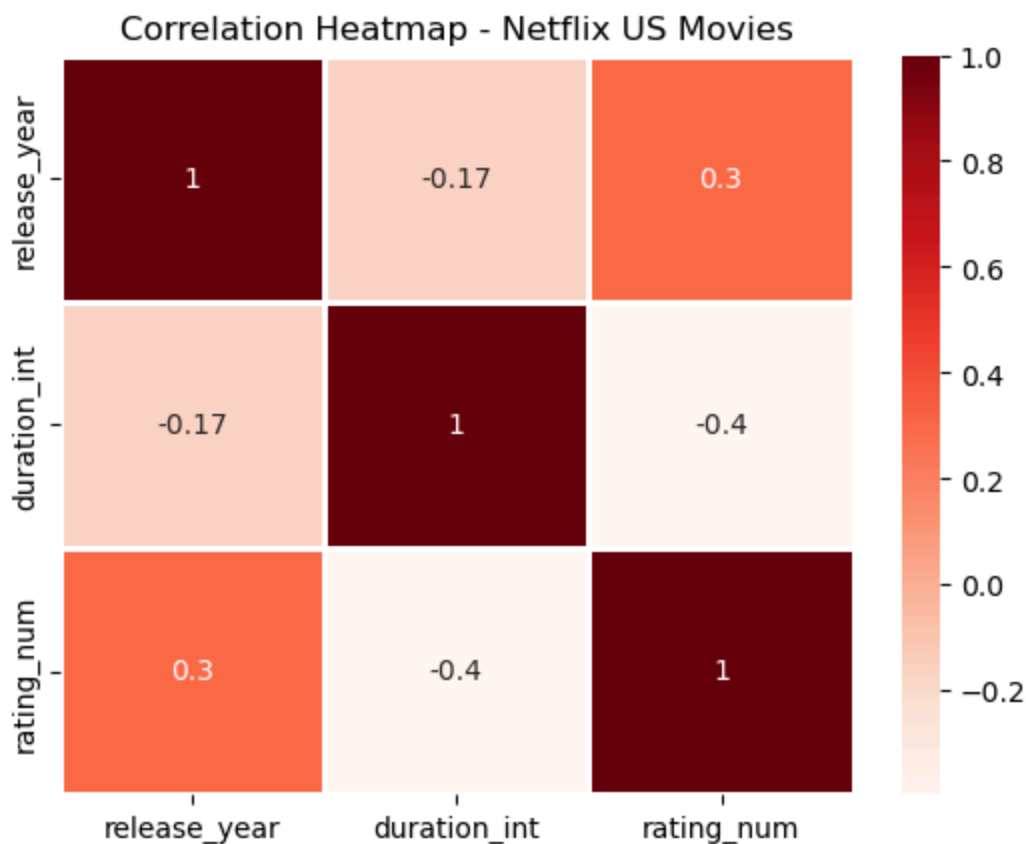
```
In [96]: print(us_movies['rating'].unique())
```

```
['PG-13' 'PG' 'TV-PG' 'TV-MA' 'TV-14' 'TV-Y' 'R' 'TV-G' 'TV-Y7' 'G'
 '74 min' '84 min' '66 min' 'NR' 0 'TV-Y7-FV' 'NC-17' 'UR']
```

```
In [97]: # CONVERT TO RATING COLUMN IN STRING
us_movies = us_movies.copy()
us_movies['rating'] = us_movies['rating'].astype(str)

# LabelEncoder
from sklearn.preprocessing import LabelEncoder
le = LabelEncoder()
us_movies['rating_num'] = le.fit_transform(us_movies['rating'].fillna('NR'))

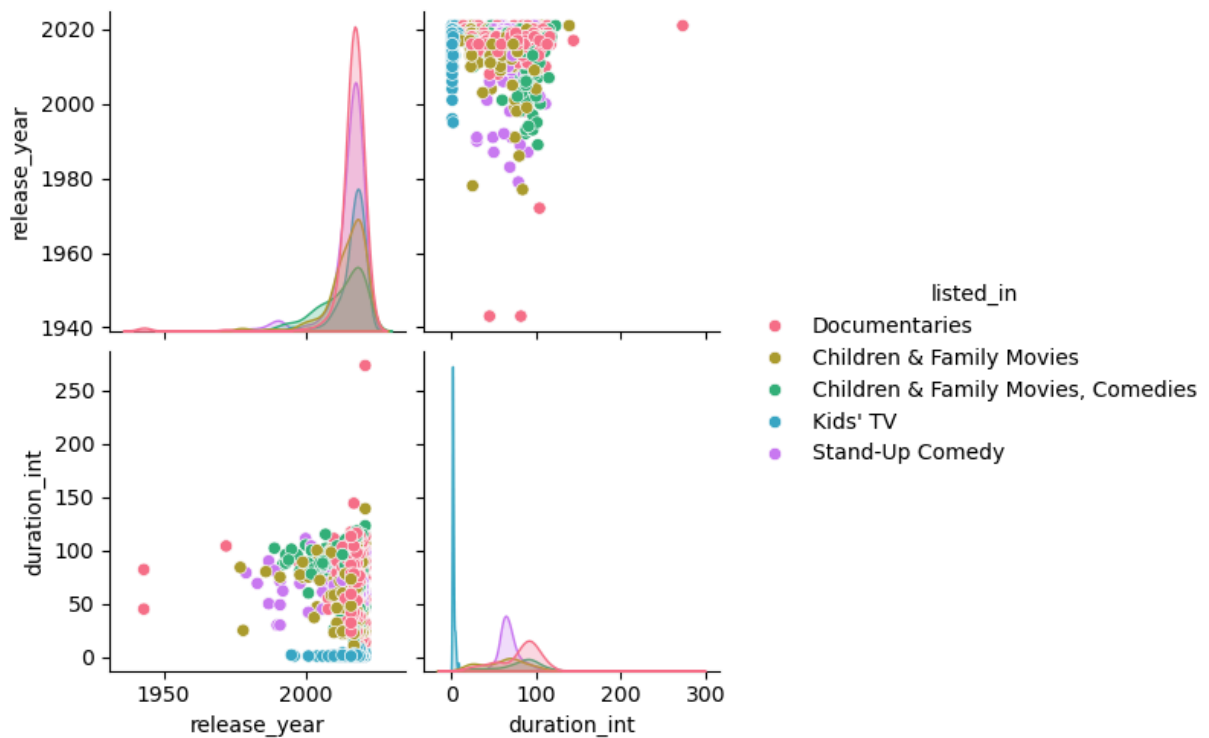
# Heatmap
netflix_corr = us_movies[['release_year', 'duration_int', 'rating_num']]
sns.heatmap(netflix_corr.corr(), linewidth=1.0, annot=True, cmap='Reds')
plt.title("Correlation Heatmap - Netflix US Movies")
plt.show()
```



```
In [98]: # More insight for correlation - Top 5 Genres on Netflix (US)
top5_genres = us_df['listed_in'].value_counts().head(5).index.tolist()

netflix_p = us_df.loc[us_df['listed_in'].isin(top5_genres)]
netflix_pair = netflix_p[['release_year', 'duration_int', 'listed_in']]

sns.pairplot(netflix_pair, hue='listed_in', palette="husl")
plt.show()
```

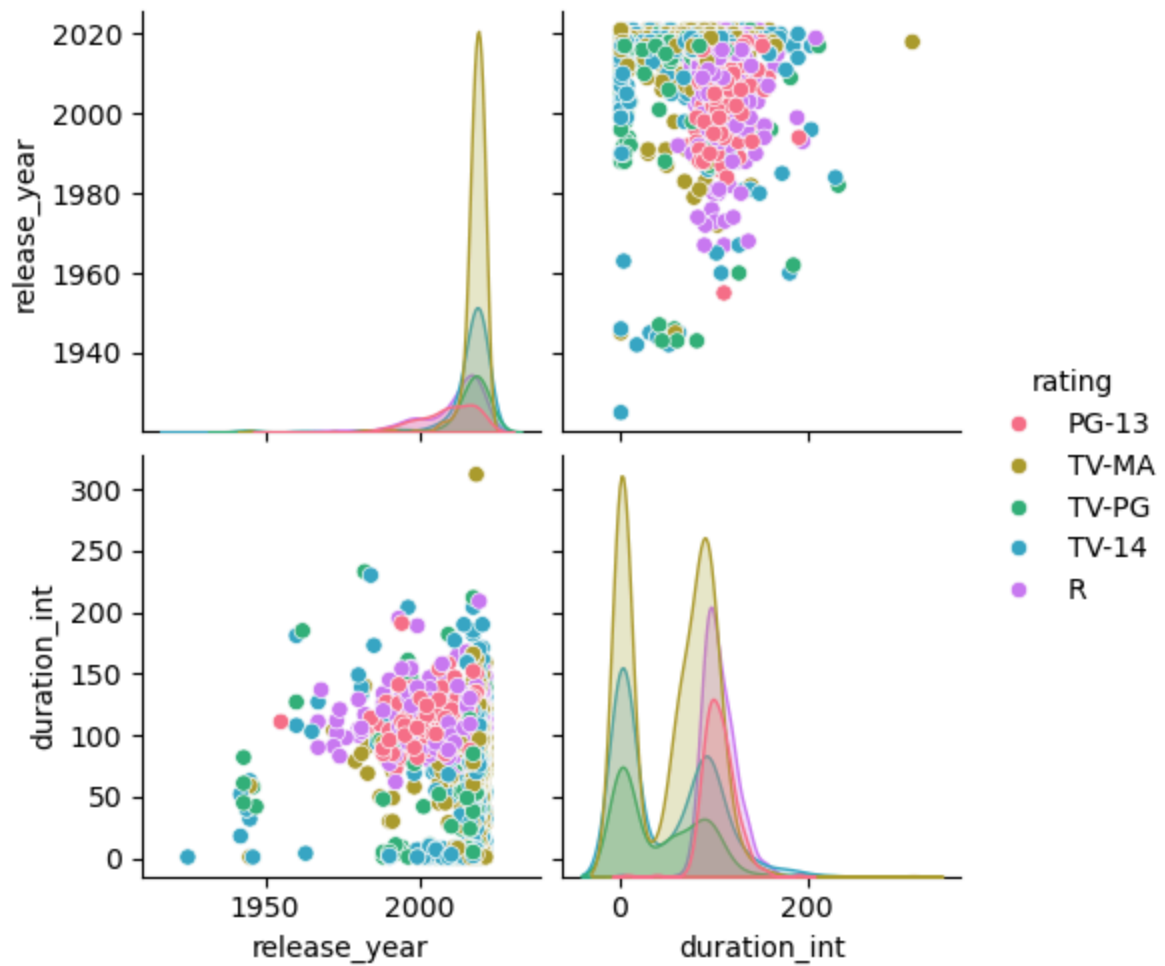


"Next lets try to find our similar relation by keeping our data limited to top 5 ratings instead of genres"

```
In [99]: # More insight for correlation - Top 5 Ratings on Netflix (US)
top5_ratings = us_df['rating'].value_counts().head(5).index.tolist()

netflix_p2 = us_df.loc[us_df['rating'].isin(top5_ratings)]
netflix_pair2 = netflix_p2[['release_year', 'duration_int', 'rating']]

sns.pairplot(netflix_pair2, hue='rating', palette="husl")
plt.show()
```



```
In [46]: us_movies
```

Out[46]:

	show_id	type	title	director	cast	country	date_added	release_year
0	s1	1	Dick Johnson Is Dead	Kirsten Johnson	Unknown	United States	September 25, 2021	2020
6	s7	1	My Little Pony: A New Generation	Robert Cullen, José Luis Ucha	Vanessa Hudgens, Kimiko Glenn, James Marsden, ...	United States	September 24, 2021	2021
9	s10	1	The Starling	Theodore Melfi	Melissa McCarthy, Chris O'Dowd, Kevin Kline, T...	United States	September 24, 2021	2021
13	s14	1	Confessions of an Invisible Girl	Bruno Garotti	Klara Castanho, Lucca Picon, Júlia Gomes, Marc...	United States	September 22, 2021	2021
16	s17	1	Europe's Most Dangerous Man: Otto Skorzeny in ...	Pedro de Echave García, Pablo Azorín Williams	Unknown	United States	September 22, 2021	2020
...	...	...	...	...	...	...	...	...
8791	s8792	1	Young Adult	Jason Reitman	Charlize Theron, Patton Oswalt, Patrick Wilson...	United States	November 20, 2019	2011
8793	s8794	1	Yours, Mine and Ours	Raja Gosnell	Dennis Quaid, Rene Russo, Sean Faris, Katija P...	United States	November 20, 2019	2005
8802	s8803	1	Zodiac	David Fincher	Mark Ruffalo, Jake Gyllenhaal,	United States	November 20, 2019	2007

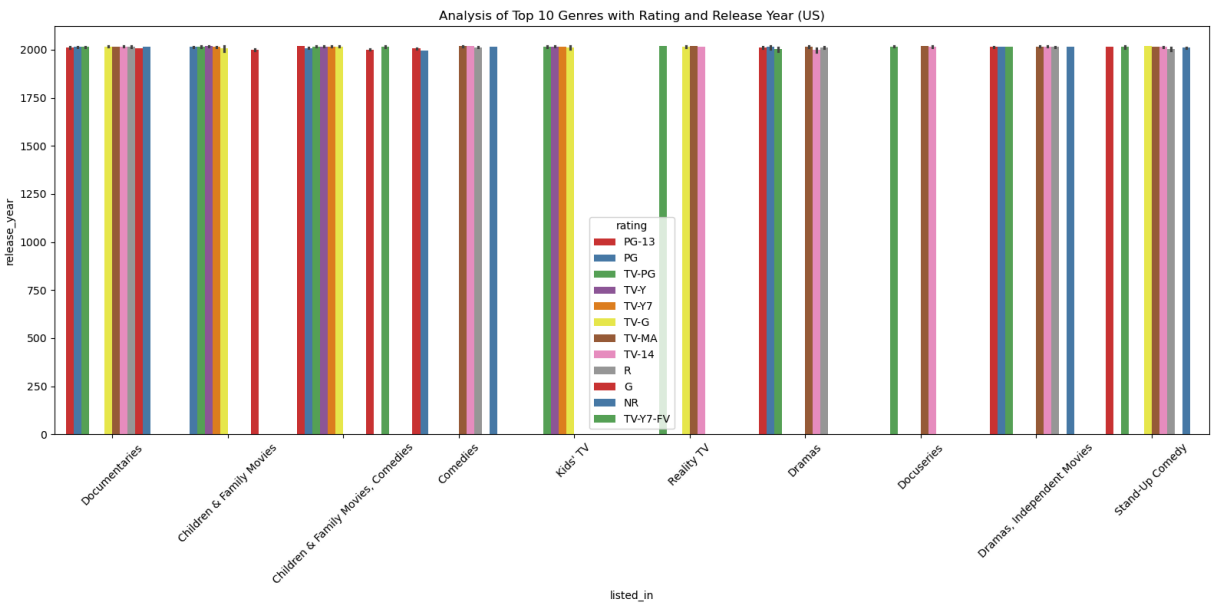
show_id	type		title	director	cast	country	date_added	release_year
					Robert Downey J...			
8804	s8805	1	Zombieland	Ruben Fleischer	Jesse Eisenberg, Woody Harrelson, Emma Stone, ...	United States	November 1, 2019	2009
8805	s8806	1	Zoom	Peter Hewitt	Tim Allen, Courteney Cox, Chevy Chase, Kate Ma...	United States	January 11, 2020	2006

2498 rows × 13 columns

```
In [100... # Analysis of Top 10 Genres with Type and Release Year (US)
top10genres_list = us_df['listed_in'].value_counts().head(10).index.tolist()

netflix_genres = us_df.loc[us_df['listed_in'].isin(top10genres_list)]
netflix_genres_data = netflix_genres[['release_year', 'duration_int', 'rating', 'li

fig, axx = plt.subplots(figsize=(16,8))
sns.barplot(x='listed_in', y='release_year', hue='rating', data=netflix_genres_data
axx.set_title("Analysis of Top 10 Genres with Rating and Release Year (US)")
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```

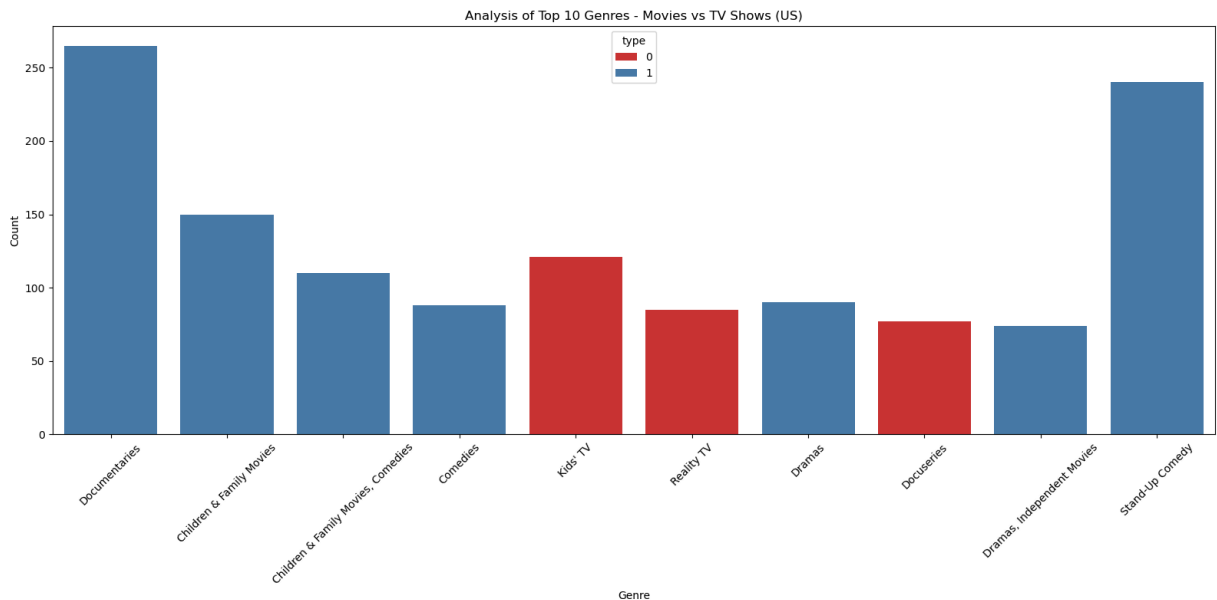


```
In [101... # Analysis of Top 10 Genres with count and Type (US)

top10genres_list = us_df['listed_in'].value_counts().head(10).index.tolist()

netflix_genres = us_df.loc[us_df['listed_in'].isin(top10genres_list)]

fig, axx = plt.subplots(figsize=(16,8))
sns.countplot(
    x='listed_in',
    hue='type',          # Sirf 2 hue – Movie & TV Show
    data=netflix_genres,
    palette="Set1"
)
axx.set_title("Analysis of Top 10 Genres - Movies vs TV Shows (US)")
plt.xlabel("Genre")
plt.ylabel("Count")
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```

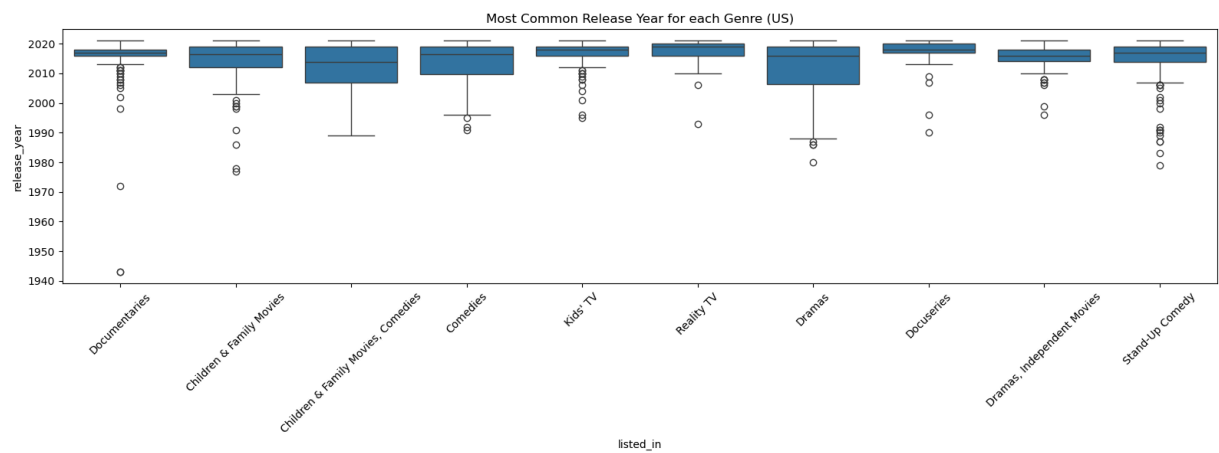


```
In [102... # Most common Release Year for each type of Genre (US)

top10genres_list = us_df['listed_in'].value_counts().head(10).index.tolist()

netflix_genres = us_df.loc[us_df['listed_in'].isin(top10genres_list)]
netflix_genres_data = netflix_genres[['release_year', 'duration_int', 'listed_in']]

fig, axx = plt.subplots(figsize=(16,6))
sns.boxplot(x='listed_in', y='release_year', data=netflix_genres_data)
plt.title("Most Common Release Year for each Genre (US)")
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```



```
In [103... top5genres_list = us_df['listed_in'].value_counts().head(5).index.tolist()
netflix_rate = us_df.loc[us_df['listed_in'].isin(top5genres_list)]

# Find total number of titles per genre
total_titles = netflix_rate.groupby(['listed_in'], as_index=False).count()[['listed_in', 'type']]
total_titles.columns = ['Genre', 'Total Titles']

# Find total count of each type (Movie/TV Show) per genre
top5genres = netflix_rate.groupby(['listed_in', 'type'], as_index=False)[['title']]
top5genres.columns = ['Genre', 'Type', 'Total Count']

# Merge both and calculate percentage
top5genres_percent = pd.merge(total_titles, top5genres, on='Genre')
top5genres_percent['Percentage'] = (
    top5genres_percent['Total Count'] /
    top5genres_percent['Total Titles']
) * 100

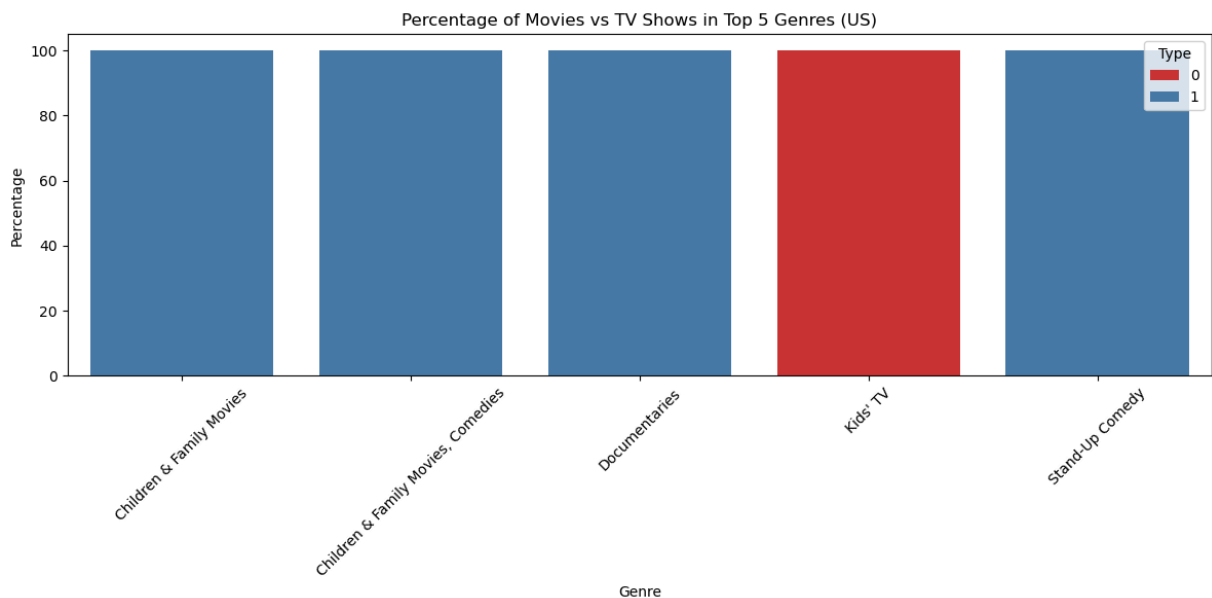
top5genres_percent
```

Out[103...]

	Genre	Total Titles	Type	Total Count	Percentage
0	Children & Family Movies	150	1	150	100.0
1	Children & Family Movies, Comedies	110	1	110	100.0
2	Documentaries	265	1	265	100.0
3	Kids' TV	121	0	121	100.0
4	Stand-Up Comedy	240	1	240	100.0

```
In [104... # Plot Type percentage of titles in Top 5 Genres (US)

fig, axx = plt.subplots(figsize=(12,6))
axx.set_title("Percentage of Movies vs TV Shows in Top 5 Genres (US)")
sns.barplot(x='Genre', y='Percentage', hue='Type', data=top5genres_percent, palette='magma')
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```



Keeping our data limited to top 3 genres with maximum titles in United States.

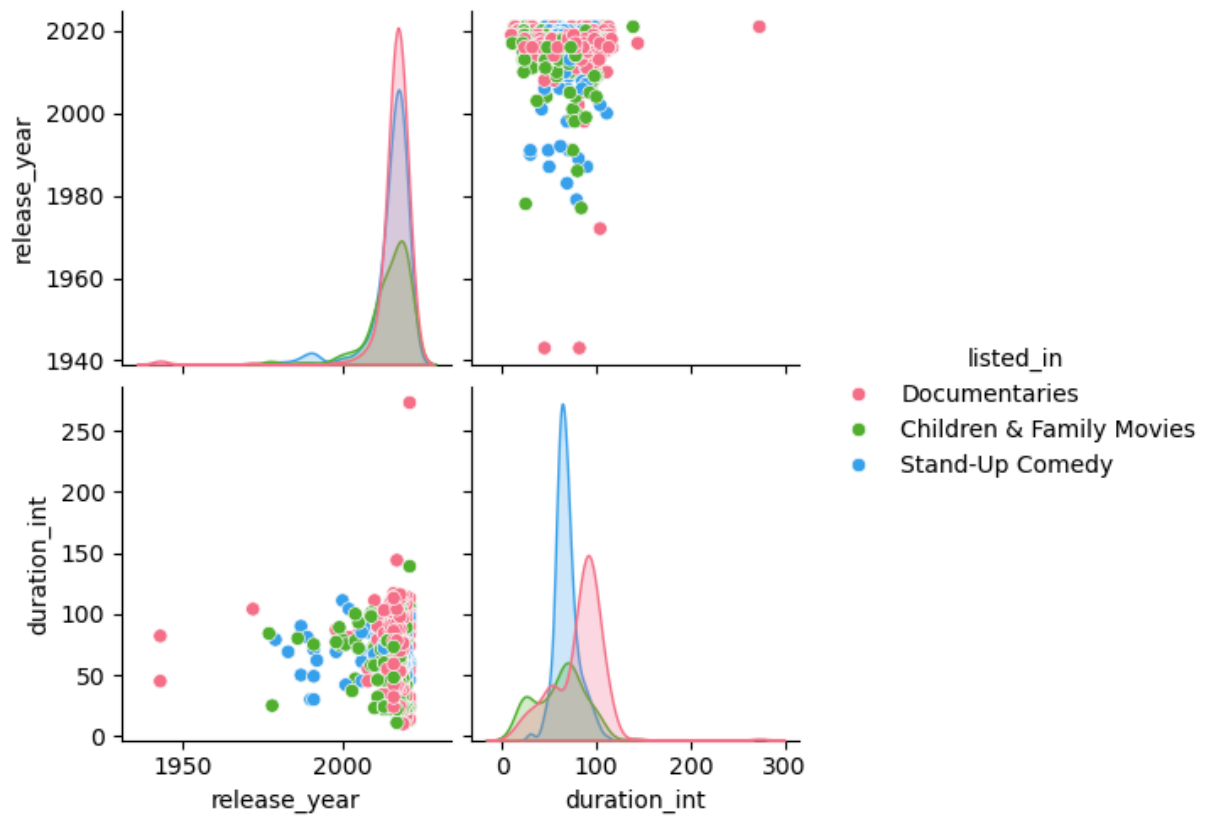
```
In [108... # duration_int column

us_df = us_df.copy()
us_df['duration_int'] = us_df['duration'].str.extract(r'(\d+)').astype(float)

# pairplot
top3genres_list = us_df['listed_in'].value_counts().head(3).index.tolist()

netflix_p = us_df.loc[us_df['listed_in'].isin(top3genres_list)]
netflix_pair = netflix_p[['release_year', 'duration_int', 'listed_in']]

sns.pairplot(netflix_pair, hue='listed_in', palette="husl")
plt.show()
```



```
In [109... us_df = df.loc[df['country'] == 'United States']
```

```
In [110... us_df
```

Out[110...

	show_id	type	title	director	cast	country	date_added	release_yea
0	s1	Movie	Dick Johnson Is Dead	Kirsten Johnson	Unknown	United States	September 25, 2021	2020
2	s3	TV Show	Ganglands	Julien Leclercq	Sami Bouajila, Tracy Gotoas, Samuel Jouy, Nabi...	United States	September 24, 2021	2021
3	s4	TV Show	Jailbirds New Orleans	Unknown	Unknown	United States	September 24, 2021	2021
5	s6	TV Show	Midnight Mass	Mike Flanagan	Kate Siegel, Zach Gilford, Hamish Linklater, H...	United States	September 24, 2021	2021
6	s7	Movie	My Little Pony: A New Generation	Robert Cullen, José Luis Ucha	Vanessa Hudgens, Kimiko Glenn, James Marsden, ...	United States	September 24, 2021	2021
...	...	...	...	...	...	...	...	...
8793	s8794	Movie	Yours, Mine and Ours	Raja Gosnell	Dennis Quaid, Rene Russo, Sean Faris, Katija P...	United States	November 20, 2019	2009
8802	s8803	Movie	Zodiac	David Fincher	Mark Ruffalo, Jake Gyllenhaal, Robert Downey J...	United States	November 20, 2019	2007

	show_id	type	title	director	cast	country	date_added	release_yea
8803	s8804	TV Show	Zombie Dumb	Unknown	Unknown	United States	July 1, 2019	2018
8804	s8805	Movie	Zombieland	Ruben Fleischer	Jesse Eisenberg, Woody Harrelson, Emma Stone, ...	United States	November 1, 2019	2009
8805	s8806	Movie	Zoom	Peter Hewitt	Tim Allen, Courteney Cox, Chevy Chase, Kate Ma...	United States	January 11, 2020	2006

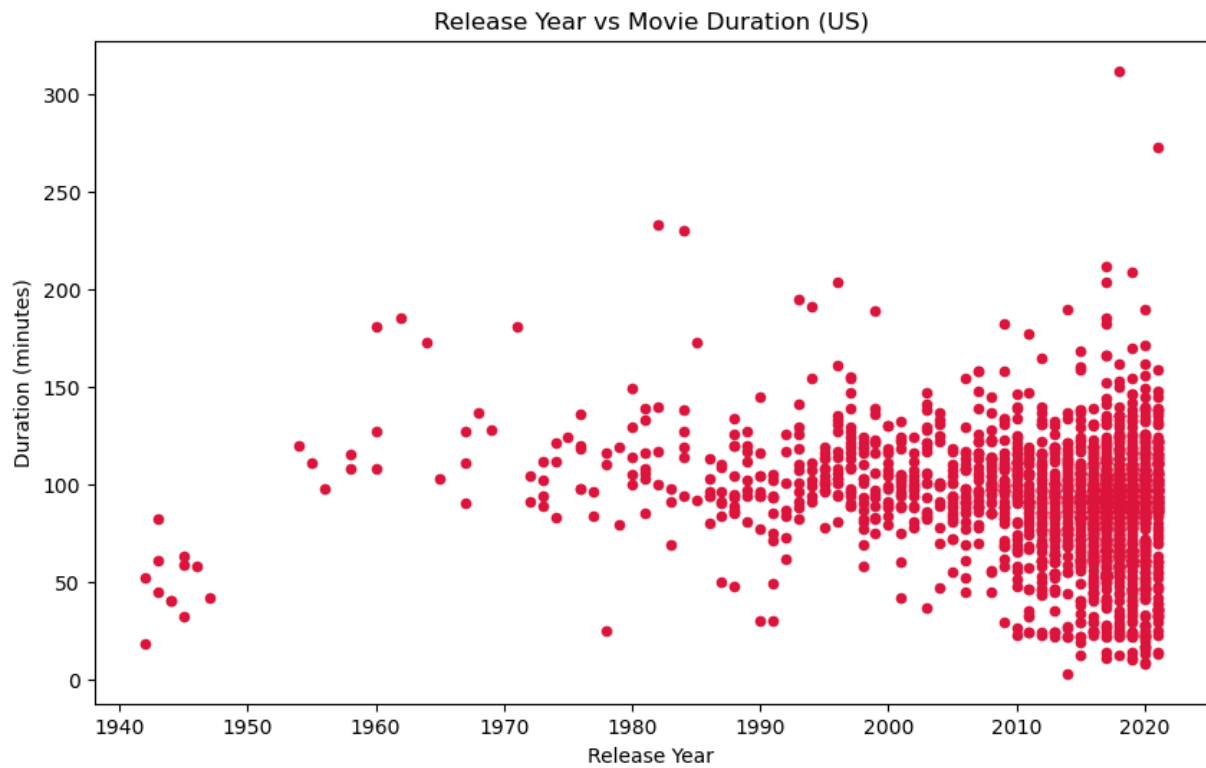
3649 rows × 13 columns

Lets try to find out if there is any relation between release year and duration of movies in United States.

In [112...

```
# Is there any relation between release year and duration of movies (US)

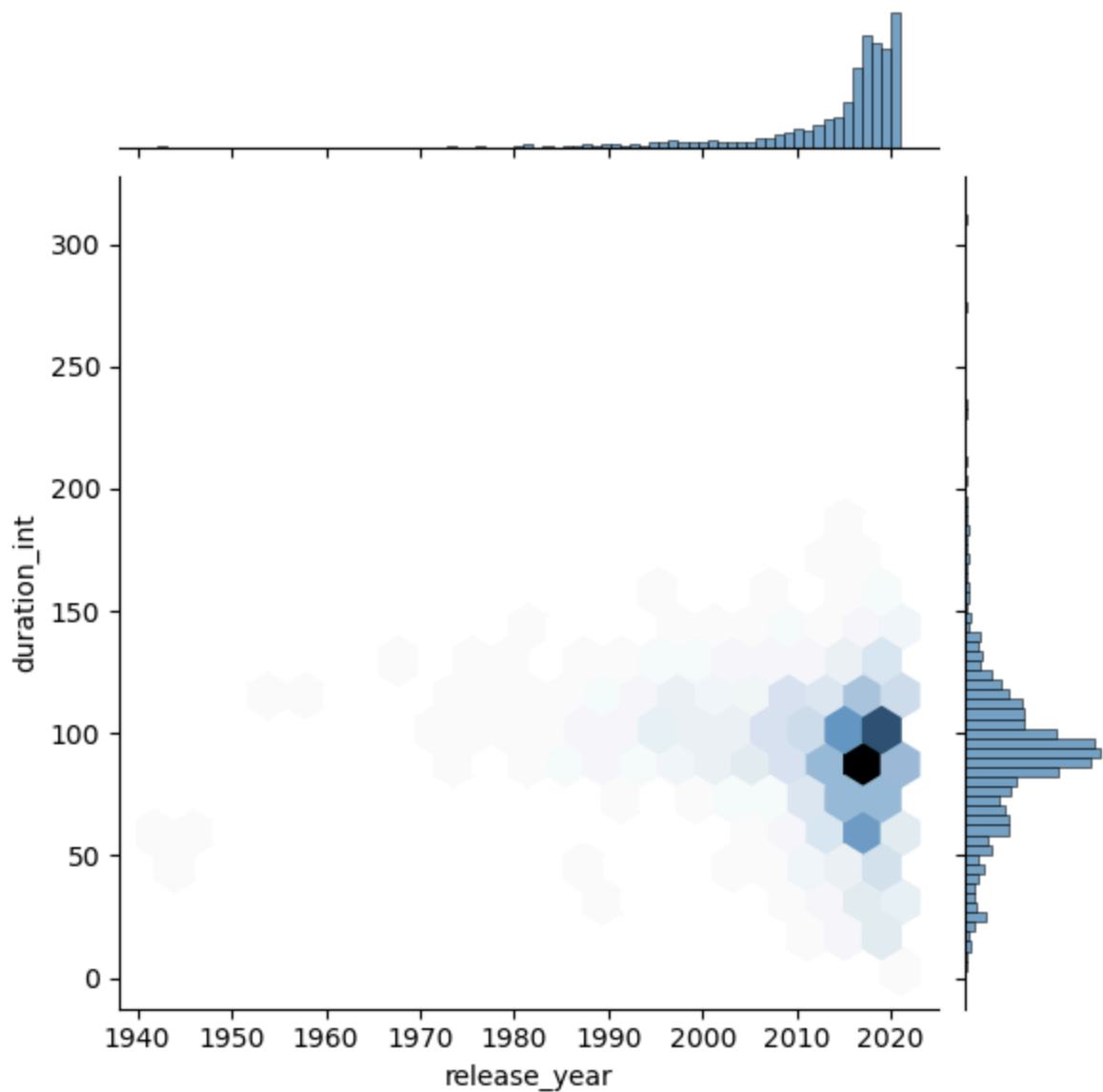
us_movies.plot.scatter(
    x='release_year',
    y='duration_int',
    figsize=(10,6),
    color='crimson',
    title='Release Year vs Movie Duration (US)'
)
plt.xlabel("Release Year")
plt.ylabel("Duration (minutes)")
plt.show()
```



In [113...

```
# Better view of relation between release year and duration of movies (US)
```

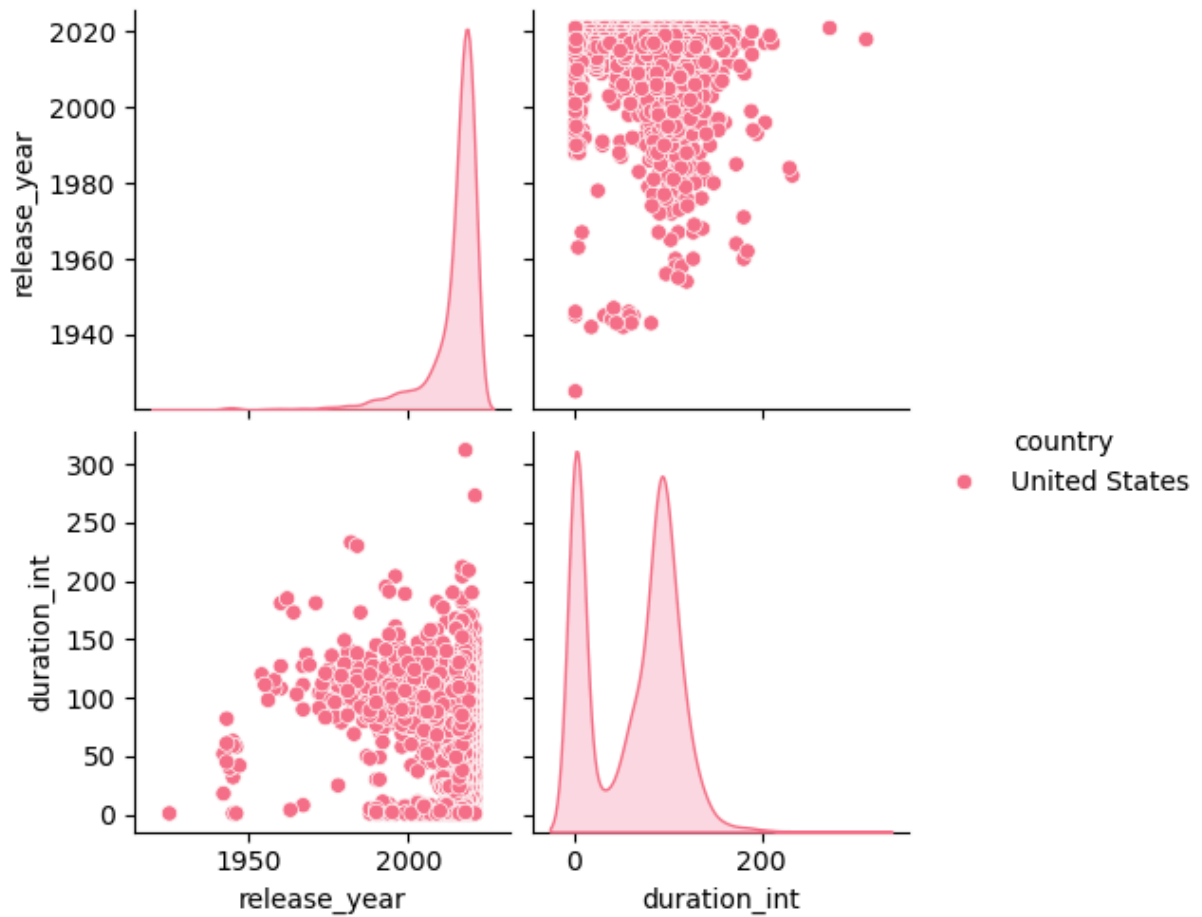
```
sns.jointplot(  
    x='release_year',  
    y='duration_int',  
    kind='hex',  
    gridsize=20,  
    data=us_movies,  
    color='steelblue'  
)  
plt.show()
```



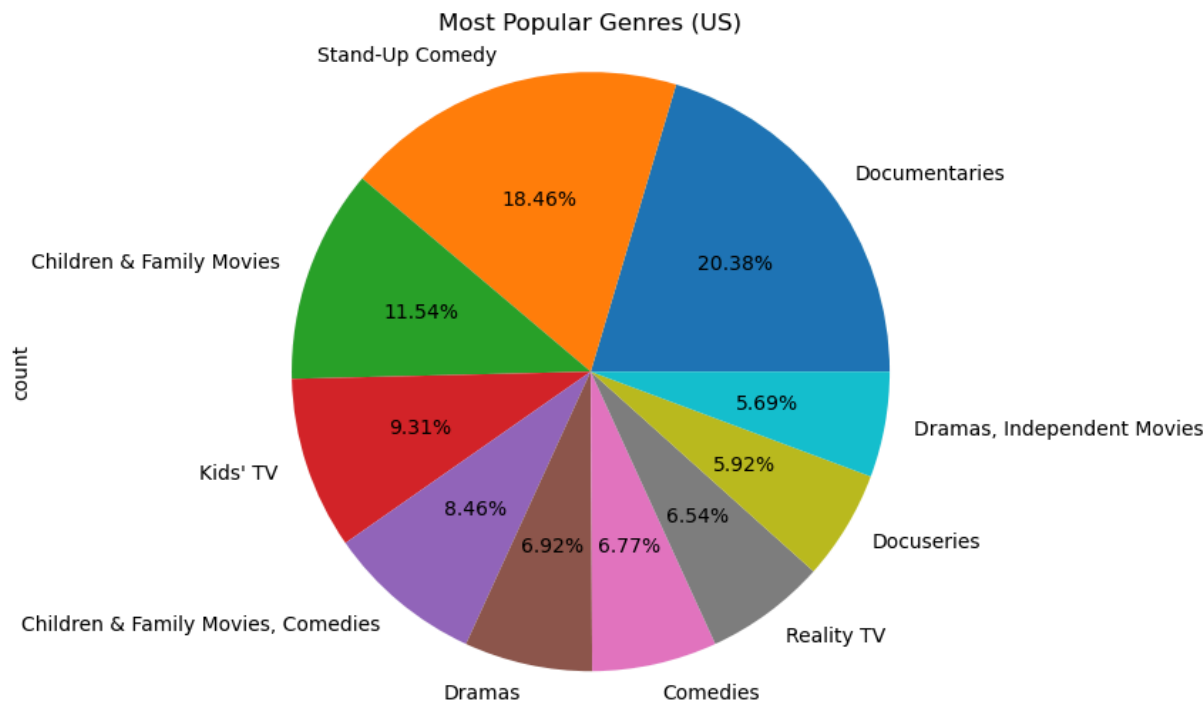
```
In [115... # duration_int column banavo
us_df = us_df.copy()
us_df['duration_int'] = us_df['duration'].str.extract(r'(\d+)').astype(float)

# Pachhi pairplot
us_country = ['United States']
netflix_v = us_df.loc[us_df['country'].isin(us_country)]
netflix_vpair = netflix_v[['release_year', 'duration_int', 'country']]

sns.pairplot(netflix_vpair, hue='country', palette="husl")
plt.show()
```



```
In [116... us_df['listed_in'].value_counts().sort_values(ascending=False).head(10).plot(
    kind='pie',
    figsize=(10,6),
    autopct='%1.2f%%'
)
plt.title("Most Popular Genres (US)")
plt.axis('equal')
plt.show()
```



```
In [117... # Title Percentage wise Type in United States
us_country = ['United States']
netflix_rate = df.loc[df['country'].isin(us_country)]

# Find total number of titles
total_titles = netflix_rate.groupby(['country'], as_index=False).count()[['country']]
total_titles.columns = ['Country', 'Total Titles']

# Find total count of each type
title_type = netflix_rate.groupby(['country', 'type'], as_index=False)[['title']].count()
title_type.columns = ['Country', 'Type', 'Total Type']

# Merge both and calculate percentage
title_type_percent = pd.merge(total_titles, title_type, on='Country')
title_type_percent['Percentage'] = (
    title_type_percent['Total Type'] /
    title_type_percent['Total Titles']
) * 100

title_type_percent
```

Out[117...

	Country	Total Titles	Type	Total Type	Percentage
0	United States	3649	Movie	2498	68.457112
1	United States	3649	TV Show	1151	31.542888

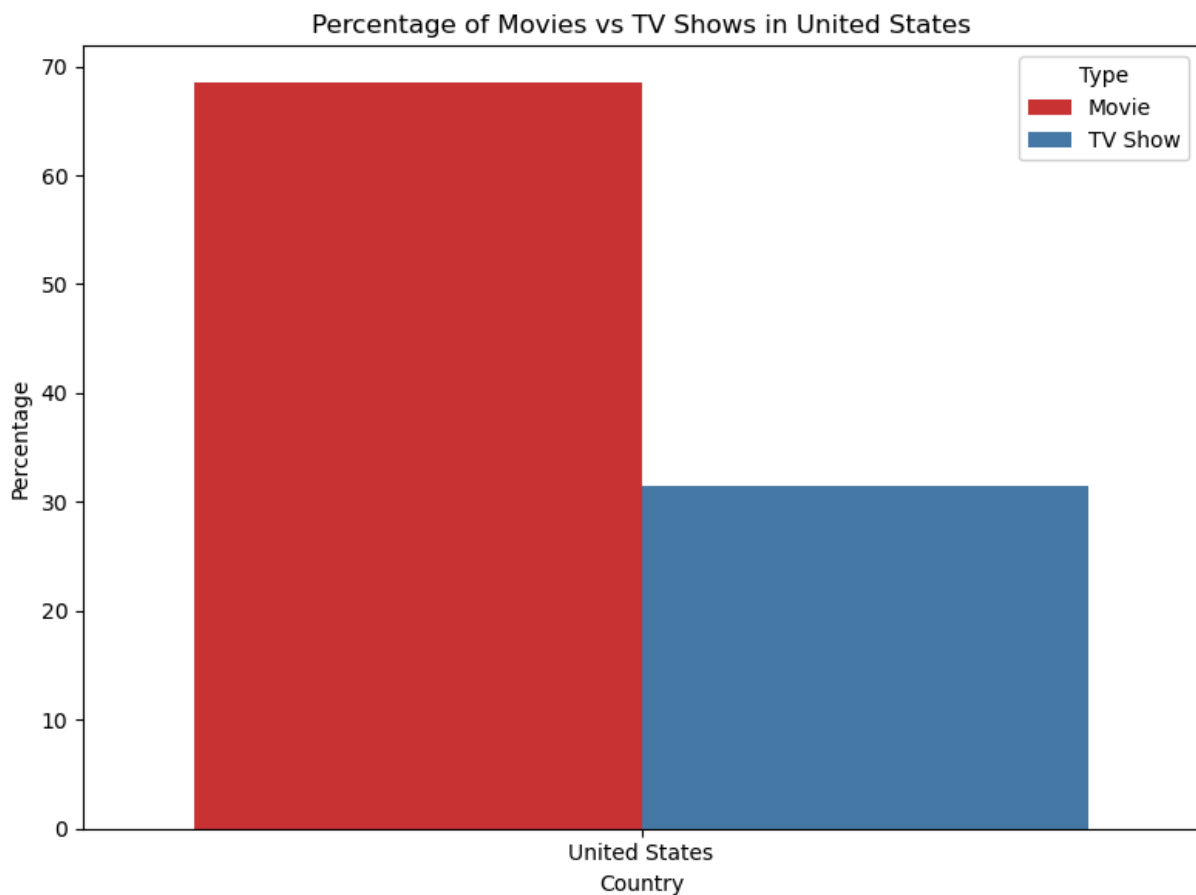
Now lets plot the above data in the form of a graph and list our findings

```
In [118... # Plot Type percentage of titles in United States
fig, axx = plt.subplots(figsize=(8,6))
axx.set_title("Percentage of Movies vs TV Shows in United States")
sns.barplot(
```

```

x='Country',
y='Percentage',
hue='Type',
data=title_type_percent,
palette='Set1'
)
plt.xlabel("Country")
plt.ylabel("Percentage")
plt.tight_layout()
plt.show()

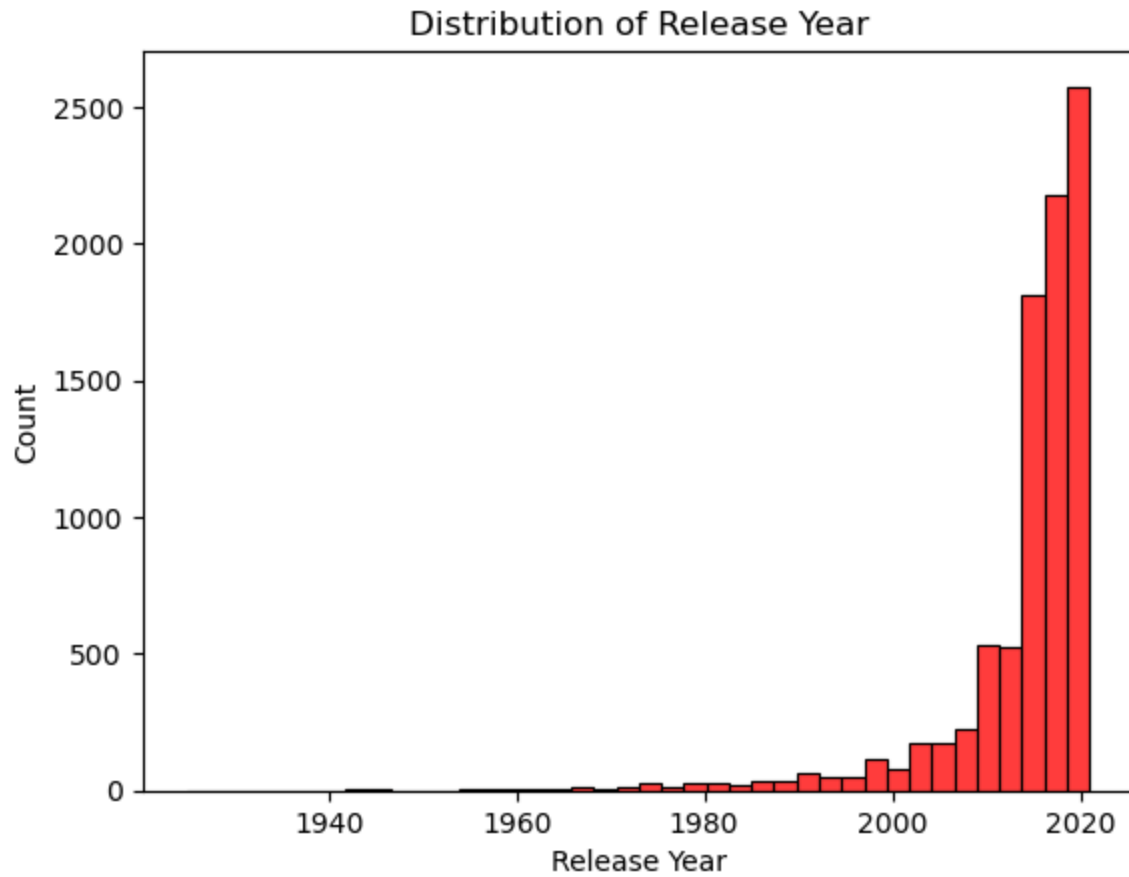
```



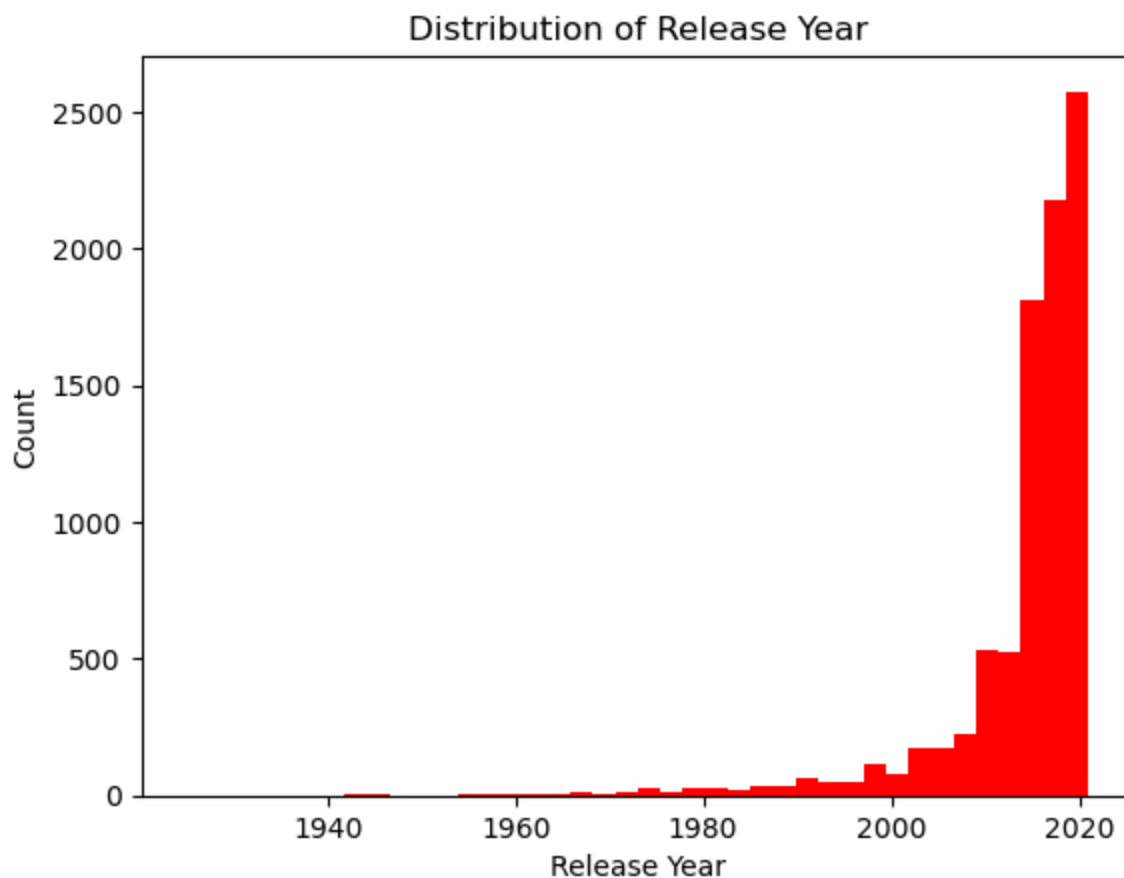
```

In [120... # Distribution of Release Year on Netflix
sns.histplot(df['release_year'].dropna(), kde=False, bins=40, color='red')
plt.title("Distribution of Release Year")
plt.xlabel("Release Year")
plt.ylabel("Count")
plt.show()

```



```
In [121... # Histogram of Release Year
plt.hist(df['release_year'], color='red', bins=40)
plt.title("Distribution of Release Year")
plt.xlabel("Release Year")
plt.ylabel("Count")
plt.show()
```

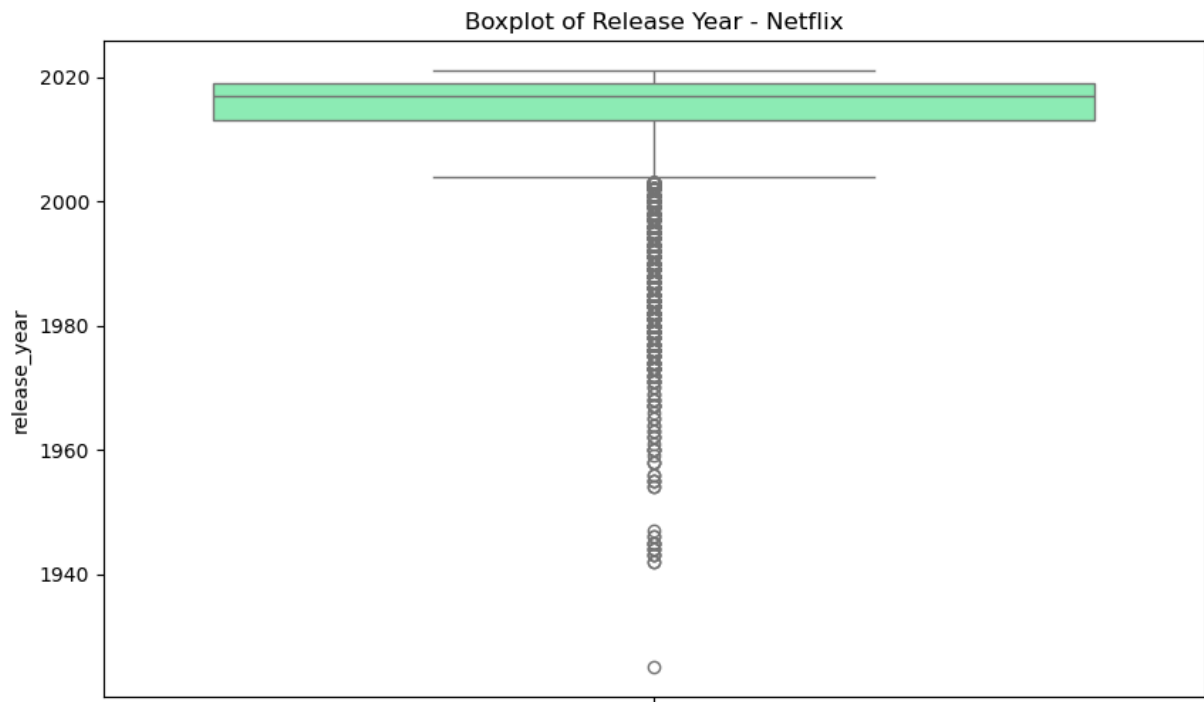


In [122...

```
# Boxplot of Release Year
plt.figure(figsize=(10,6))
sns.boxplot(data=df, y='release_year', palette='rainbow')
plt.title("Boxplot of Release Year - Netflix")
plt.show()
```

C:\Users\HP\AppData\Local\Temp\ipykernel_21952\359934396.py:3: FutureWarning:

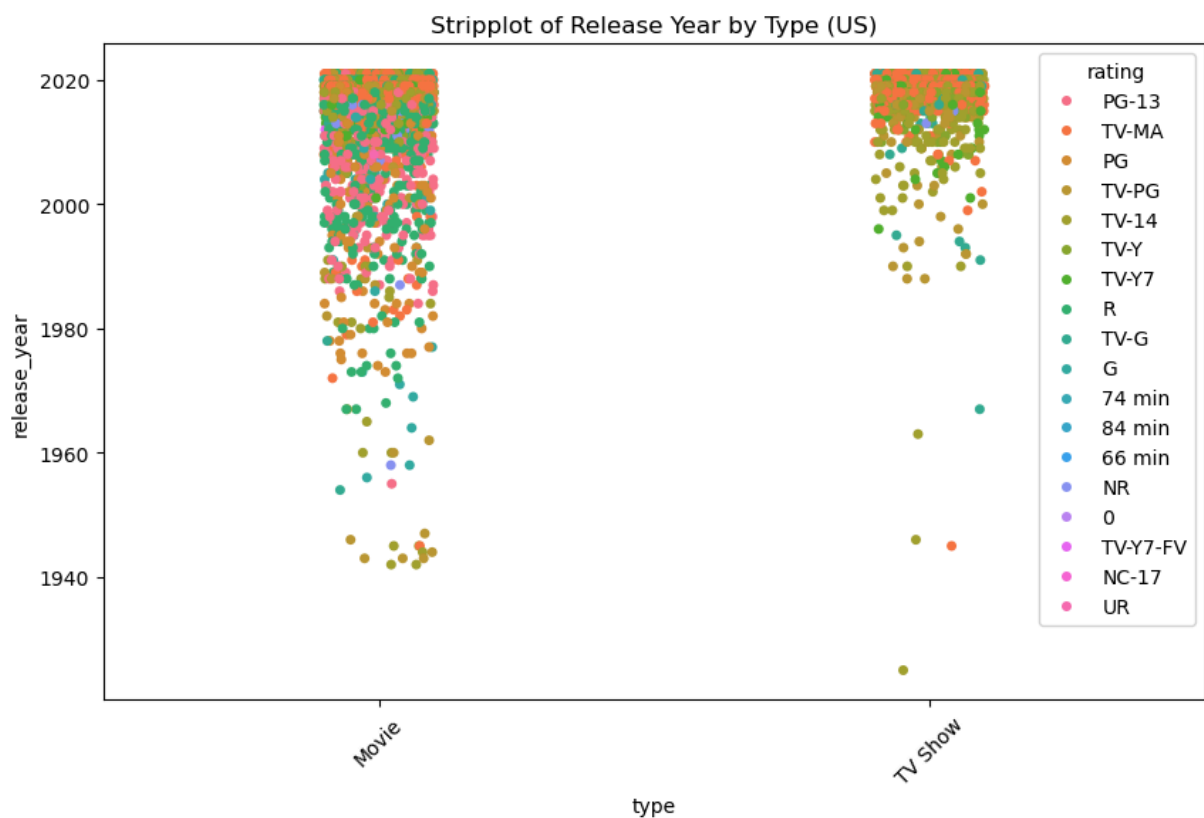
Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effect.



In [124...

Stripplot of Duration by Type hue Rating

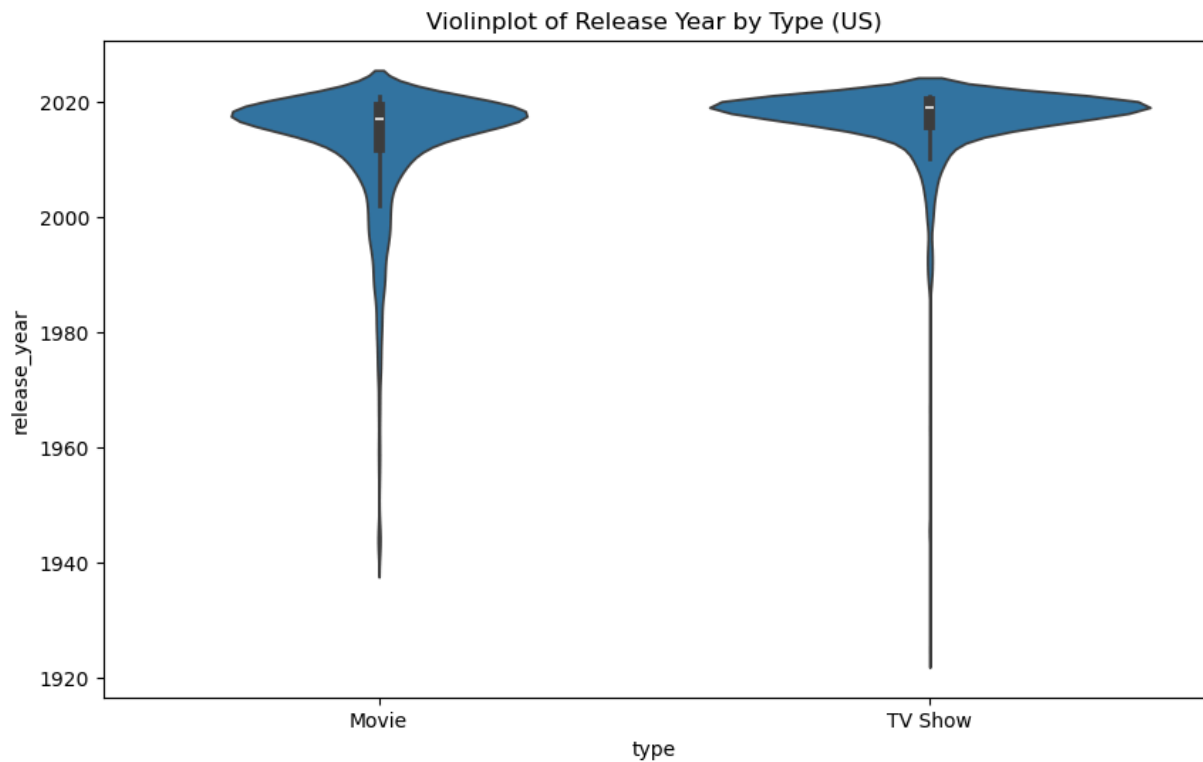
```
plt.figure(figsize=(10,6))
sns.stripplot(y='release_year', x='type', hue='rating', data=us_df)
plt.title("Stripplot of Release Year by Type (US)")
plt.xticks(rotation=45)
plt.show()
```



In [125...

```
# Violinplot of Release Year by Type (US)

plt.figure(figsize=(10,6))
sns.violinplot(y='release_year', x='type', data=us_df)
plt.title("Violinplot of Release Year by Type (US)")
plt.show()
```

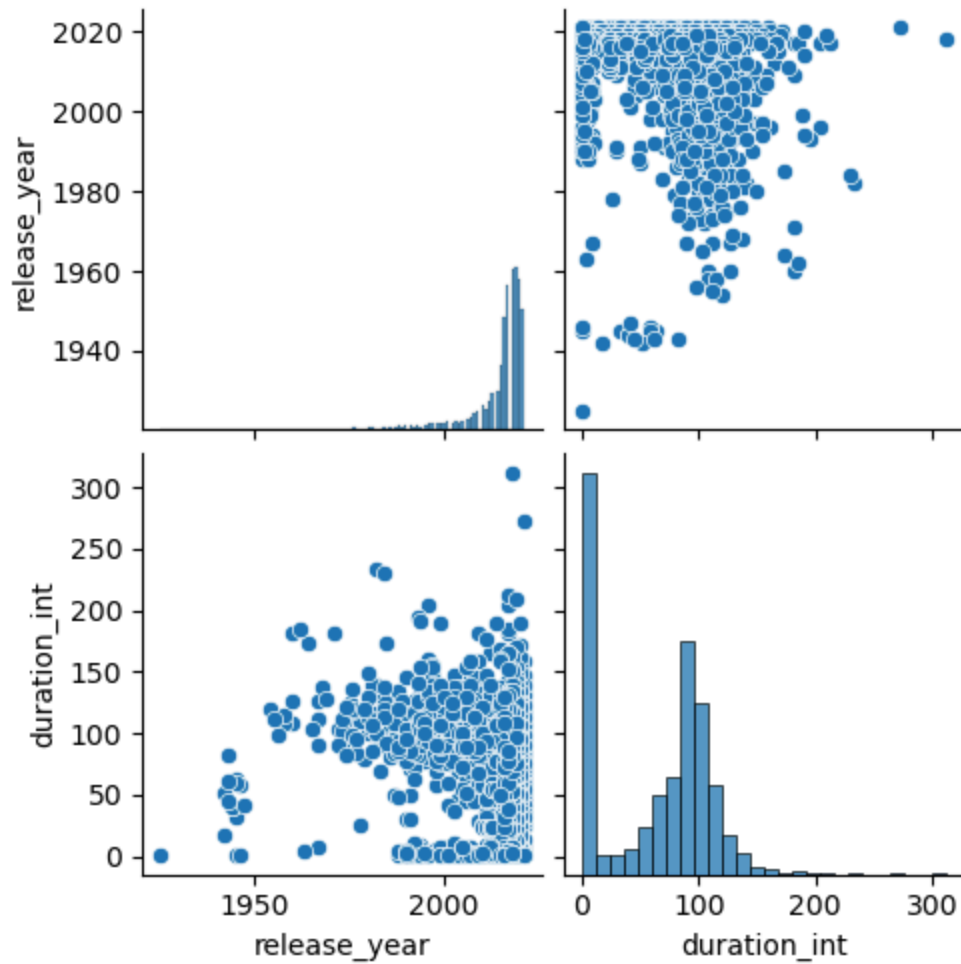


In [126...

```
# Pairplot of Netflix Dataset (US)

plt.figure(figsize=(12,8))
sns.pairplot(us_df[['release_year', 'duration_int']])
plt.show()
```

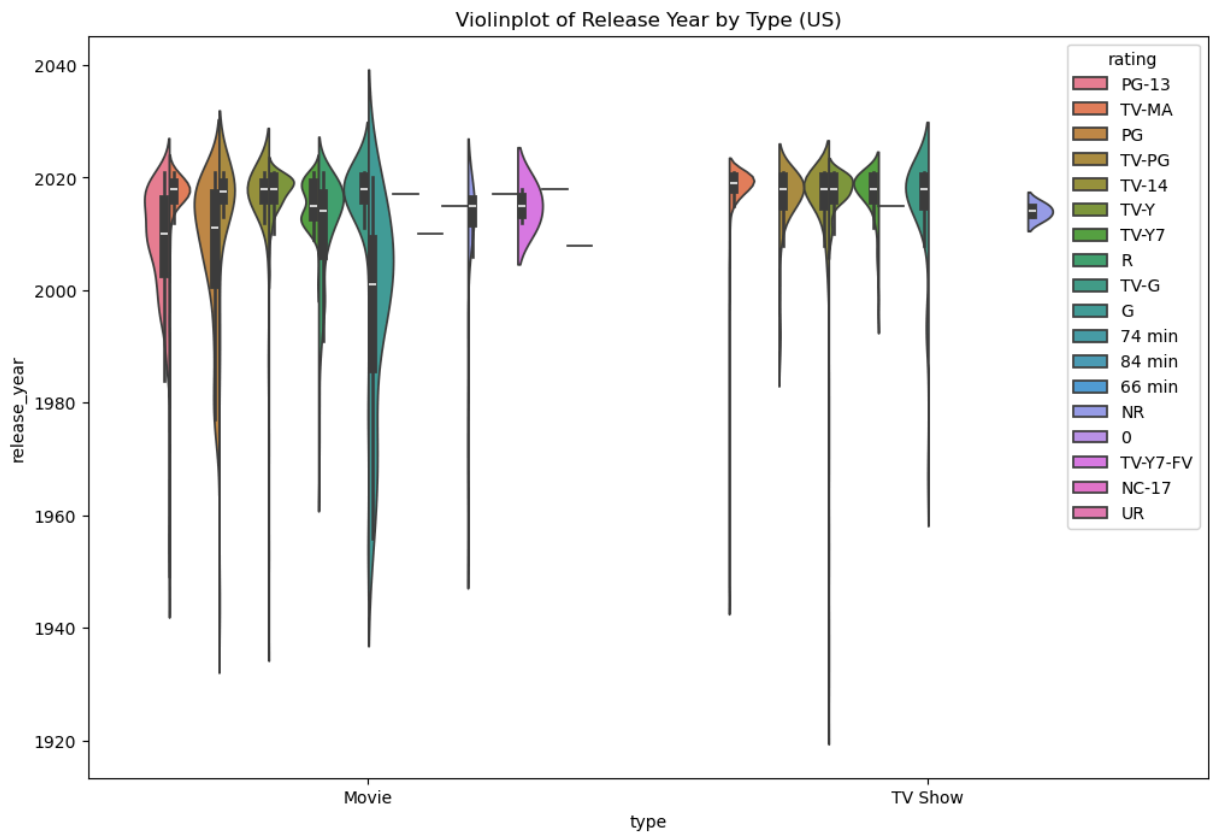
<Figure size 1200x800 with 0 Axes>



In [127...

```
# Violinplot of Release Year by Type with hue Rating (US)
```

```
plt.figure(figsize=(12,8))
sns.violinplot(x='type', y='release_year', data=us_df, hue='rating', split=True)
plt.title("Violinplot of Release Year by Type (US)")
plt.show()
```

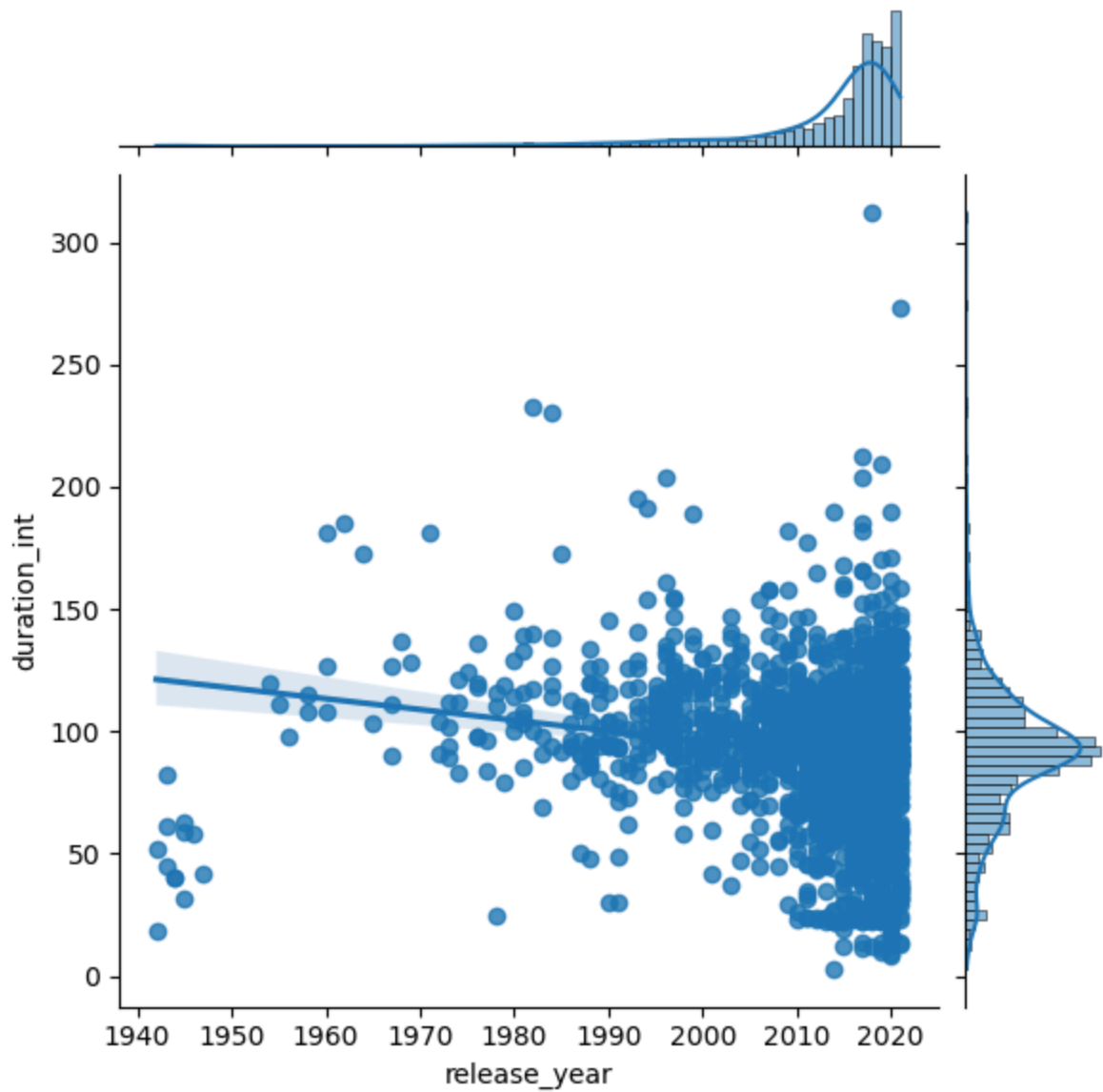


In [128...

Jointplot of Release Year vs Duration (US)

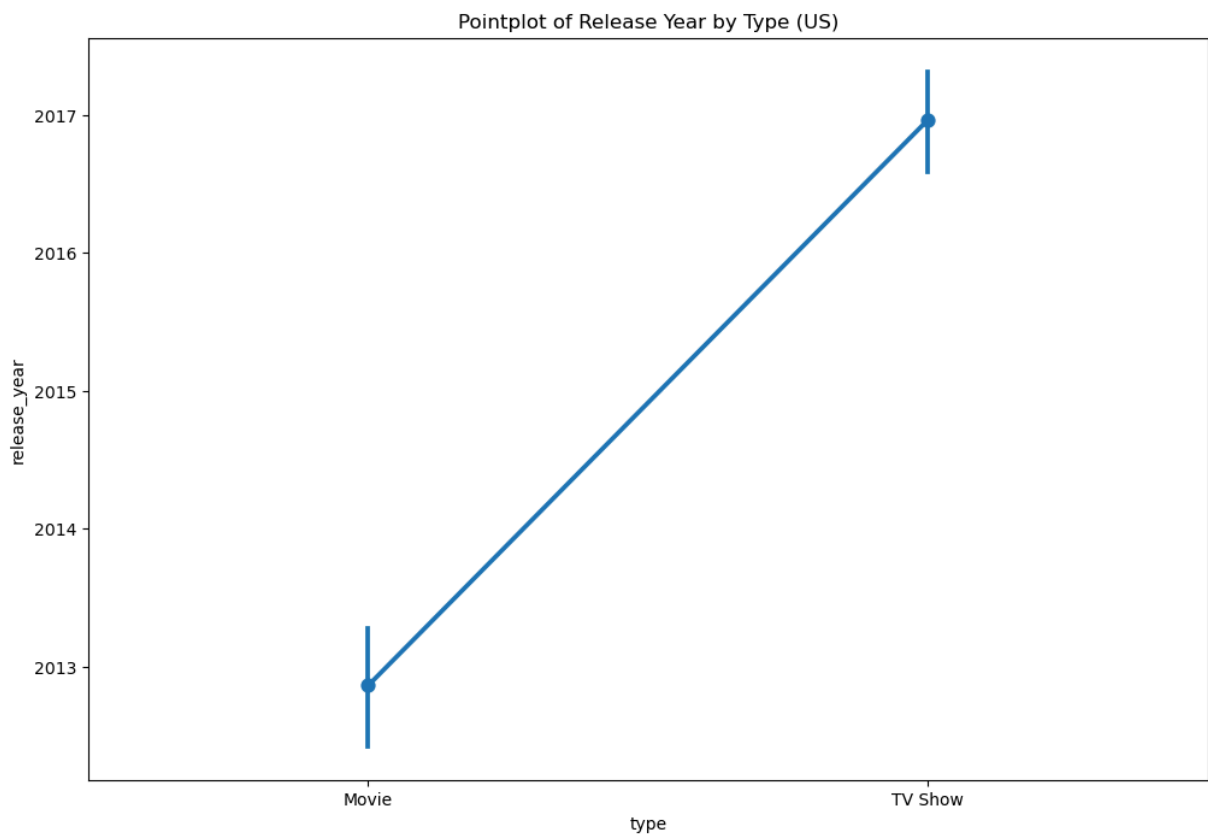
```
plt.figure(figsize=(12,8))
sns.jointplot(data=us_movies, x='release_year', y='duration_int', kind='reg')
plt.show()
```

<Figure size 1200x800 with 0 Axes>



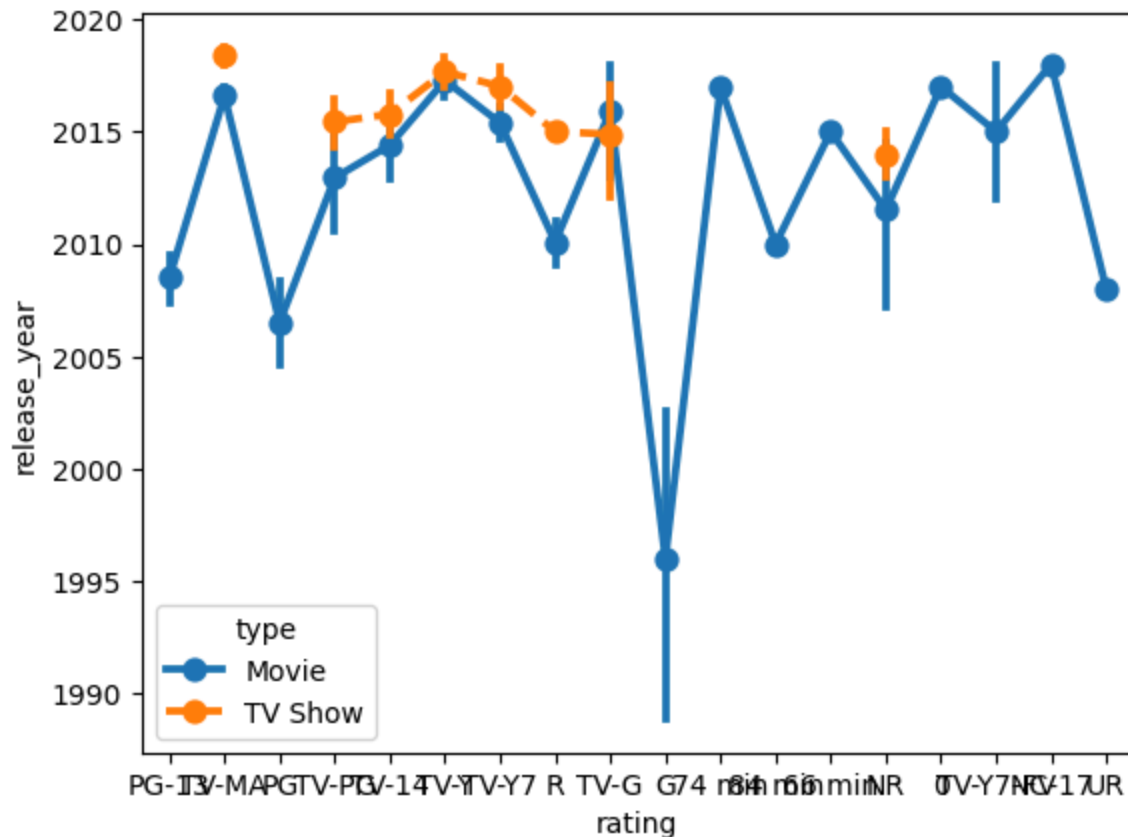
```
In [139... # Pointplot of Release Year by Type (US)

plt.figure(figsize=(12,8))
sns.pointplot(x='type', y='release_year', data=us_df)
plt.title("Pointplot of Release Year by Type (US)")
plt.show()
```



```
In [142... # Sirf Movie/TV Show 2 categories
sns.pointplot(
    x='rating',
    y='release_year',
    hue='type',
    linestyle=["-", "--"],
    data=us_df
)
```

```
Out[142... <Axes: xlabel='rating', ylabel='release_year'>
```



India filter data

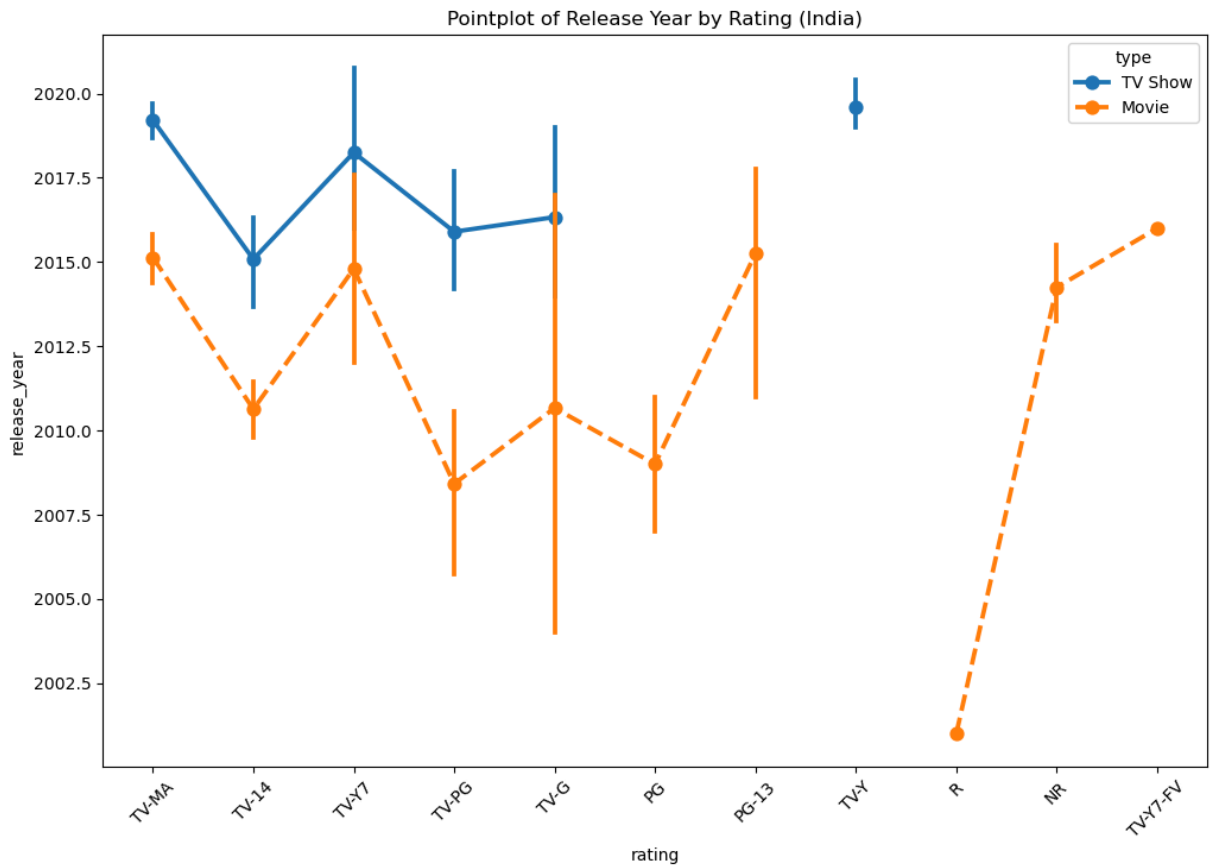
```
In [144...] df.loc[df['country'] == 'India']['type'].value_counts()
```

```
Out[144...] type
Movie      893
TV Show     79
Name: count, dtype: int64
```

```
In [143...] # India filtered data

india_df = df.loc[df['country'] == 'India'].copy()
india_df['duration_int'] = india_df['duration'].str.extract(r'(\d+)').astype(float)

# Pointplot of Release Year by Rating hue Type (India)
plt.figure(figsize=(12,8))
sns.pointplot(
    x='rating',
    y='release_year',
    hue='type',
    linestyles=["-", "--"],
    data=india_df
)
plt.title("Pointplot of Release Year by Rating (India)")
plt.xticks(rotation=45)
plt.show()
```



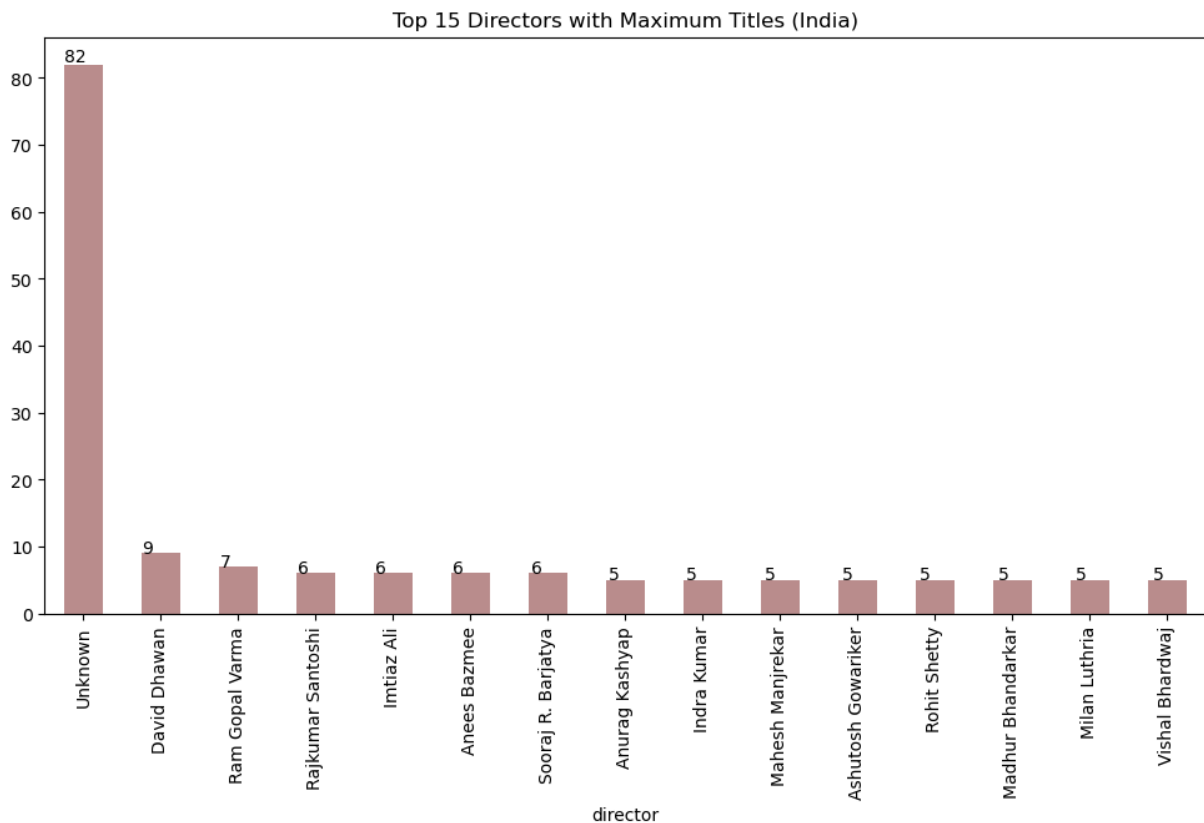
In [145...

Top 15 Directors with maximum number of titles on Netflix (India)

```

ax = india_df['director'].value_counts().head(15).plot.bar(figsize=(12,6), color="r")
ax.set_title("Top 15 Directors with Maximum Titles (India)")
for i in ax.patches:
    ax.annotate(i.get_height(), (i.get_x() * 1.005, i.get_height() * 1.005))
plt.show()

```

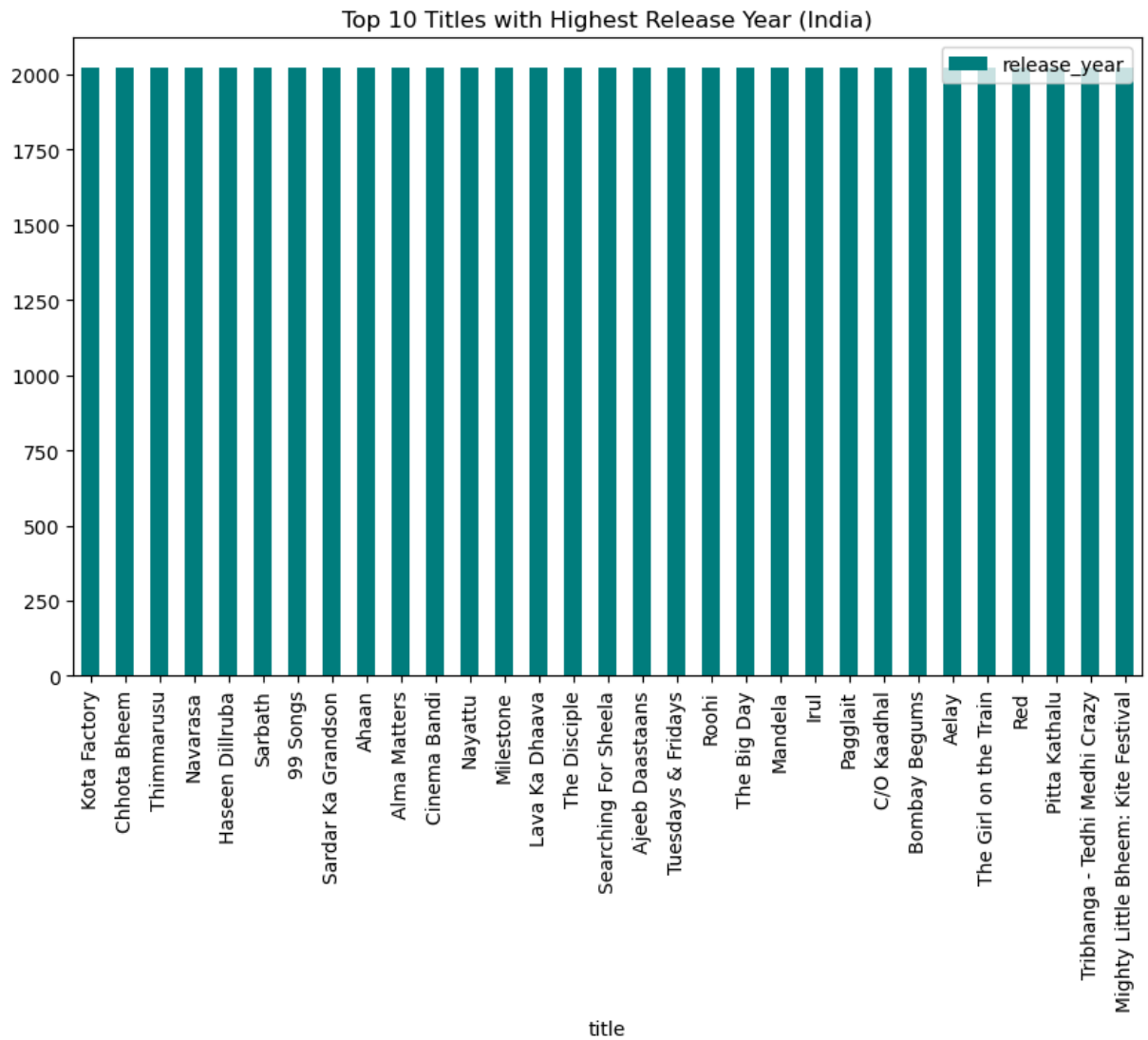


```
In [146... # Top 10 latest released titles on Netflix (India)
max_votes = india_df['release_year'].sort_values(ascending=False).head(10)
max_votes
```

```
Out[146... 4      2021
1037    2021
735     2021
850     2021
871     2021
873     2021
876     2021
903     2021
909     2021
918     2021
Name: release_year, dtype: int64
```

```
In [147... # Top 10 Titles with highest release year (India)

max_year = india_df['release_year'].sort_values(ascending=False).head(10)
india_df.loc[india_df['release_year'].isin(max_year)][['title', 'release_year']].pl
    x='title',
    y='release_year',
    figsize=(10,6),
    color="teal"
)
plt.title("Top 10 Titles with Highest Release Year (India)")
plt.show()
```



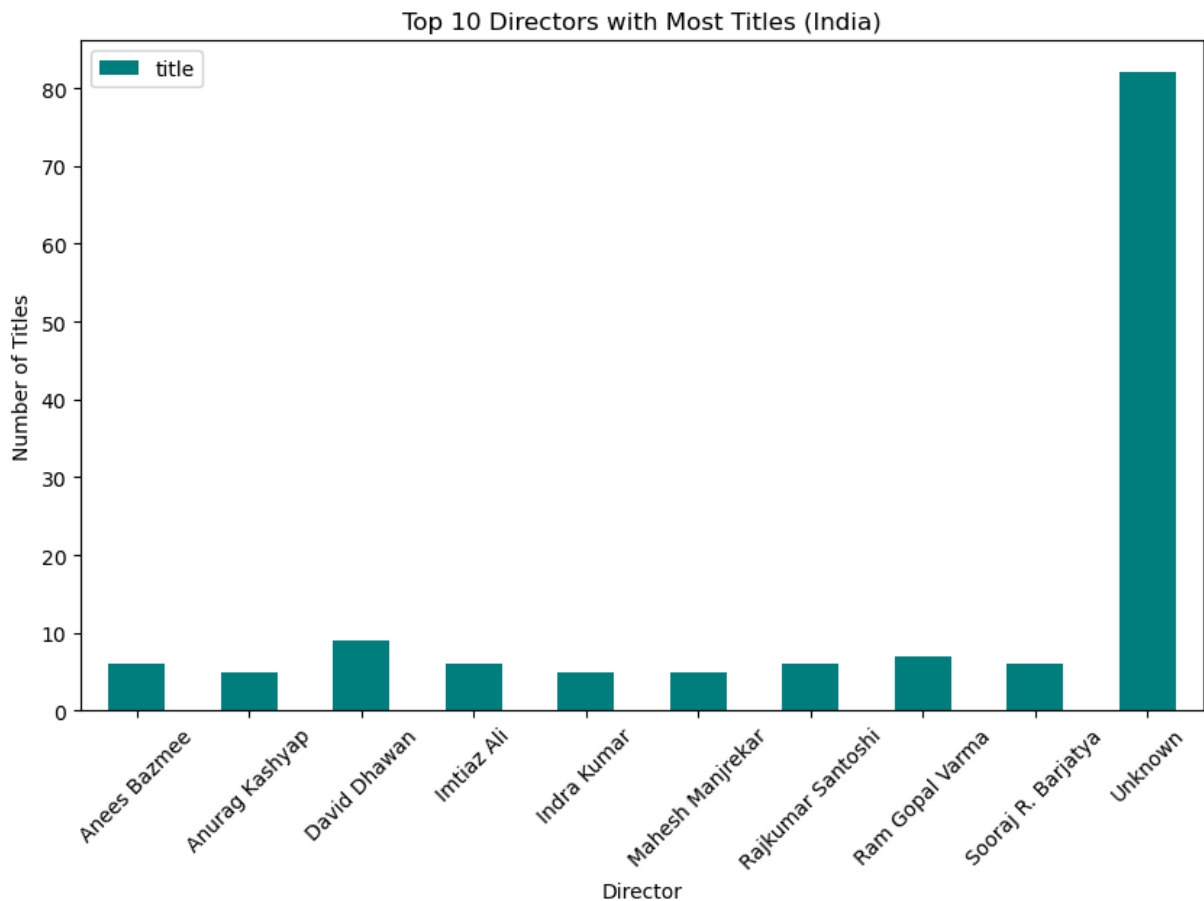
In [148...

Top 10 Directors with most titles on Netflix (India)

```

top_directors = india_df['director'].value_counts().head(10)
india_df.loc[india_df['director'].isin(top_directors.index)][['title', 'director']]
    figsize=(10,6),
    color="teal"
)
plt.title("Top 10 Directors with Most Titles (India)")
plt.xlabel("Director")
plt.ylabel("Number of Titles")
plt.xticks(rotation=45)
plt.show()

```



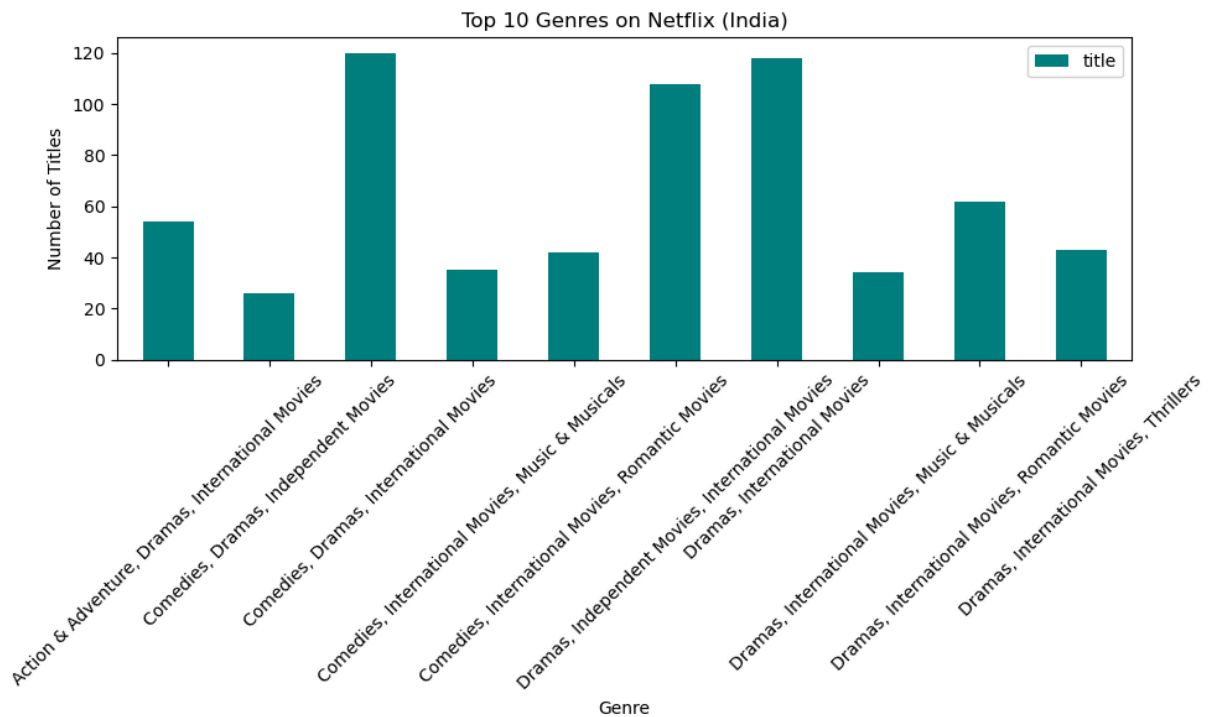
In [149...

Top 10 Genres on Netflix (India)

```

max_genres = india_df['listed_in'].value_counts().sort_values(ascending=False).head
india_df.loc[india_df['listed_in'].isin(max_genres.index)][['title', 'listed_in']].
    y='title',
    figsize=(10,6),
    color="teal"
)
plt.title("Top 10 Genres on Netflix (India)")
plt.xlabel("Genre")
plt.ylabel("Number of Titles")
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()

```

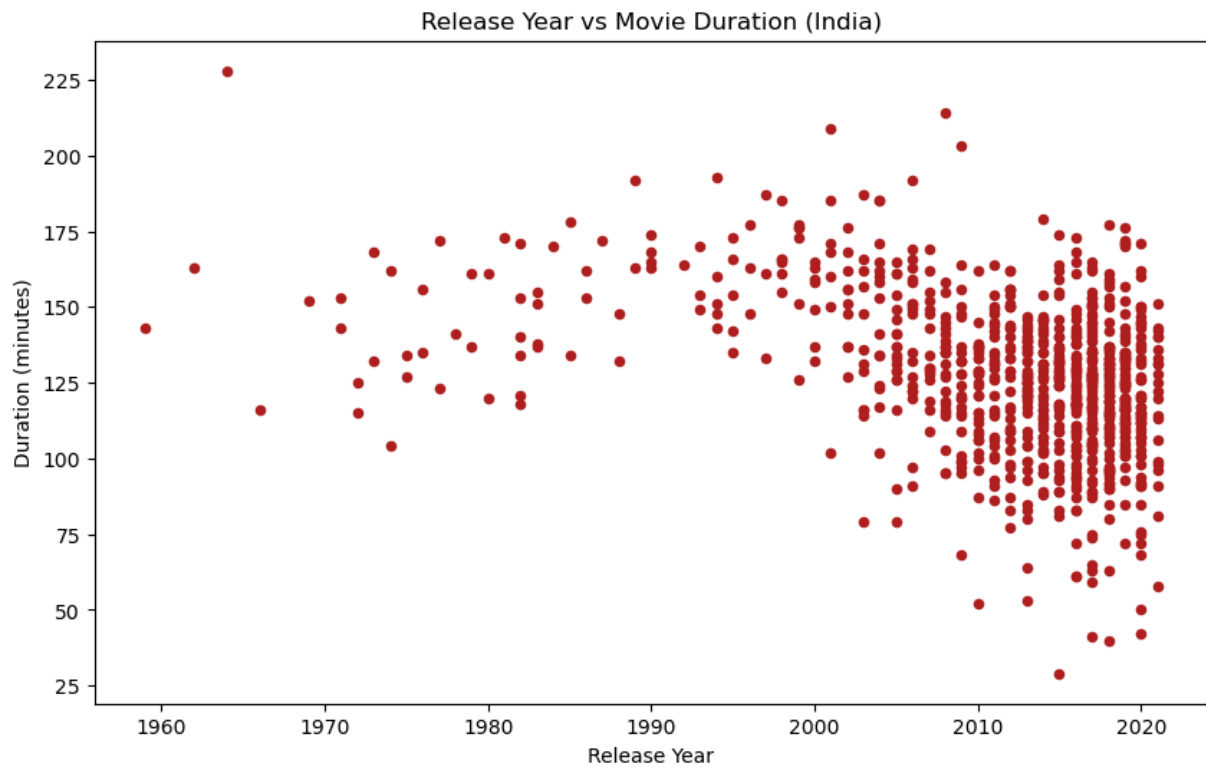


```
In [150... india_movies = india_df[india_df['type'] == 1] # 1 = Movie
```

```
In [158... india_df['duration_int'] = india_df['duration'].str.extract(r'(\d+)').astype(float)
india_movies = india_df[india_df['type'] == 'Movie'].copy()
print(india_movies[['title', 'duration', 'duration_int']].head())
```

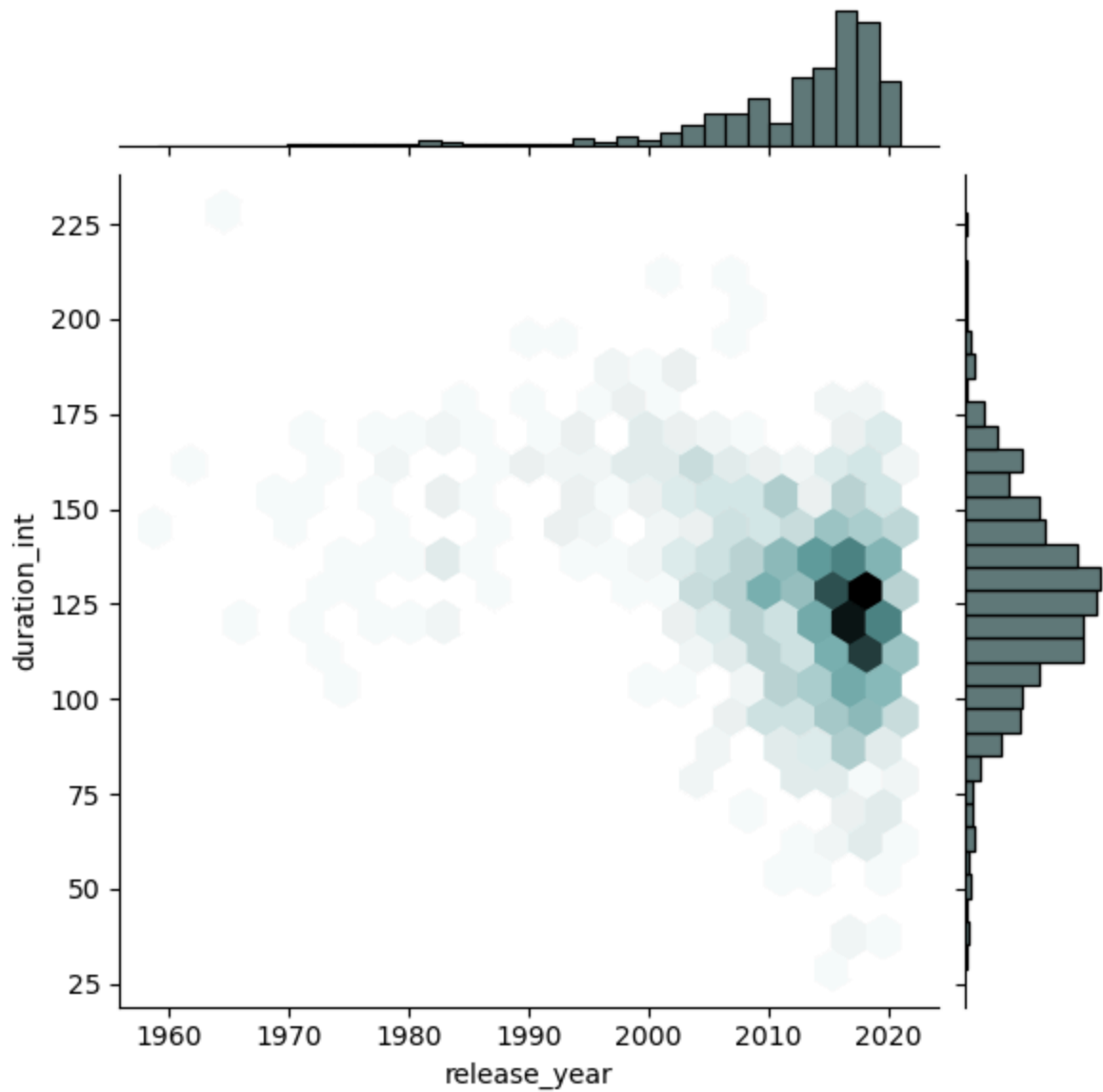
	title	duration	duration_int
24	Jeans	166 min	166.0
105	Angamaly Diaries	128 min	128.0
114	Anjaam	143 min	143.0
116	Dhanak	114 min	114.0
118	Gurgaon	106 min	106.0

```
In [159... india_movies.plot.scatter(
    x='release_year',
    y='duration_int',
    figsize=(10,6),
    color='firebrick',
    title='Release Year vs Movie Duration (India)'
)
plt.xlabel("Release Year")
plt.ylabel("Duration (minutes)")
plt.show()
```

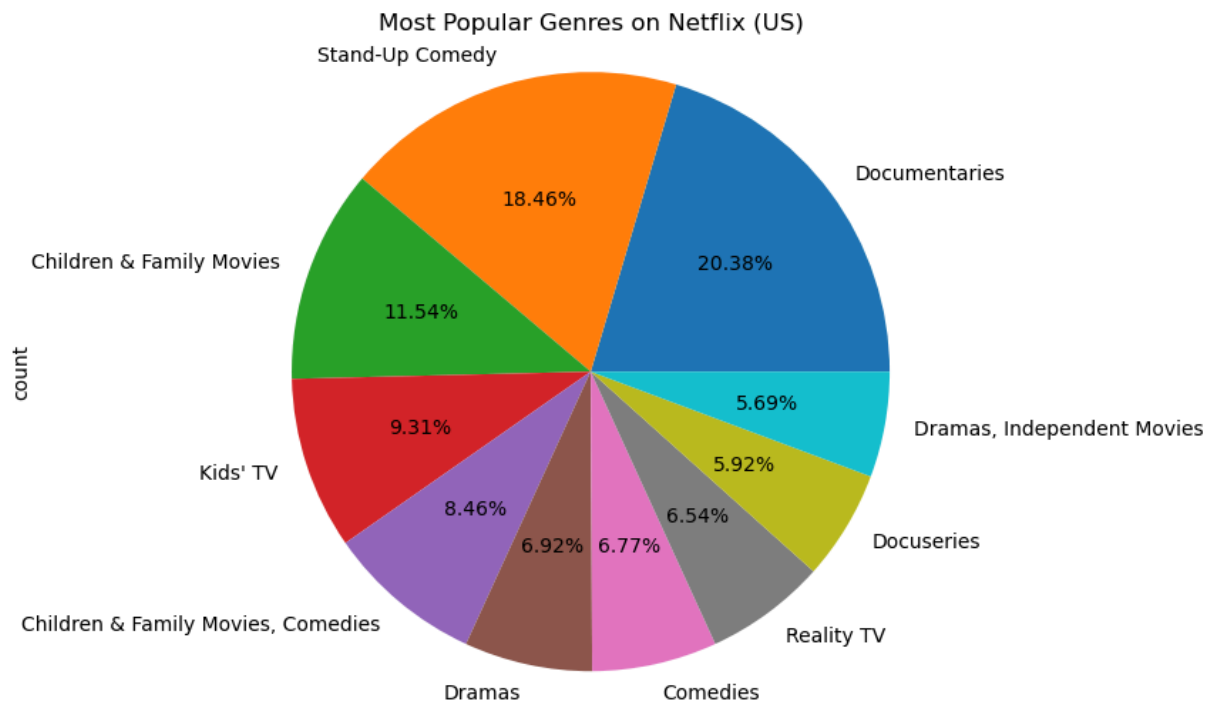


In [160... *# Better view of relation between Release Year and Movie Duration (India)*

```
sns.jointplot(  
    x='release_year',  
    y='duration_int',  
    kind='hex',  
    gridsize=22,  
    data=india_movies,  
    color='darkslategrey'  
)  
plt.show()
```



```
In [161... # Top 10 Genres on Netflix (US)
us_df['listed_in'].value_counts().sort_values(ascending=False).head(10).plot(
    kind='pie',
    figsize=(10,6),
    title="Most Popular Genres on Netflix (US)",
    autopct='%1.2f%%'
)
plt.axis('equal')
plt.show()
```

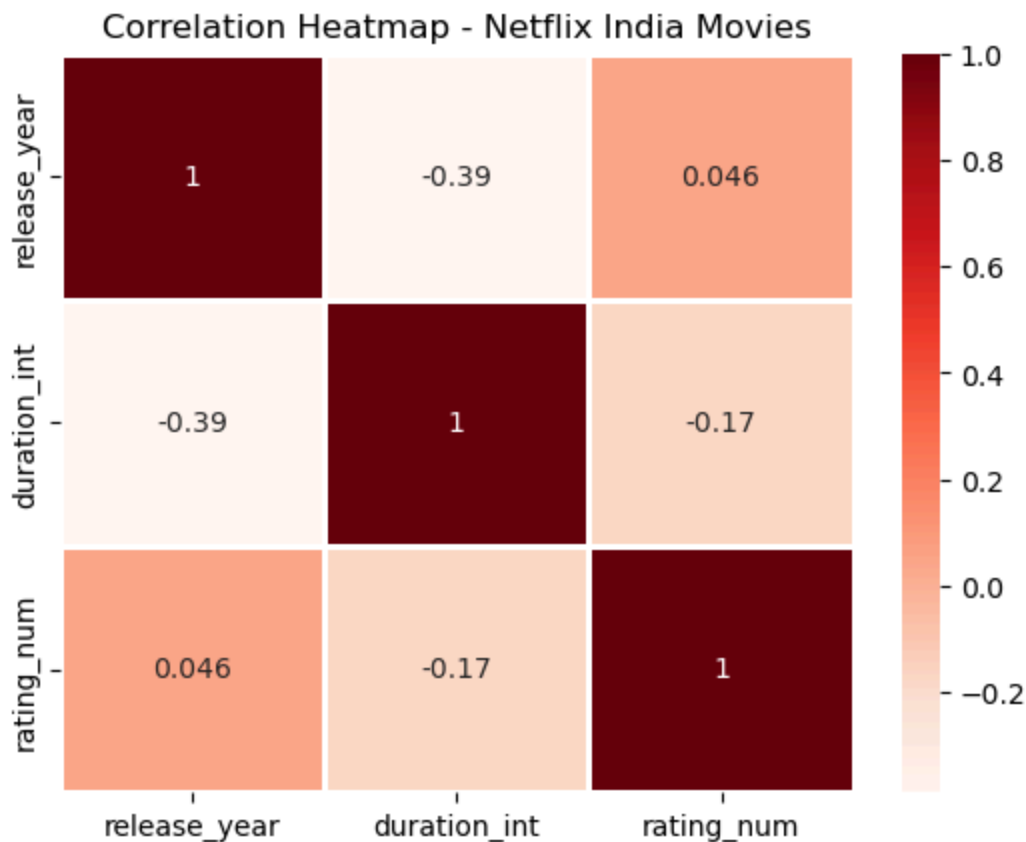


```
In [162... print(india_movies['rating'].unique())
```

```
['TV-14' 'TV-PG' 'TV-MA' 'TV-Y7' 'PG' 'PG-13' 'TV-G' 'R' 'NR' 'TV-Y7-FV']
```

```
In [163... # CONVERT TO RATING COLUMN IN STRING

india_movies = india_movies.copy()
india_movies['rating'] = india_movies['rating'].astype(str)
# LabelEncoder
from sklearn.preprocessing import LabelEncoder
le = LabelEncoder()
india_movies['rating_num'] = le.fit_transform(india_movies['rating'].fillna('NR'))
# Heatmap
netflix_corr = india_movies[['release_year', 'duration_int', 'rating_num']]
sns.heatmap(netflix_corr.corr(), linewidth=1.0, annot=True, cmap='Reds')
plt.title("Correlation Heatmap - Netflix India Movies")
plt.show()
```



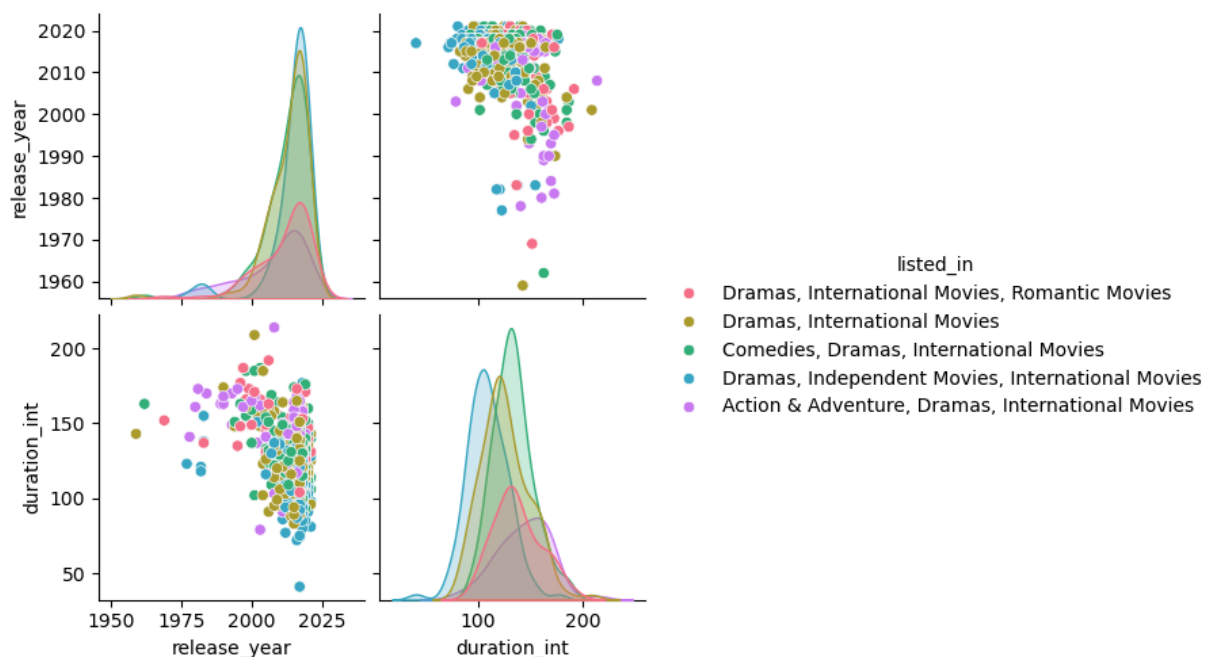
In [164...

More insight for correlation - Top 5 Genres on Netflix (India)

```

top5_genres = india_df['listed_in'].value_counts().head(5).index.tolist()
netflix_p = india_df.loc[india_df['listed_in'].isin(top5_genres)]
netflix_pair = netflix_p[['release_year', 'duration_int', 'listed_in']]
sns.pairplot(netflix_pair, hue='listed_in', palette="husl")
plt.show()

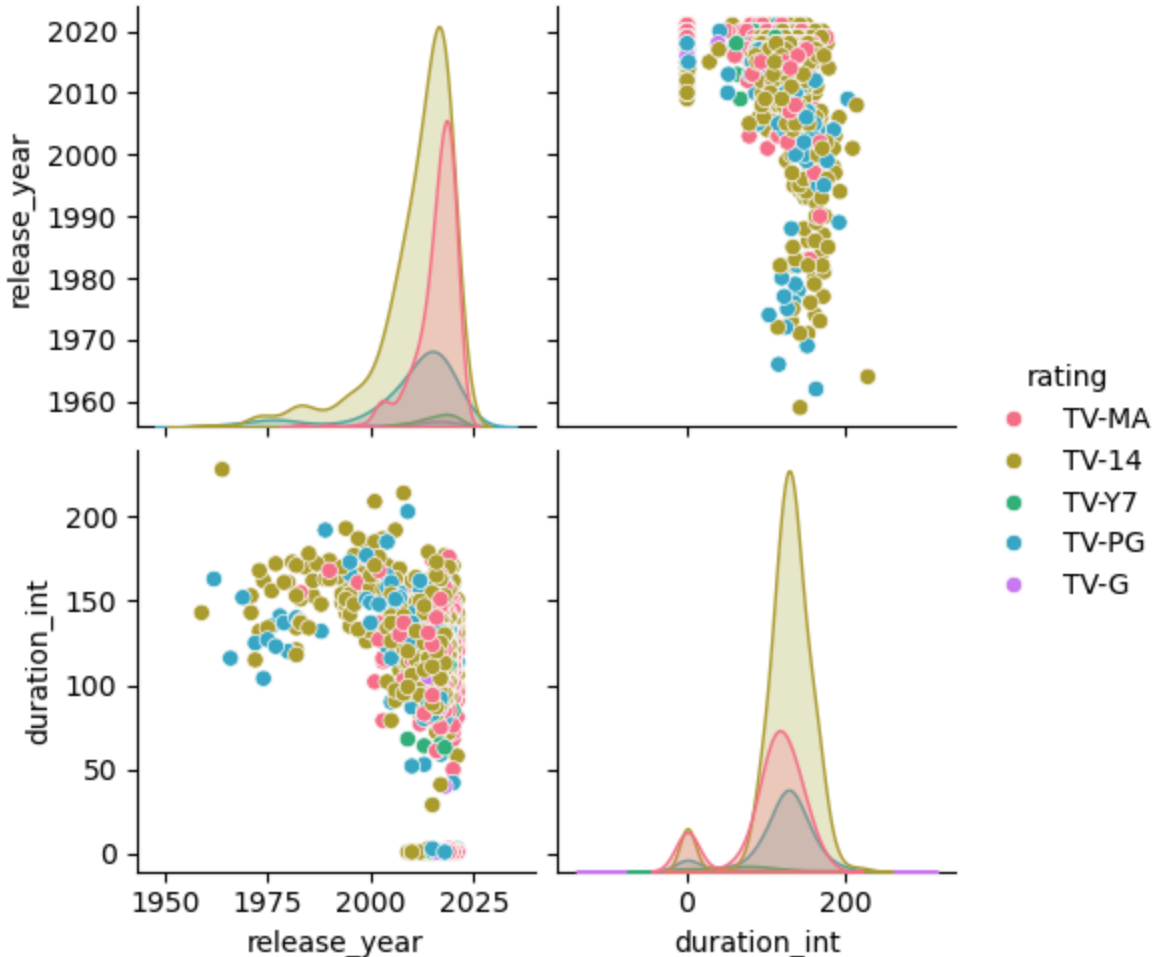
```



In [166...

```
# More insight for correlation - Top 5 Ratings on Netflix (India)

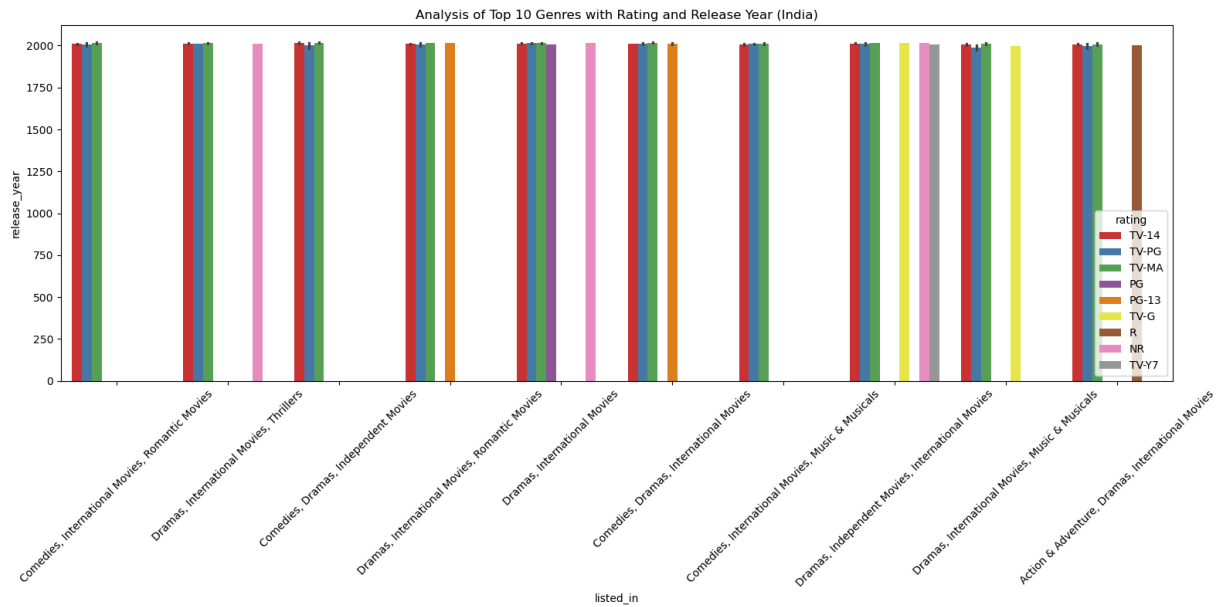
top5_ratings = india_df['rating'].value_counts().head(5).index.tolist()
netflix_p2 = india_df.loc[india_df['rating'].isin(top5_ratings)]
netflix_pair2 = netflix_p2[['release_year', 'duration_int', 'rating']]
sns.pairplot(netflix_pair2, hue='rating', palette="husl")
plt.show()
```



In [167...

```
# Analysis of Top 10 Genres with Type and Release Year (India)

top10genres_list = india_df['listed_in'].value_counts().head(10).index.tolist()
netflix_genres = india_df.loc[india_df['listed_in'].isin(top10genres_list)]
netflix_genres_data = netflix_genres[['release_year', 'duration_int', 'rating', 'li
fig, axx = plt.subplots(figsize=(16,8))
sns.barplot(x='listed_in', y='release_year', hue='rating', data=netflix_genres_data
axx.set_title("Analysis of Top 10 Genres with Rating and Release Year (India)")
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```



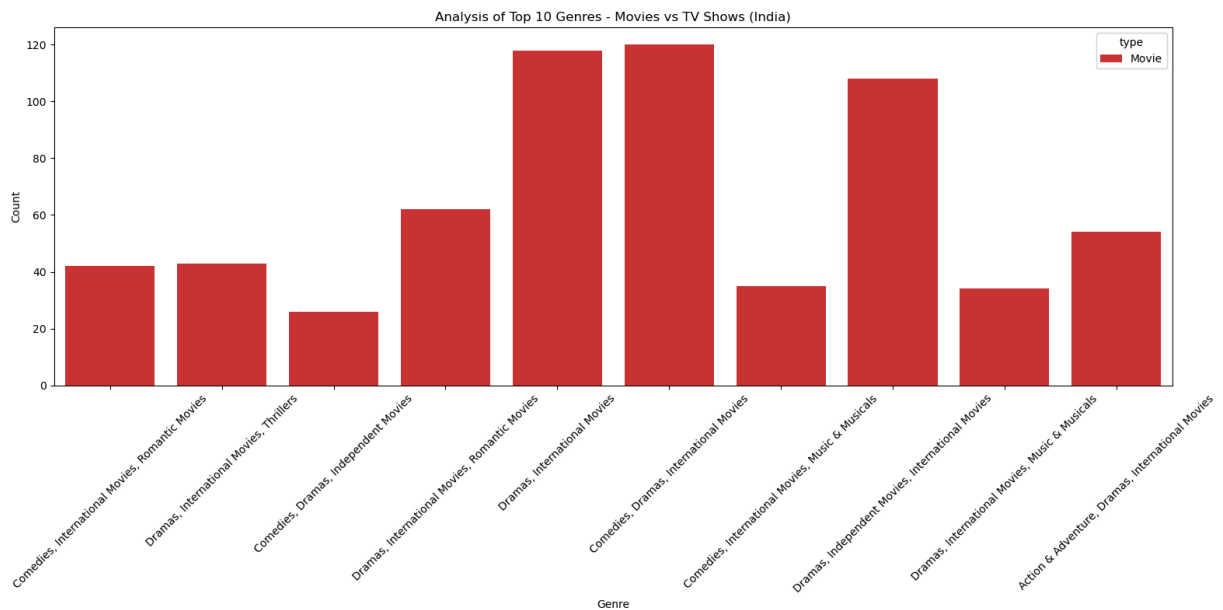
In [168...

Analysis of Top 10 Genres with count and Type (India)

```

top10genres_list = india_df['listed_in'].value_counts().head(10).index.tolist()
netflix_genres = india_df.loc[india_df['listed_in'].isin(top10genres_list)]
fig, axx = plt.subplots(figsize=(16,8))
sns.countplot(
    x='listed_in',
    hue='type',
    data=netflix_genres,
    palette="Set1"
)
axx.set_title("Analysis of Top 10 Genres - Movies vs TV Shows (India)")
plt.xlabel("Genre")
plt.ylabel("Count")
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()

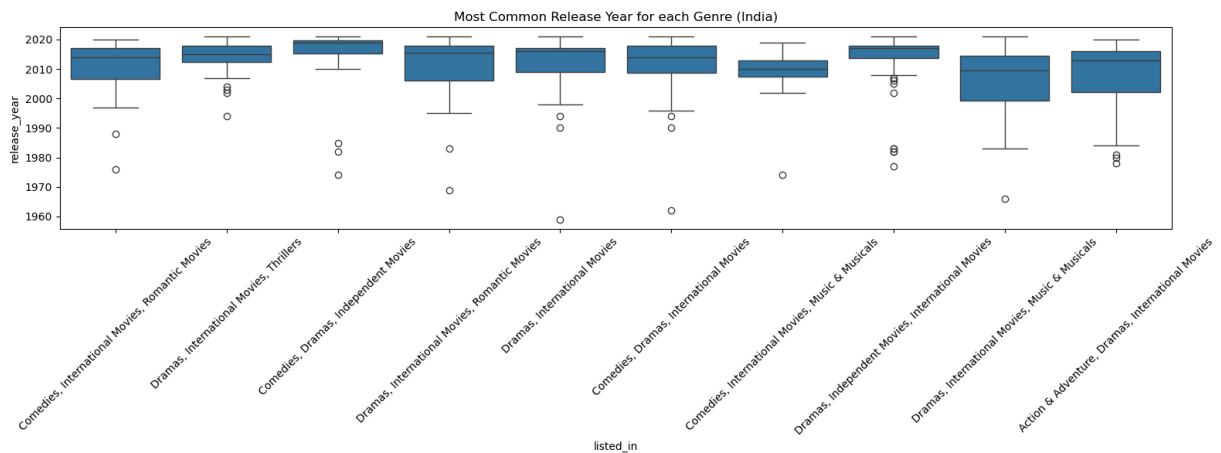
```



In [169...

```
# Most common Release Year for each type of Genre (India)

top10genres_list = india_df['listed_in'].value_counts().head(10).index.tolist()
netflix_genres = india_df.loc[india_df['listed_in'].isin(top10genres_list)]
netflix_genres_data = netflix_genres[['release_year', 'duration_int', 'listed_in']]
fig, axx = plt.subplots(figsize=(16,6))
sns.boxplot(x='listed_in', y='release_year', data=netflix_genres_data)
plt.title("Most Common Release Year for each Genre (India)")
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```

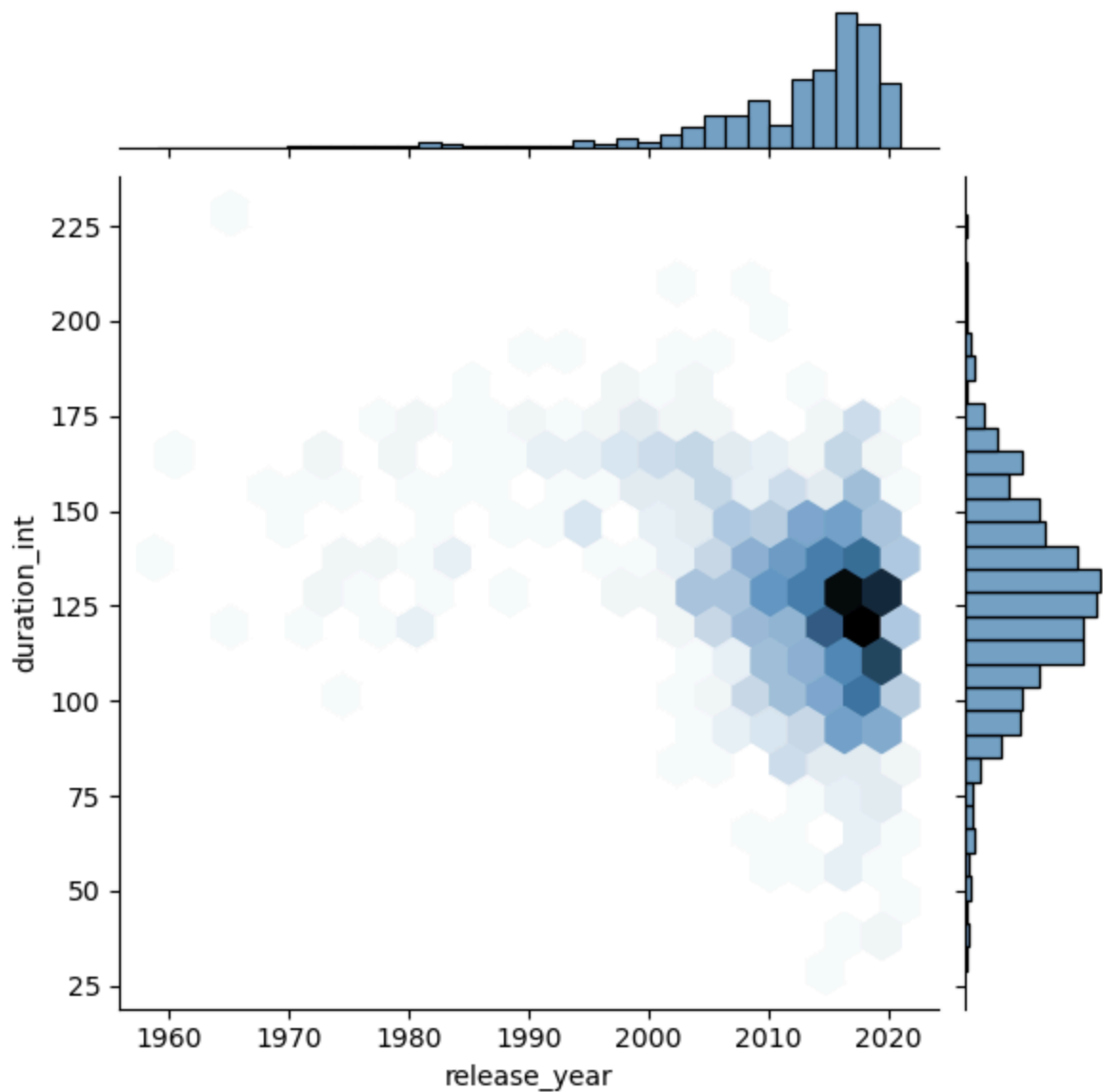


In [170...

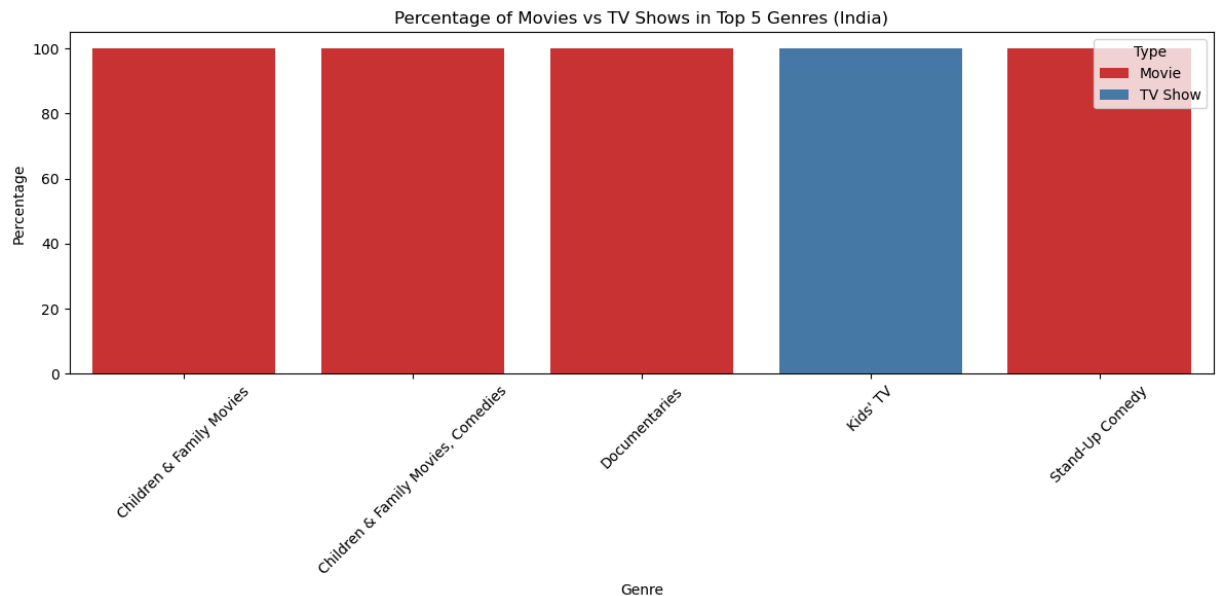
```
top5genres_list = india_df['listed_in'].value_counts().head(5).index.tolist()
netflix_rate = india_df.loc[india_df['listed_in'].isin(top5genres_list)]
# Find total number of titles per genre
total_titles = netflix_rate.groupby(['listed_in'], as_index=False).count()[['listed_in', 'duration_int']]
total_titles.columns = ['Genre', 'Total Titles']
# Find total count of each type
```

In [171...

```
# Better view of relation between release year and duration of movies (India)
sns.jointplot(
    x='release_year',
    y='duration_int',
    kind='hex',
    gridsize=20,
    data=india_movies,
    color='steelblue'
)
plt.show()
```



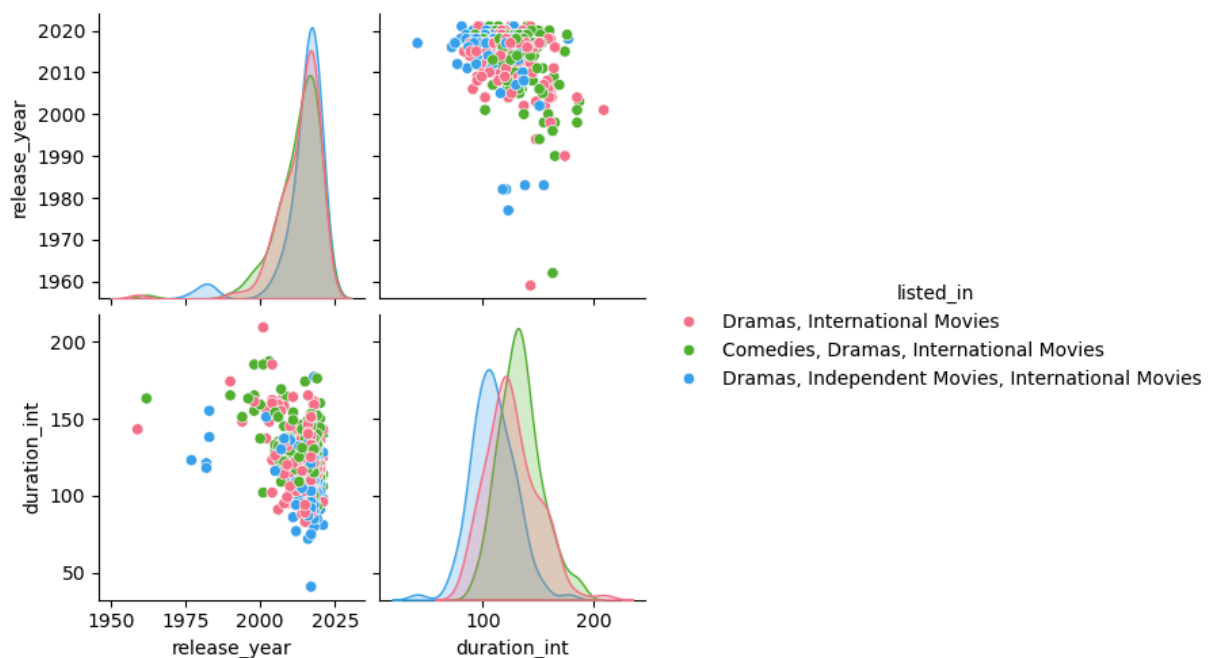
```
In [172... # Plot Type percentage of titles in Top 5 Genres (India)
fig, axx = plt.subplots(figsize=(12,6))
axx.set_title("Percentage of Movies vs TV Shows in Top 5 Genres (India)")
sns.barplot(x='Genre', y='Percentage', hue='Type', data=top5genres_percent, palette=
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```



Keeping our data limited to top 3 genres with maximum titles is India.

```
In [174... # duration_int column

india_df = india_df.copy()
india_df['duration_int'] = india_df['duration'].str.extract(r'(\d+)').astype(float)
# pairplot
top3genres_list = india_df['listed_in'].value_counts().head(3).index.tolist()
netflix_p = india_df.loc[india_df['listed_in'].isin(top3genres_list)]
netflix_pair = netflix_p[['release_year', 'duration_int', 'listed_in']]
sns.pairplot(netflix_pair, hue='listed_in', palette="husl")
plt.show()
```



```
In [175... india_df = df.loc[df['country'] == 'India']
```

In [176...

india_df

Out[176...

	show_id	type	title	director	cast	country	date_added	release_
4	s5	TV Show	Kota Factory	Unknown	Mayur More, Jitendra Kumar, Ranjan Raj, Alam K...	India	September 24, 2021	
24	s25	Movie	Jeans	S. Shankar	Prashanth, Aishwarya Rai Bachchan, Sri Lakshmi...	India	September 21, 2021	
39	s40	TV Show	Chhota Bheem	Unknown	Vatsal Dubey, Julie Tejwani, Rupa Bhimani, Jig...	India	September 16, 2021	
50	s51	TV Show	Dharmakshetra	Unknown	Kashmira Irani, Chandan Anand, Dinesh Mehta, A...	India	September 15, 2021	
66	s67	TV Show	Raja Rasoi Aur Anya Kahaniyan	Unknown	Unknown	India	September 15, 2021	
...	...	...	...	...	...	...	...	...
8773	s8774	Movie	Yanda Kartavya Aahe	Kedar Shinde	Ankush Choudhary, Smita Shewale, Mohan Joshi, ...	India	January 1, 2018	
8775	s8776	TV Show	Yeh Meri Family	Unknown	Vishesh Bansal, Mona Singh, Akarsh Khurana, Ah...	India	August 31, 2018	

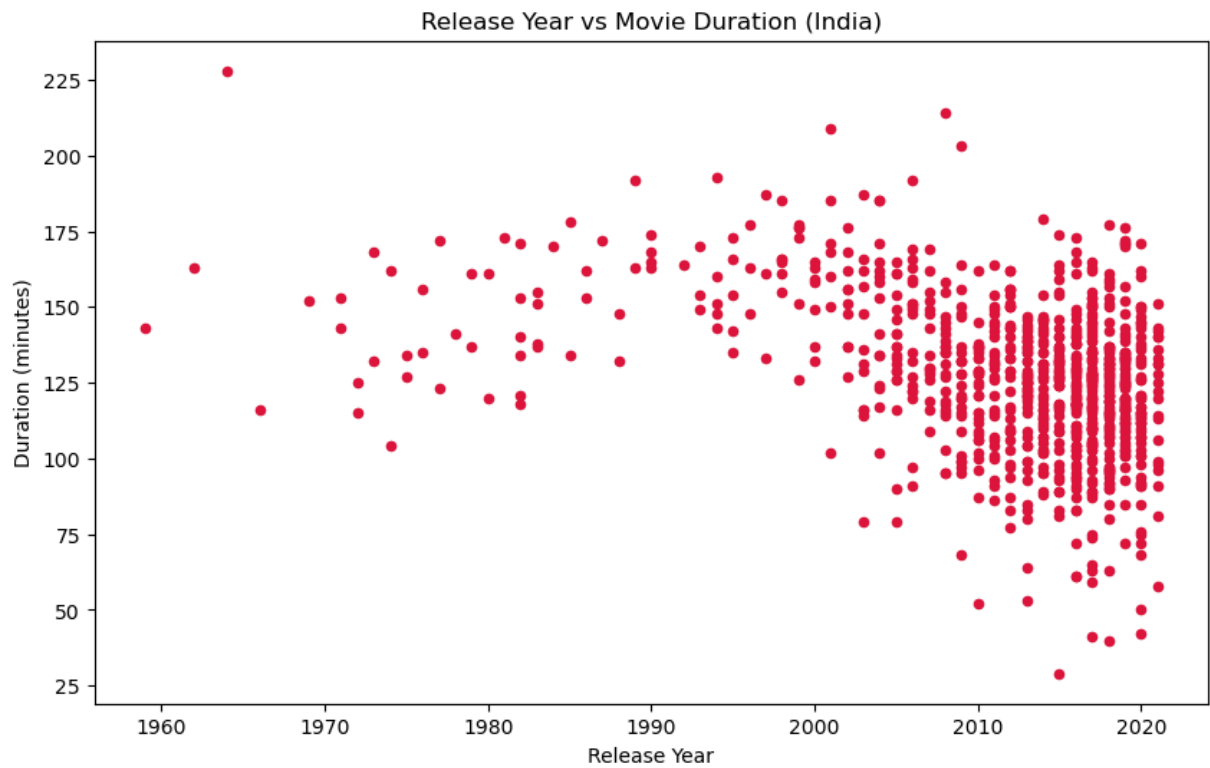
	show_id	type	title	director	cast	country	date_added	release_
8798	s8799	Movie	Zed Plus	Chandra Prakash Dwivedi	Adil Hussain, Mona Singh, K.K. Raina, Sanjay M...	India	December 31, 2019	
8799	s8800	Movie	Zenda	Avadhoot Gupte	Santosh Juvekar, Siddharth Chandekar, Sachit P...	India	February 15, 2018	
8806	s8807	Movie	Zubaan	Mozez Singh	Vicky Kaushal, Sarah-Jane Dias, Raaghav Chanan...	India	March 2, 2019	

972 rows × 13 columns

Lets try to find out if there is any relation between release year and duration of movies in India.

```
In [177... # Is there any relation between release year and duration of movies (India)

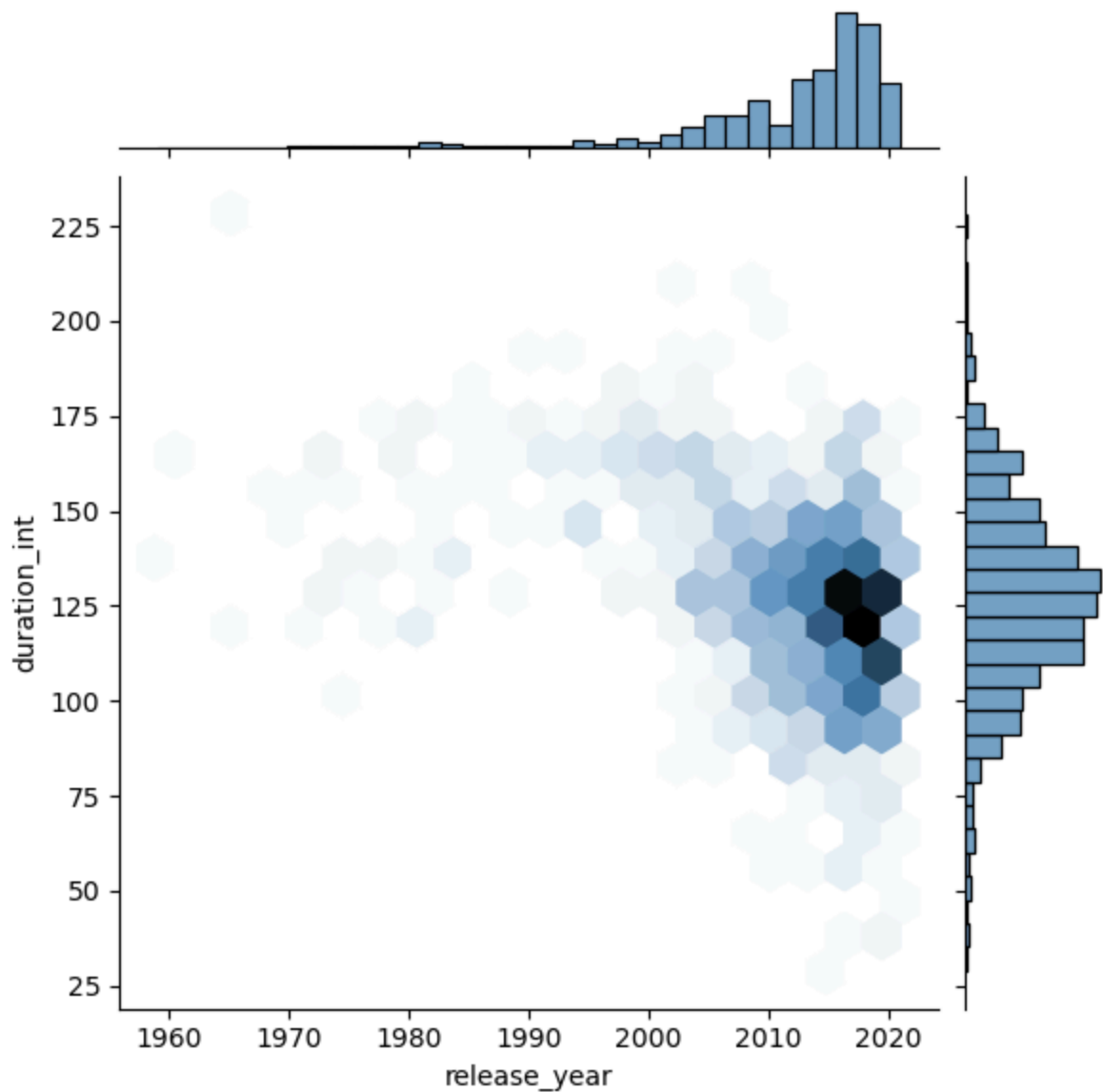
india_movies.plot.scatter(
    x='release_year',
    y='duration_int',
    figsize=(10,6),
    color='crimson',
    title='Release Year vs Movie Duration (India)'
)
plt.xlabel("Release Year")
plt.ylabel("Duration (minutes)")
plt.show()
```



In [178...

```
# Better view of relation between release year and duration of movies (India)
```

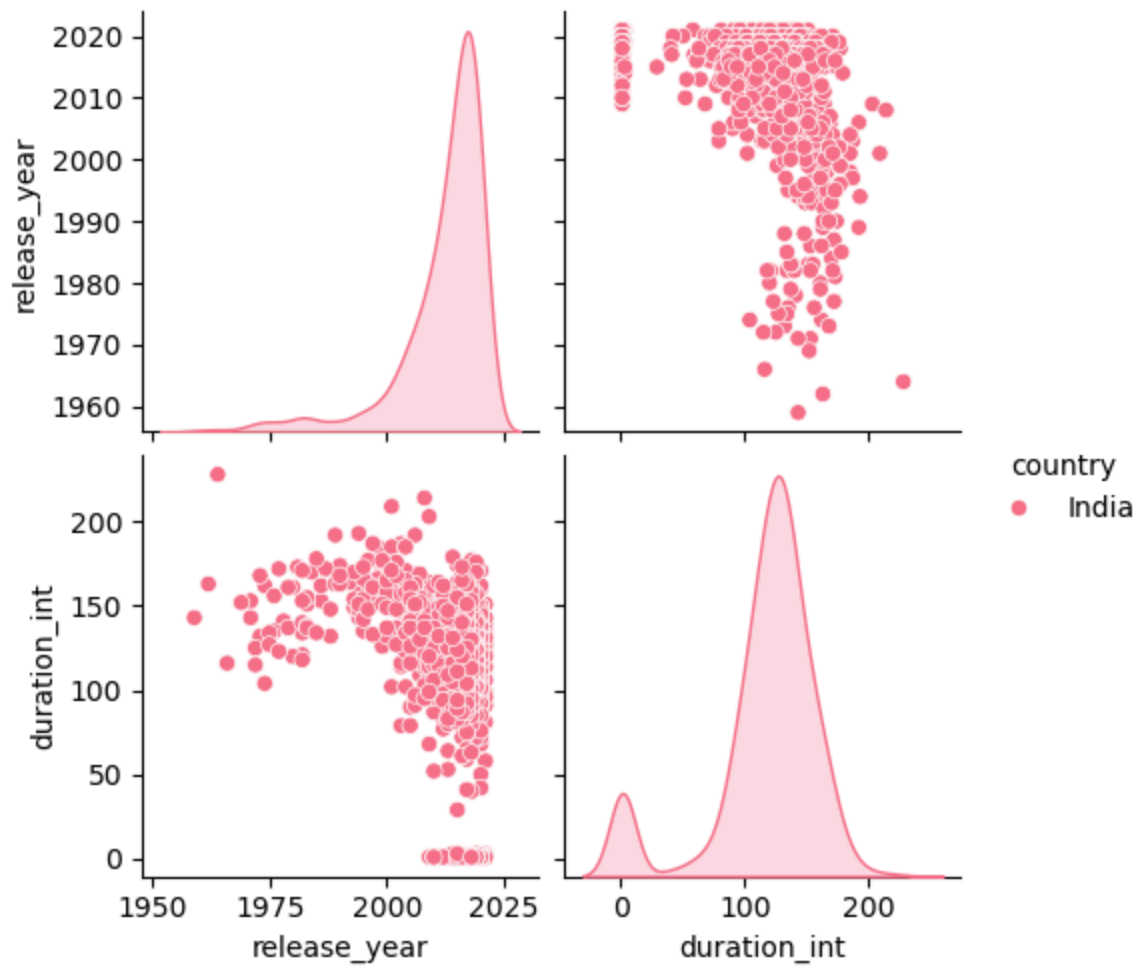
```
sns.jointplot(  
    x='release_year',  
    y='duration_int',  
    kind='hex',  
    gridsize=20,  
    data=india_movies,  
    color='steelblue'  
)  
plt.show()
```



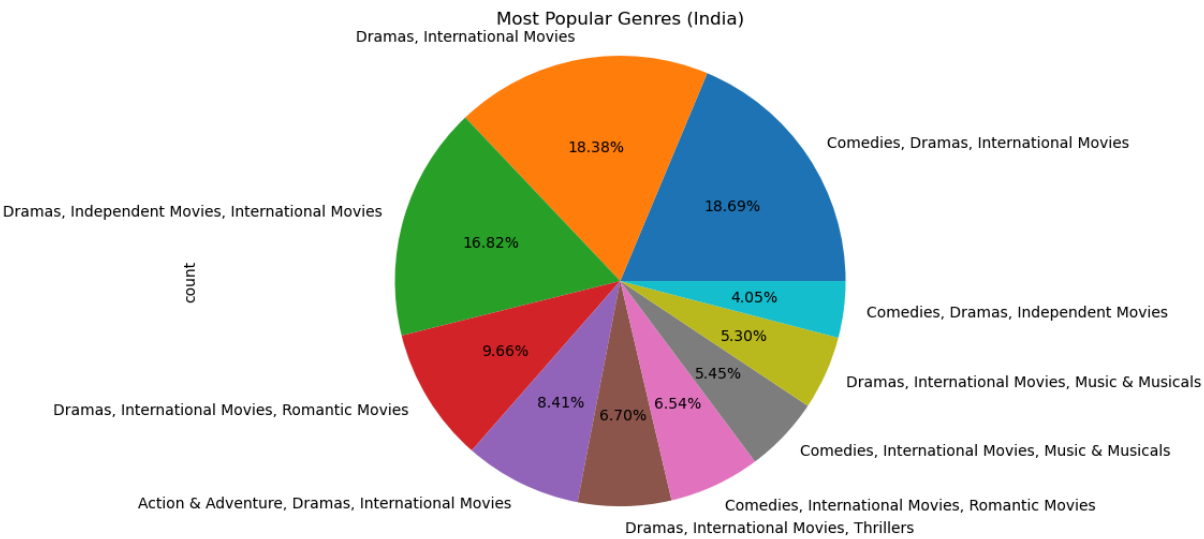
```
In [179... # duration_int

india_df = india_df.copy()
india_df['duration_int'] = india_df['duration'].str.extract(r'(\d+)').astype(float)
# pairplot

india_country = ['India']
netflix_v = india_df.loc[india_df['country'].isin(india_country)]
netflix_vpair = netflix_v[['release_year', 'duration_int', 'country']]
sns.pairplot(netflix_vpair, hue='country', palette="husl")
plt.show()
```



```
In [180... india_df['listed_in'].value_counts().sort_values(ascending=False).head(10).plot(
    kind='pie',
    figsize=(10,6),
    autopct='%1.2f%%'
)
plt.title("Most Popular Genres (India)")
plt.axis('equal')
plt.show()
```



```

In [181... # Title Percentage wise Type in India

india_country = ['India']
netflix_rate = df.loc[df['country'].isin(india_country)]
# Find total number of titles
total_titles = netflix_rate.groupby(['country'], as_index=False).count()[['country']]
total_titles.columns = ['Country', 'Total Titles']
# Find total count of each type
title_type = netflix_rate.groupby(['country', 'type'], as_index=False)[['title']].count()
title_type.columns = ['Country', 'Type', 'Total Type']
# Merge both and calculate percentage
title_type_percent = pd.merge(total_titles, title_type, on='Country')
title_type_percent['Percentage'] = (
    title_type_percent['Total Type'] /
    title_type_percent['Total Titles']
) * 100
title_type_percent

```

```

Out[181...

```

	Country	Total Titles	Type	Total Type	Percentage
0	India	972	Movie	893	91.872428
1	India	972	TV Show	79	8.127572

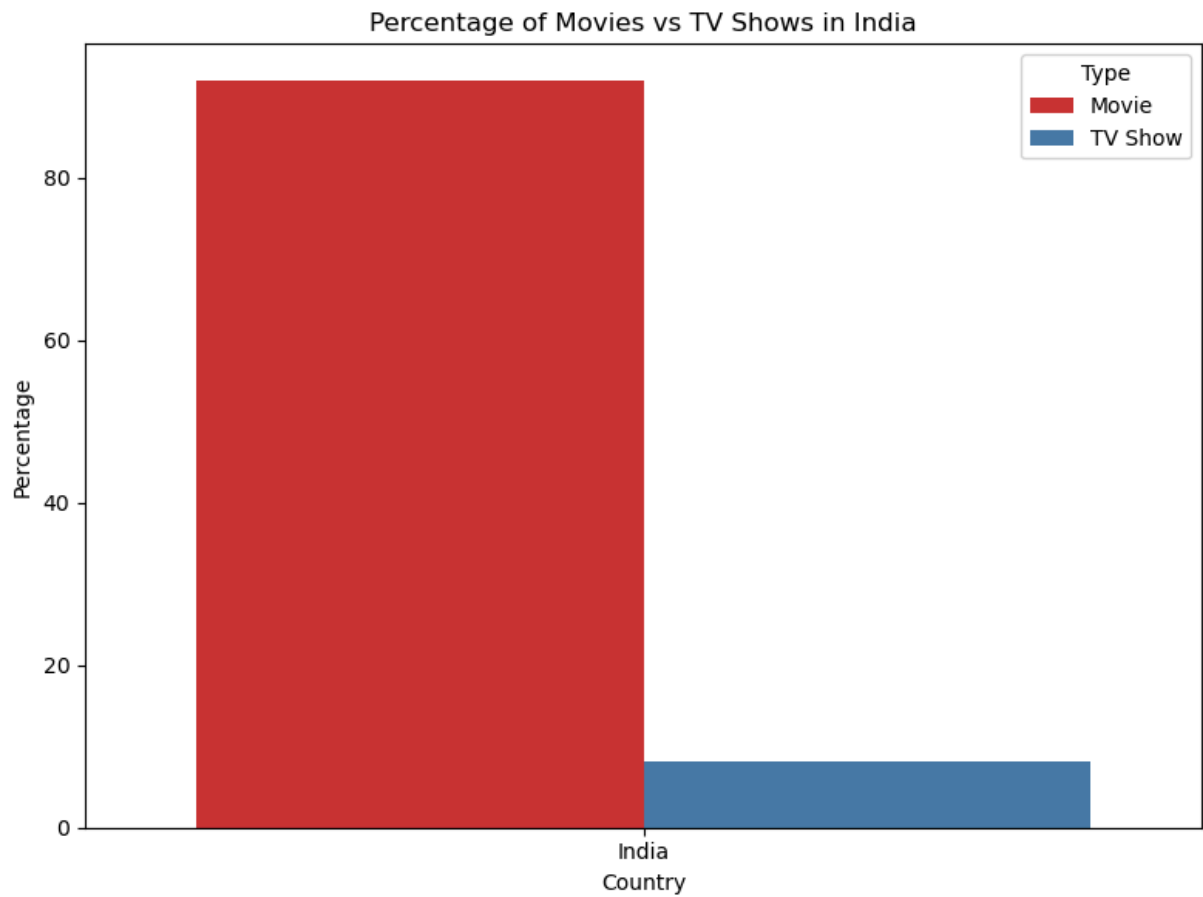
Now lets plot the above data in the form of a graph and list our findings

```

In [182... # Plot Type percentage of titles in India

fig, axx = plt.subplots(figsize=(8,6))
axx.set_title("Percentage of Movies vs TV Shows in India")
sns.barplot(
    x='Country',
    y='Percentage',
    hue='Type',
    data=title_type_percent,
    palette='Set1'
)
plt.xlabel("Country")
plt.ylabel("Percentage")
plt.tight_layout()
plt.show()

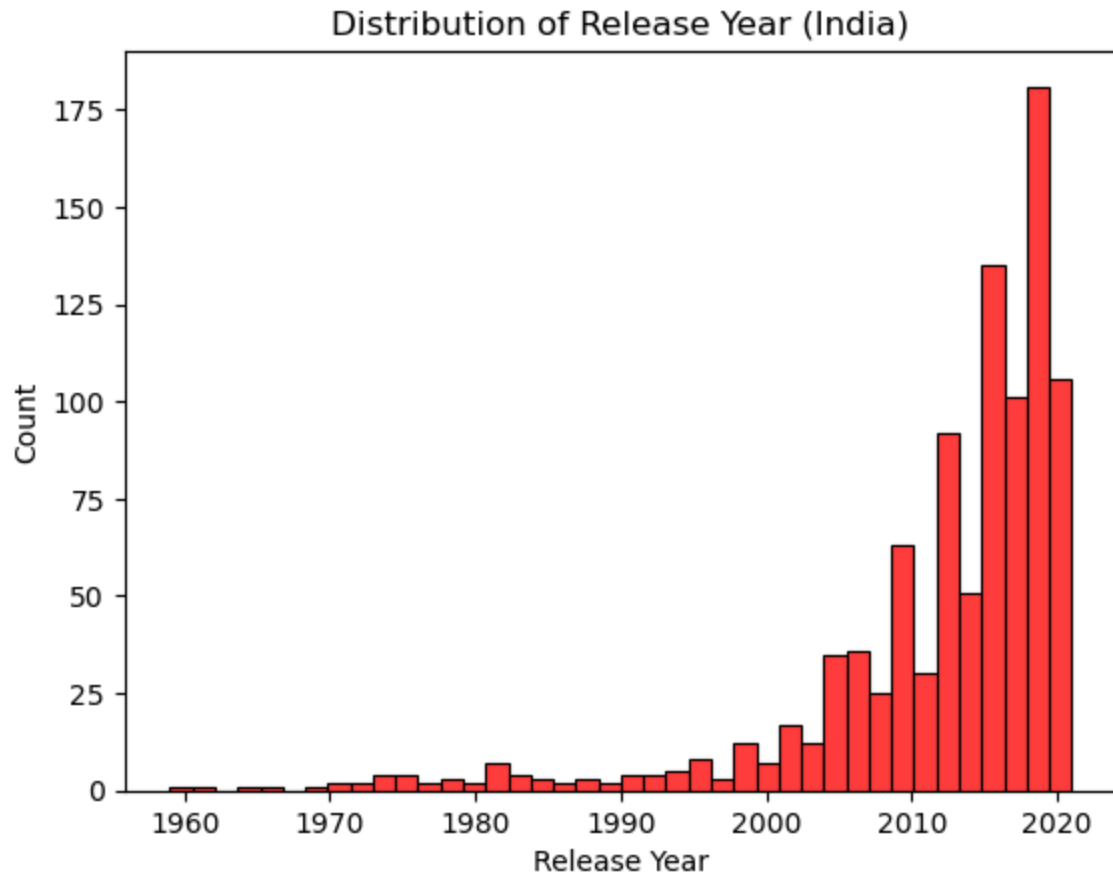
```



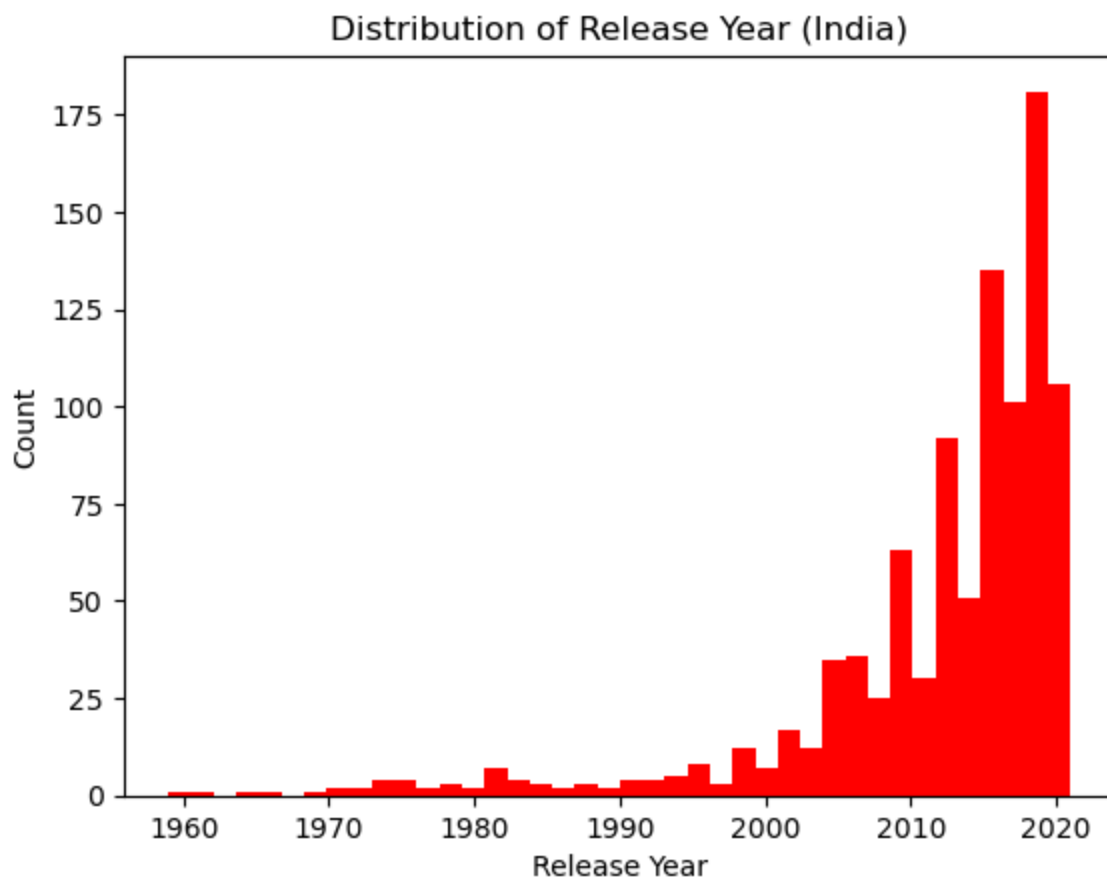
In [183...

Distribution of Release Year on Netflix (India)

```
sns.histplot(india_df['release_year'].dropna(), kde=False, bins=40, color='red')  
plt.title("Distribution of Release Year (India)")  
plt.xlabel("Release Year")  
plt.ylabel("Count")  
plt.show()
```



```
In [184... # Histogram of Release Year (India)
plt.hist(india_df['release_year'], color='red', bins=40)
plt.title("Distribution of Release Year (India)")
plt.xlabel("Release Year")
plt.ylabel("Count")
plt.show()
```



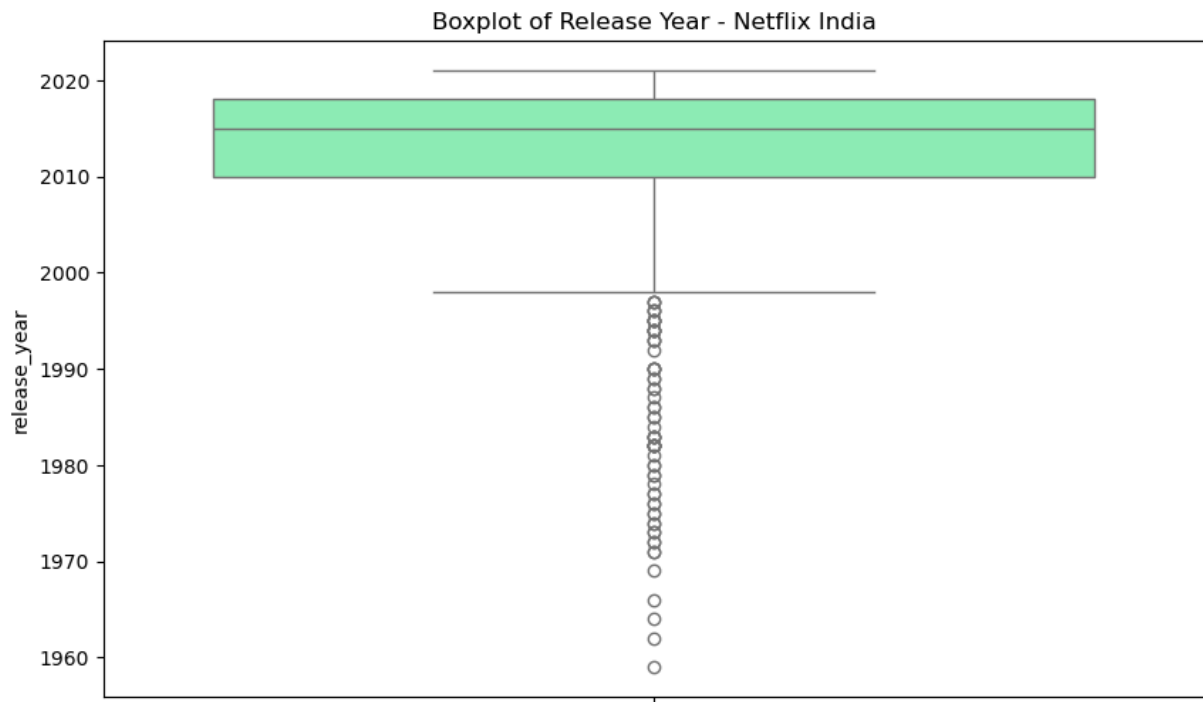
In [185...

Boxplot of Release Year (India)

```
plt.figure(figsize=(10,6))
sns.boxplot(data=india_df, y='release_year', palette='rainbow')
plt.title("Boxplot of Release Year - Netflix India")
plt.show()
```

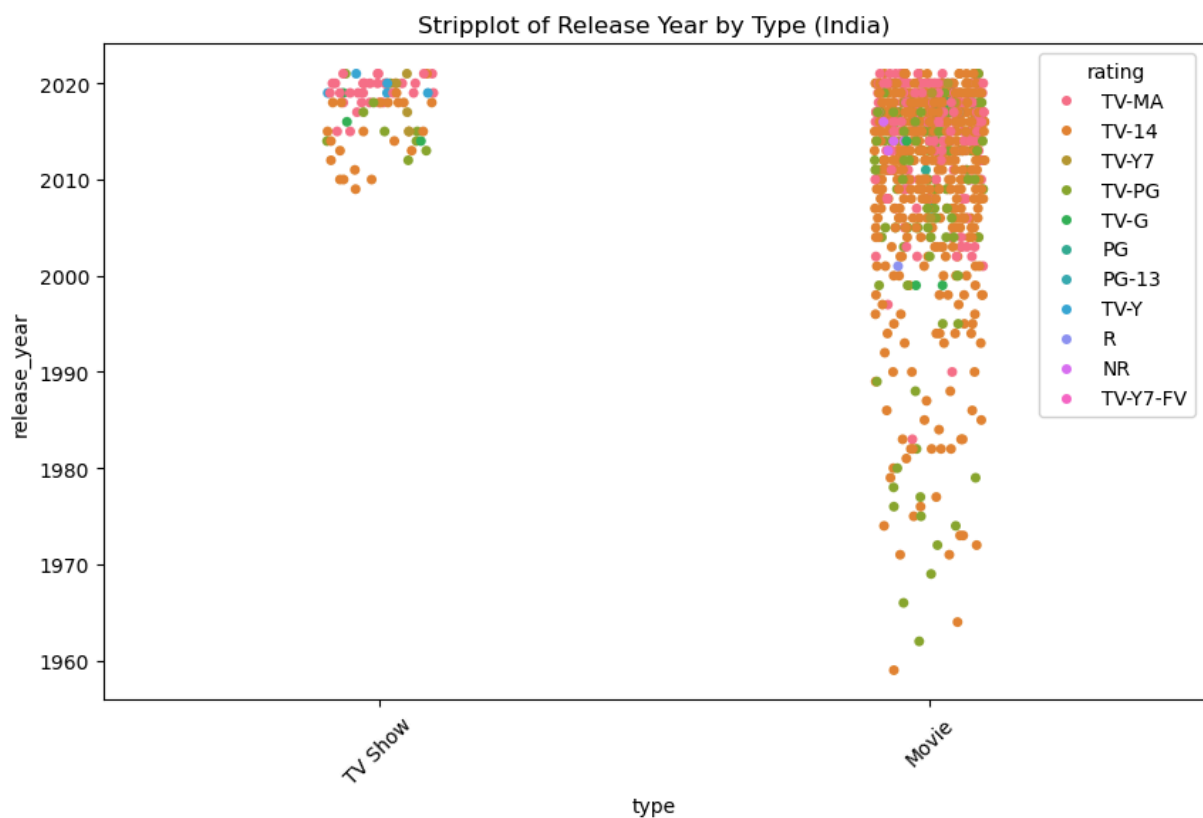
C:\Users\HP\AppData\Local\Temp\ipykernel_21952\2879505984.py:4: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effect.

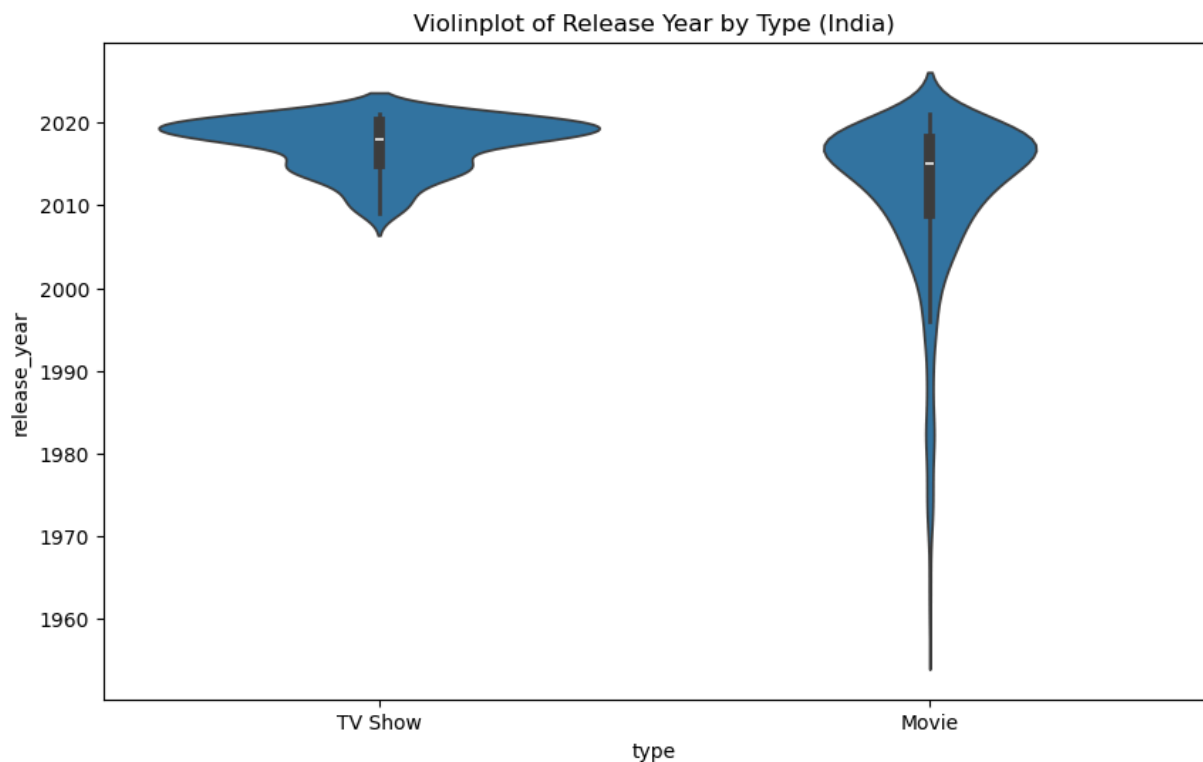


```
In [186... # Stripplot of Duration by Type hue Rating (India)

plt.figure(figsize=(10,6))
sns.stripplot(y='release_year', x='type', hue='rating', data=india_df)
plt.title("Stripplot of Release Year by Type (India)")
plt.xticks(rotation=45)
plt.show()
```

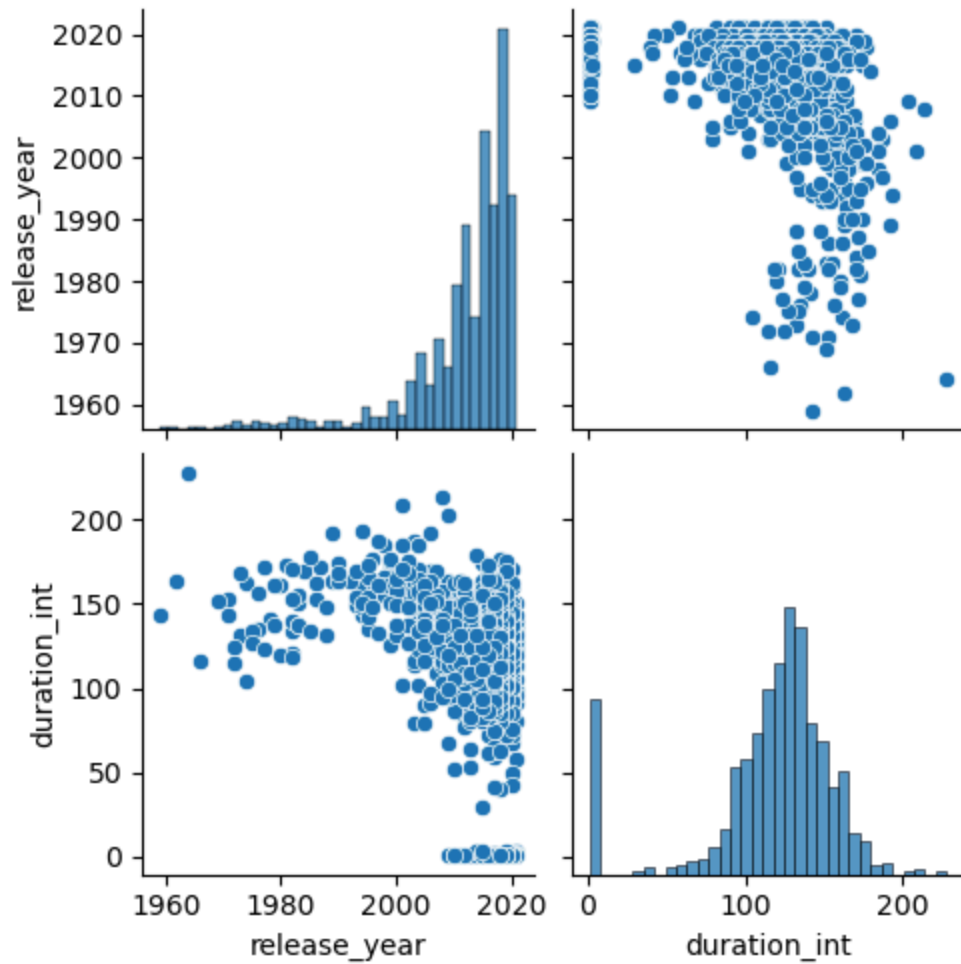


```
In [187... # Violinplot of Release Year by Type (India)
plt.figure(figsize=(10,6))
sns.violinplot(y='release_year', x='type', data=india_df)
plt.title("Violinplot of Release Year by Type (India)")
plt.show()
```

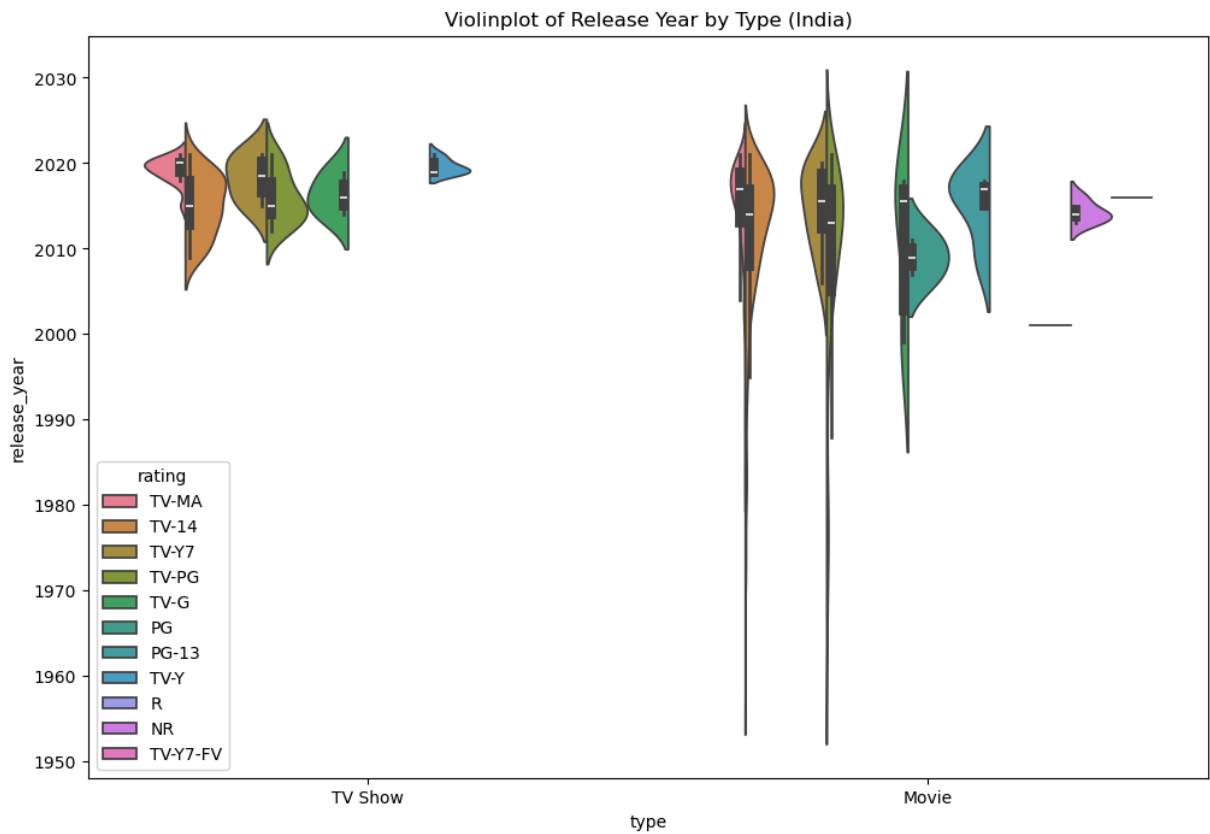


```
In [188... # Pairplot of Netflix Dataset (India)
plt.figure(figsize=(12,8))
sns.pairplot(india_df[['release_year', 'duration_int']])
plt.show()
```

<Figure size 1200x800 with 0 Axes>



```
In [189... # Violinplot of Release Year by Type with hue Rating (India)
plt.figure(figsize=(12,8))
sns.violinplot(x='type', y='release_year', data=india_df, hue='rating', split=True)
plt.title("Violinplot of Release Year by Type (India)")
plt.show()
```

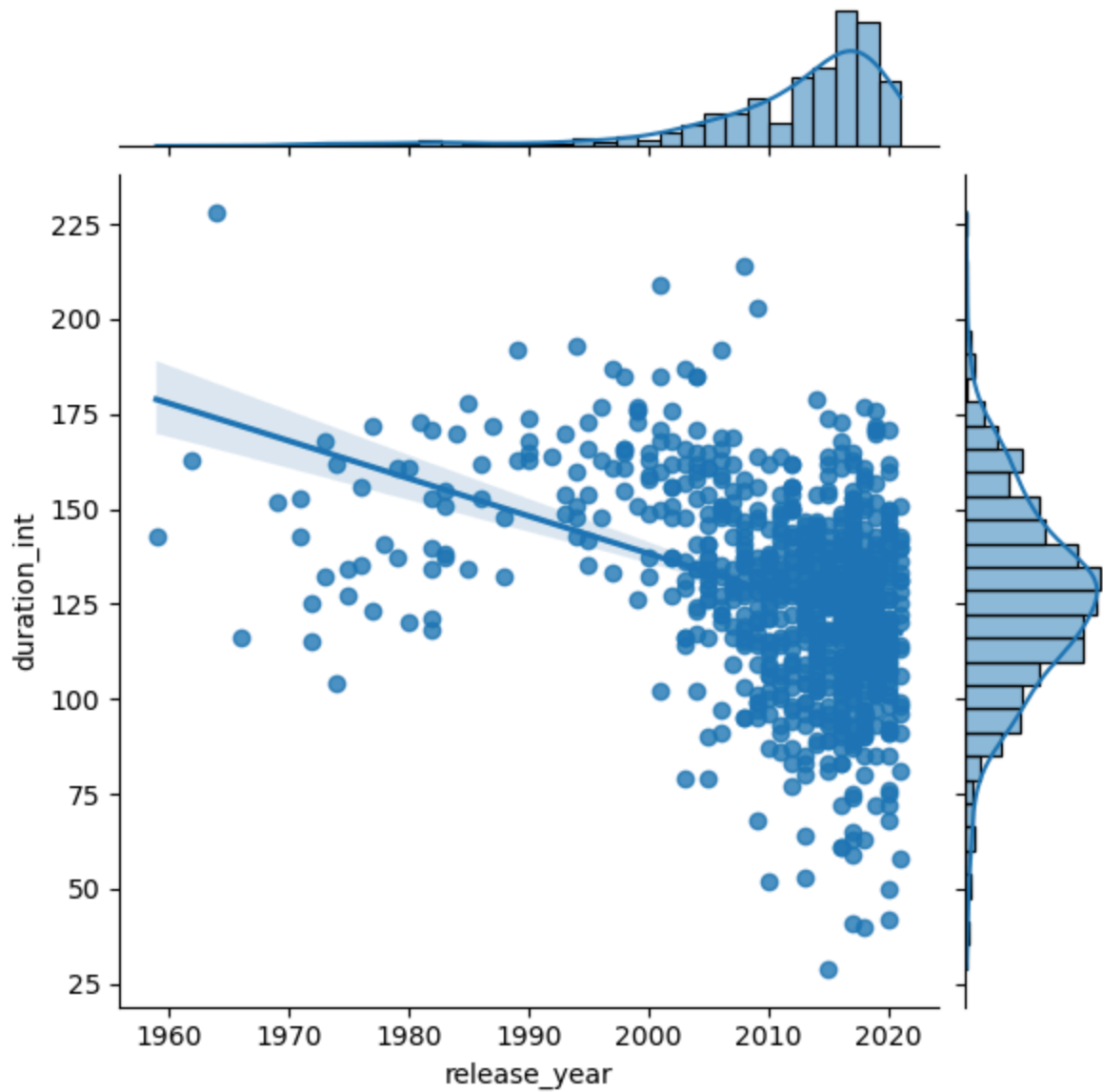


In [190...

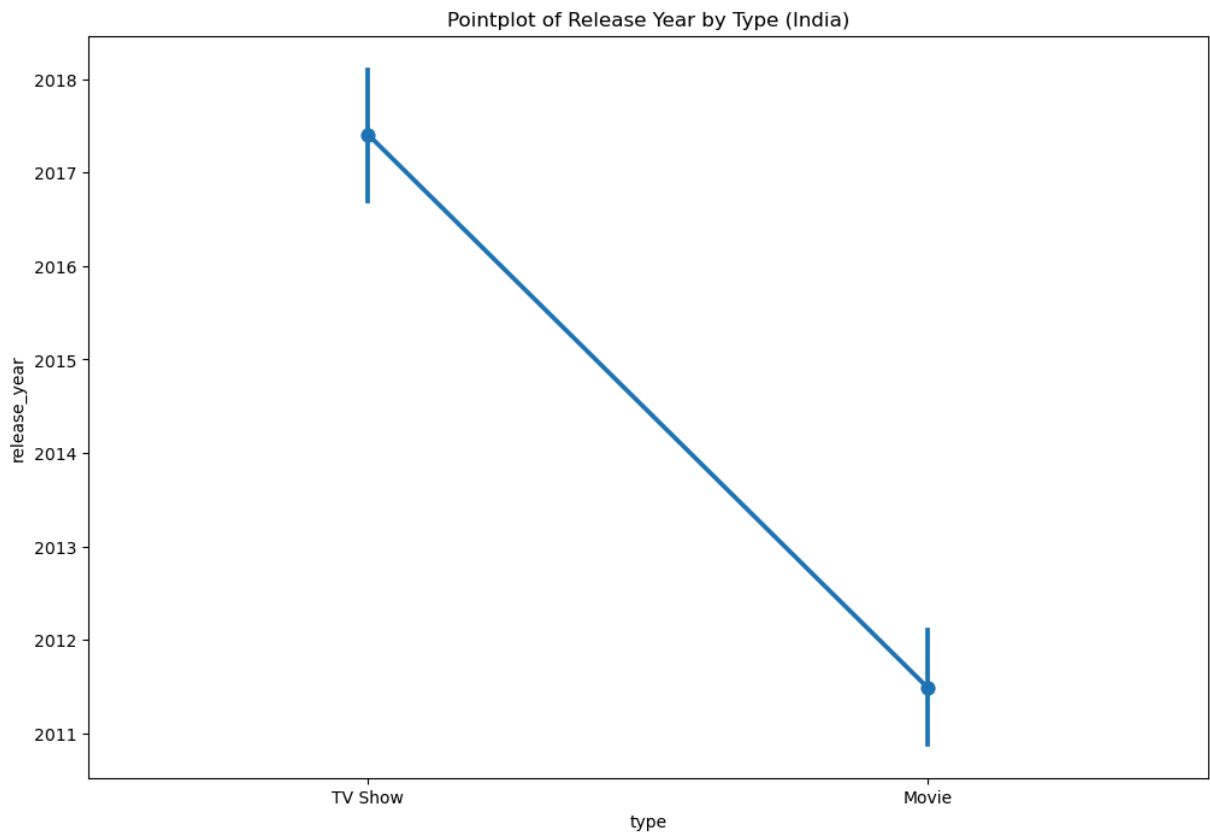
Jointplot of Release Year vs Duration (India)

```
plt.figure(figsize=(12,8))
sns.jointplot(data=india_movies, x='release_year', y='duration_int', kind='reg')
plt.show()
```

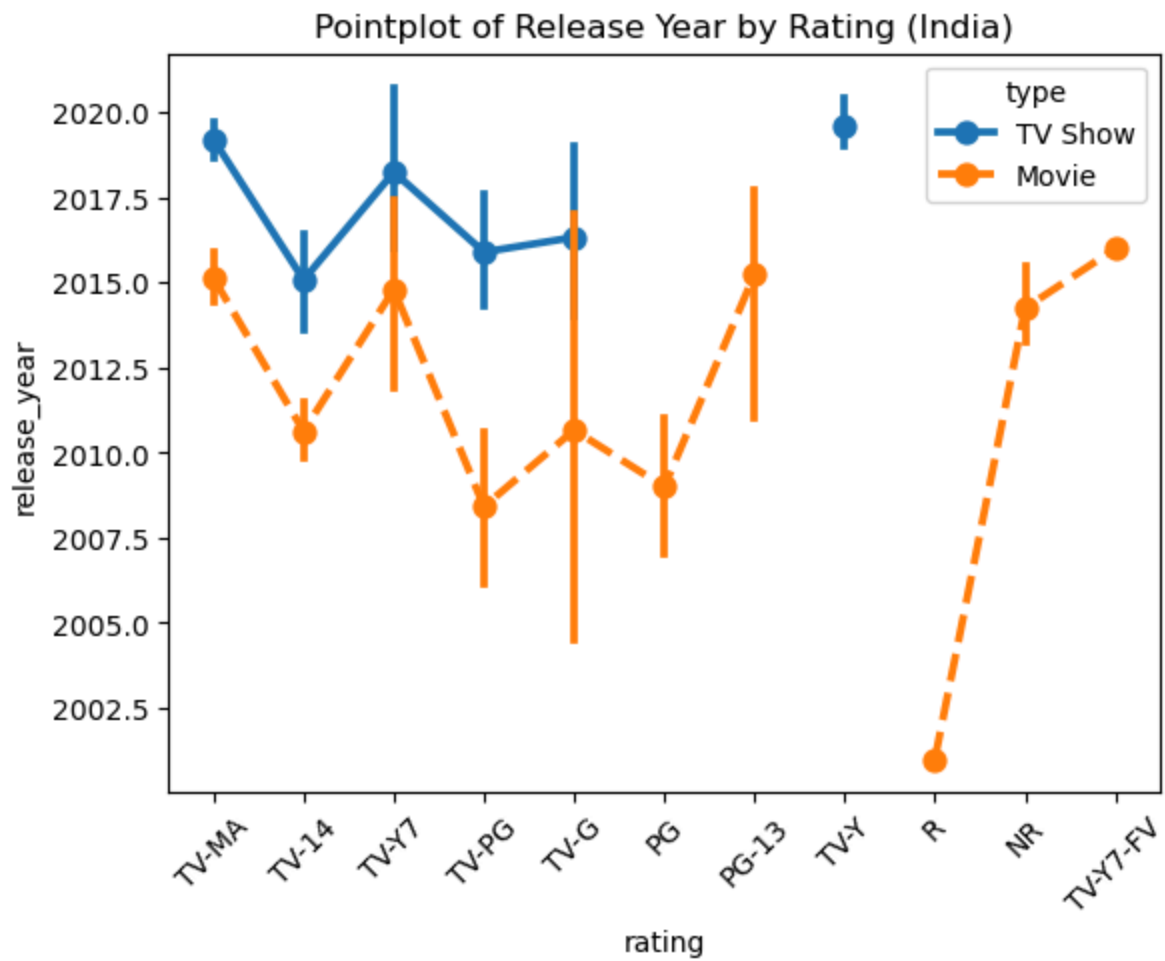
<Figure size 1200x800 with 0 Axes>



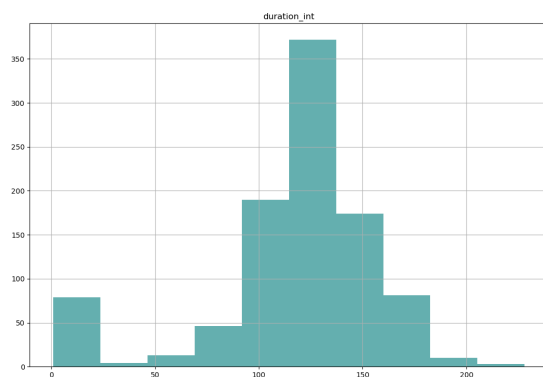
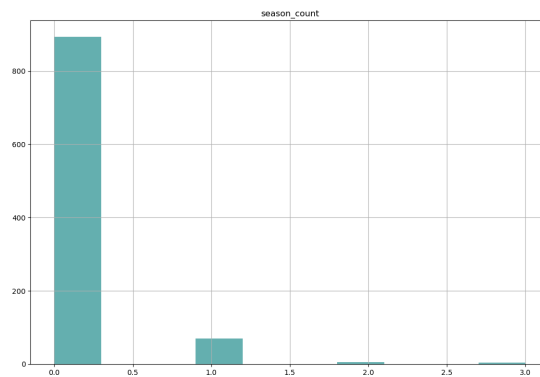
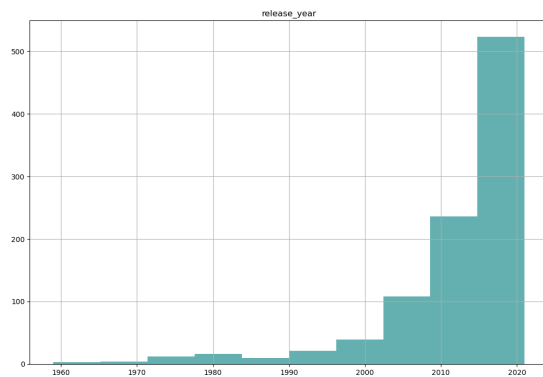
```
In [191... # Pointplot of Release Year by Type (India)
plt.figure(figsize=(12,8))
sns.pointplot(x='type', y='release_year', data=india_df)
plt.title("Pointplot of Release Year by Type (India)")
plt.show()
```



```
In [192... # Sirf Movie/TV Show 2 categories (India)
sns.pointplot(
    x='rating',
    y='release_year',
    hue='type',
    linestyle=["-", "--"],
    data=india_df
)
plt.title("Pointplot of Release Year by Rating (India)")
plt.xticks(rotation=45)
plt.show()
```

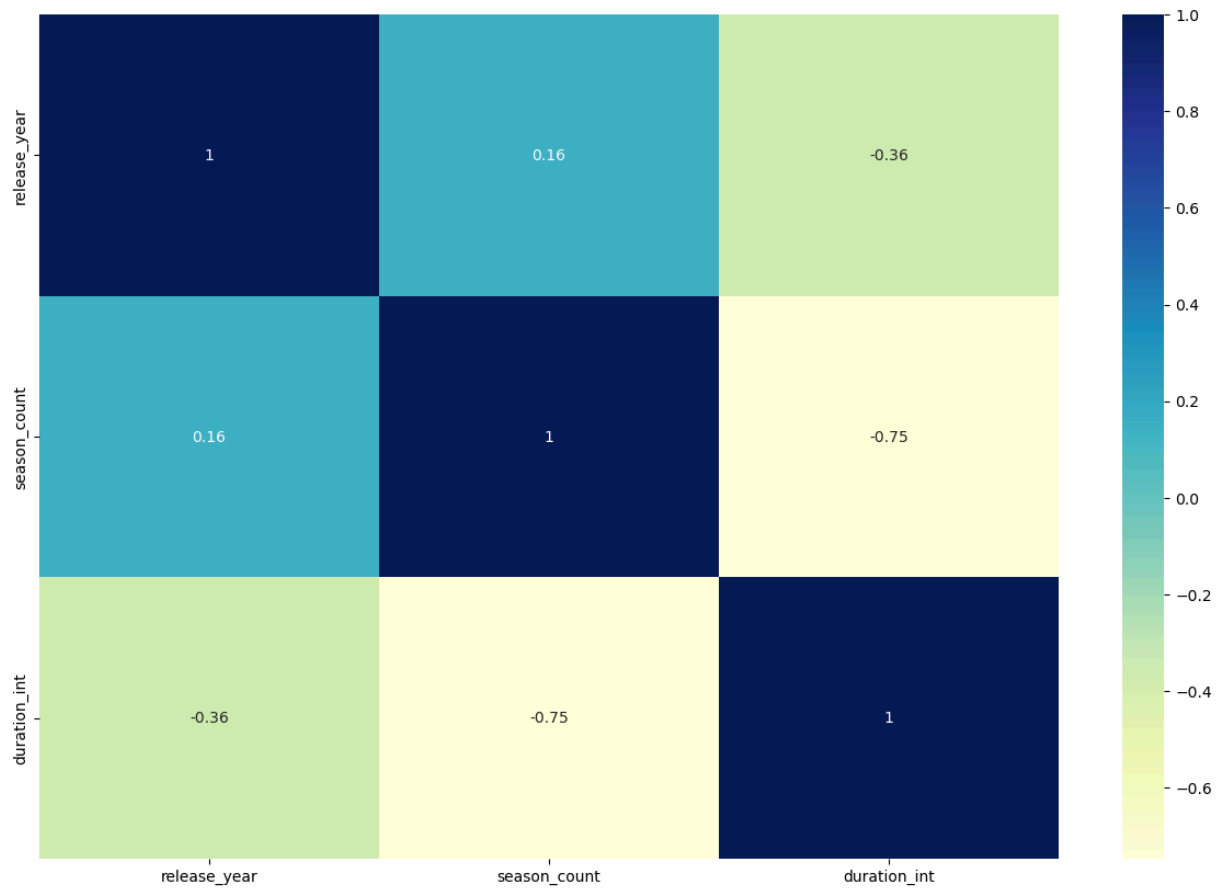


```
In [193... india_df.hist(figsize=(30,20), color='teal', alpha=0.6)
plt.show()
```



In [194...

```
corr_matrix = india_df.corr(numeric_only=True)
plt.figure(figsize=(15,10))
sns.heatmap(corr_matrix, annot=True, cmap="YlGnBu")
plt.show()
```



```
In [ ]:
```